

Intel® 10G PON Chipset

PRX126 (PRX126A1BI)

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Preliminary Data Sheet

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1 Overview

1.1 General Description

The Intel® 10G PON Chipset is a highly integrated, cost optimized and low power consumption PON/XGS-PON/NG-PON2/EPON/10G-EPON SFP+/SFP ONU system solution. On the optical side the device connects to an optical transceiver on board via the Serdes interface. On the SFP connector side it provides one high speed Serdes interface supporting XFI/SFI/SGMII. The device features real wire-speed packet switching capabilities with carrier grade QoS management independent of the packet size and application processing performed by the on-chip CPU.

1.2 Block Diagram

The block diagram of PRX126 is shown in [Figure 1](#).

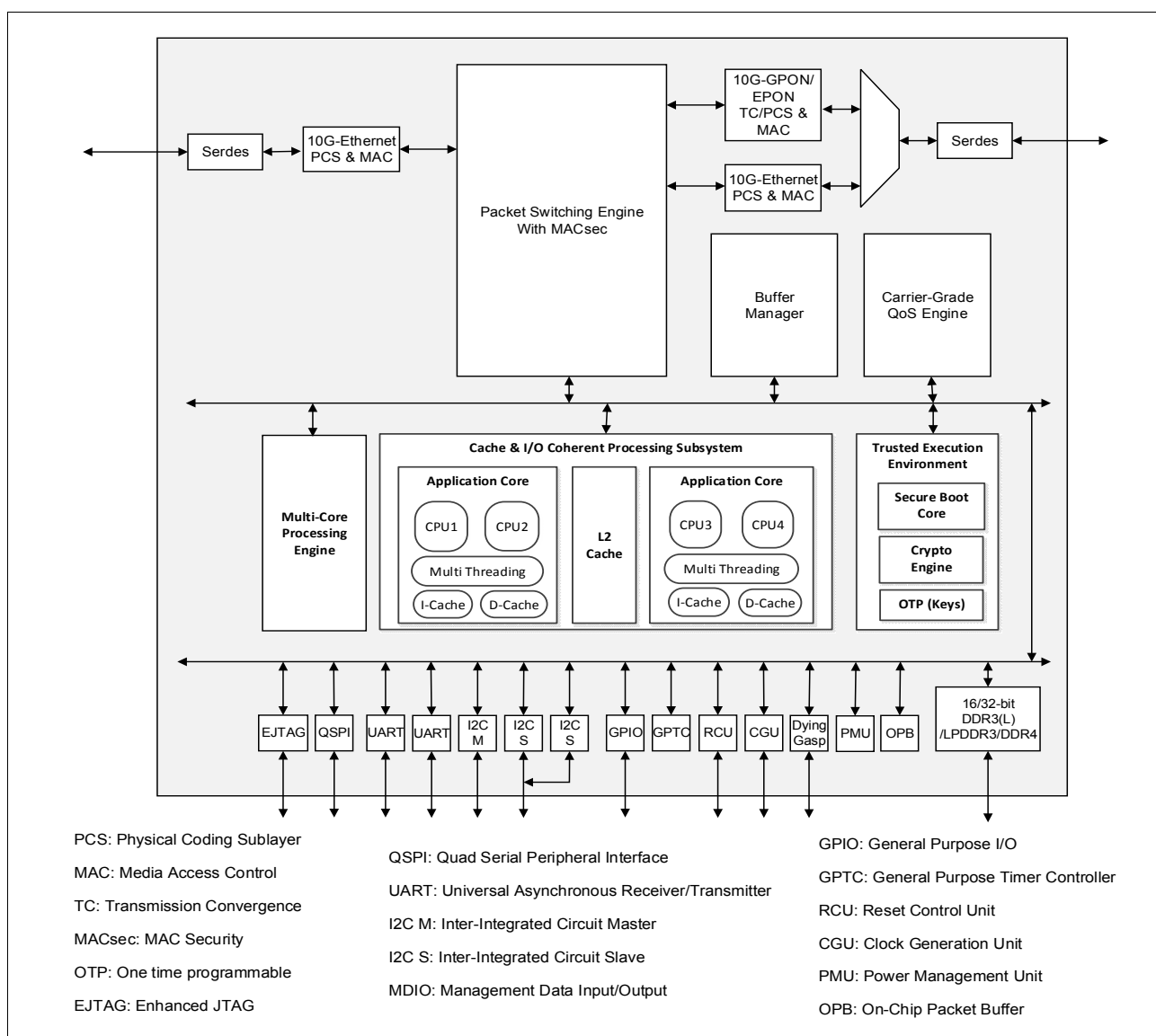


Figure 1 PRX126 Block Diagram



1.3 Features

PRX126 supports the following features:

- Line/Optical Interface with one of the following mode:
 - PON (Passive Optical Network) Mode: Integrated Transmission Convergence or Physical Coding Sublayer, MAC and Serdes connecting to an external optical transceiver; Support PON/XGS-PON/NG-PON2/EPON/10G-EPON SFP+/SFP standards, 256 GEM/XGEM ports, 64 T-CONTs and 16 LLIDs;
 - AON (Active Optical Network) Mode: Integrated Ethernet Physical Coding Sublayer, MAC and Serdes connecting to an external optical transceiver supporting 10GBASE-R and 1000BASE-X
 - Copper Wire Mode: Integrated Ethernet Physical Coding Sublayer, MAC and Serdes connecting to an external Ethernet transceiver supporting 10GBASE-T or lower speed grade
 - The optical transceiver or Ethernet transceiver is device on board.
- SFP Connector Side:
 - One Ethernet Serdes interface supporting 10G-XFI/SFI, 2.5G or 1G-SGMII
- Packet Switching Engine
 - Support wirespeed bridging, VLAN classification and modification, filtering and flow based classification for all valid packet sizes
 - 10 Gbit/s in-line MACsec encryption and decryption
- Carrier grade wirespeed QoS engine for all valid packet sizes
 - RED/WRED drop polices, hierarchical shaping and scheduling
- Cache and I/O Coherence Processor Subsystem
 - Dual core CPU (four virtual CPUs) running at 800 MHz
 - 32KB L1 instruction cache and 32KB data cache per core
 - 256KB L2 cache with I/O coherency
- Trusted Execution Environment
 - One dedicated secure boot core
 - On chip OTP module to store the Root of Trust assets
 - Integrated Crypto engine to support encryption, decryption and true random number generation
 - Access protection
- Multiple-Core Processing Engine
 - Integrated hardware offloading logics to enable flexible and high speed packet processing, for example, routing and tunneling
- High speed 16 or 32-bit 2133-DDR3, DDR3L, LPDDR3 and 2666-DDR4 interface
- Integrated Power on reset and Dying gasp detection logics
- Advanced power management capabilities
- G.8275/Y.1369/IEEE 1588v2 precise time protocol support and Synchronous Ethernet support
- Accurate IEEE 802.1ag and ITU-T Y.1731 Ethernet OAM delay and loss measurement hardware support

1.4 Ordering Information

This section shows the product and packaging information of PRX126 device.

Table 1 Product and Package Naming

Product Name	Ordering Code	S-Spec#	Package
PRX126	PRX126A1BI	t.b.d	PG-VF2RBGA-287-9.4x8.6-C-1A

1.5 Typical Application

This section shows the typical application of PRX126 device.

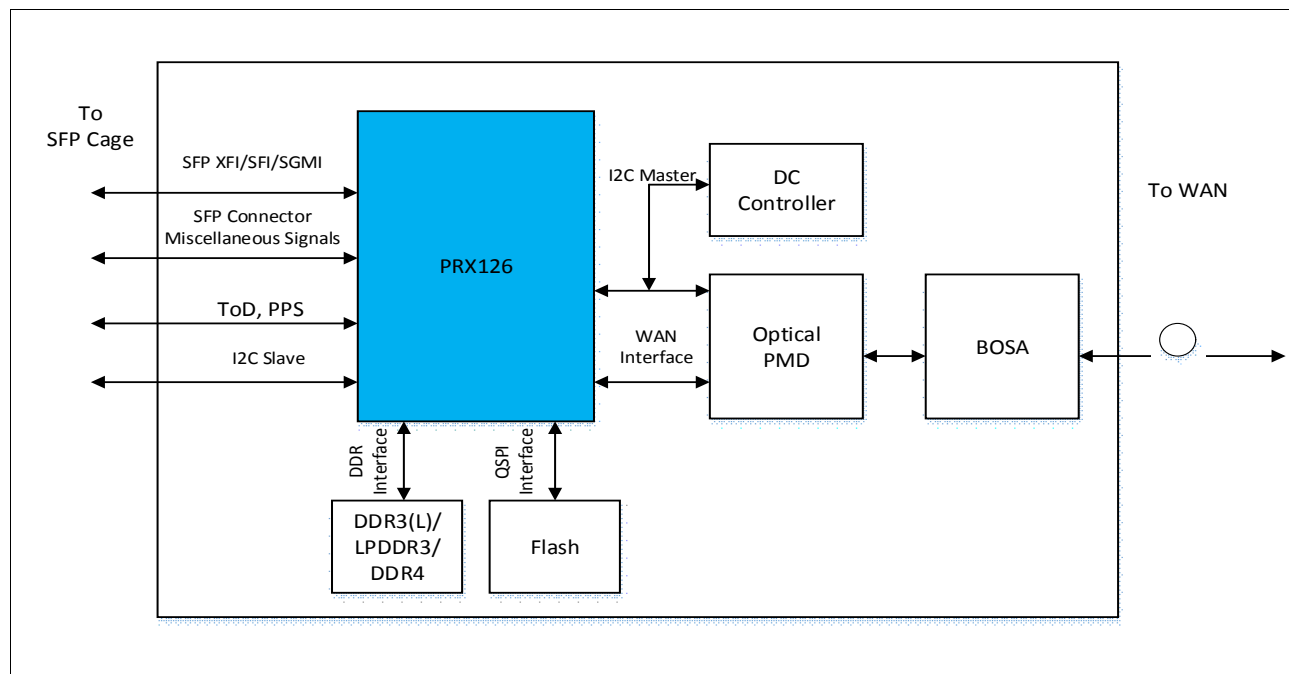


Figure 2 PRX126 Typical Application



2 Pin Descriptions

This chapter describes the pin diagram and pin assignments in PRX126.

2.1 List of Interfaces

Table 2 summarizes the supported interfaces.

Table 2 External Signals

Interface	Link
WAN Signals	Table 5
SFP XFI/SFI/SGMII Signals	Table 6
SFP Connector Miscellaneous Signals	Table 7
UART Signals	Table 8
QSPI Signals	Table 9
I2C Signals	Table 10
Clock Signals	Table 11
GPIO Signals	Table 12
EJTAG Signals	Table 13
16/32-bit DDR SDRAM Signals	Table 14
Reset Signals	Table 15
Power	Table 16
Ground	Table 17
Fuse	Table 18
Dying Gasp	Table 19
Reserved	Table 20



2.2 Ball Diagram

The ball diagram is shown in [Figure 3](#). The color code means:

- White in outer ring = Unpopulated balls
- Green = DDR Interface
- Orange = Power
- Red = WAN Signals
- Blue = SFP Connector XFI/SFI/SGMII Signals
- Pink = Clock, Reset, Dying Gasp
- Yellow = GPIO, EJTAG Interface
- Grey = Ground, Fuse, Reserved
- White = NC

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
A		VSS	DDR_DM0	DDR_DQ5	DDR_CS0	DDR_BA0	DDR_BG1	DDR_BA1	DDR_CK_T	DDR_CK_C	DDR_A8	DDR_A7	DDR_A5	DDR_A0	DDR_DQ20	DDR_DQ16	DDR_DQ22	VSS	A
B	DDR_DQS0_C	DDR_DQ3	DDR_DQ7	DDR_DQ1	VSS	DDR_CS1	DDR_ODT0	DDR_A15	VSS	DDR_A11	DDR_A9	DDR_A6	DDR_A2	DDR_A1	VSS	DDR_DQ18	DDR_DQ17	DDR_DQS2_T	B
C	DDR_DQS0_T	DDR_DQ6	DDR_DQ4	VSS	VSS	DDR_ODT1	DDR_CKE0	DDR_BG0	DDR_A16	DDR_A12	DDR_A10	DDR_A3	DDR_A4	VSS	VSS	DDR_DQ19	DDR_DQ23	DDR_DQS2_C	C
D	DDR_DQ2	DDR_DQ0	VSS	VDDQ	VSS	VDDQ	DDR_CKE1	VSS	DDR_A14	DDR_A13	VSS	DDR_RST	VDDQ	VSS	VDDQ	VSS	DDR_DQ21	DDR_DM2	D
E	DDR_DM1	DDR_DQ12	DDR_DQ14	VSS	VDDQ	VDDQ	DDR_ZQ	VDDQ	VSS	VSS	VDDQ	DDR_ATO	VDDQ	VDDQ	VSS	DDR_DQ29	DDR_DQ25	DDR_DQ31	E
F	DDR_DQS1_C	DDR_DQ8	DDR_DQ10	VDDQ	VSS	VSS	VAA1_V8_D_DRP*	DDR_ALERT	DDR_ACT	VAA1_V8_D_DRP*	DDR_DTO0	VAA1_V8_D_DRP*	VSS	VSS	VDDQ	DDR_DQ30	DDR_DQ27	DDR_DQS3_C	F
G	DDR_DQS1_T	DDR_DQ15	DDR_DQ9	VSS	VDD0_V9	DDR_VREF_DQ0	VSS	DDR_PAR	DDR_VREF_CA	VSS	VDD0_V9	DDR_DTO1	DDR_VREF_DQ1	VDD0_V9	VSS	DDR_DQ26	DDR_DM3	DDR_DQS3_T	G
H	VSS	DDR_DQ11	DDR_DQ13	VDD_I_O_0	VDD_I_O_0	VSS	VDD0_V9	VSS	VDD0_V9	VDD0_V9	VSS	VDD0_V9	VSS	VDD0_V9	VSS	DDR_DQ24	DDR_DQ28	VSS	H
J	UTXD0	URXD0	TRST	VSS	VAA1_V8_P_VT_L	VSS	VSS	VDD0_V9	VSS	VSS	VDD0_V9	VSS	VDD0_V9	VSS	VSS	Reserved	VSS	VSSA_XO	J
K	VSSA_SPH_Y1	POR	HRST	TDI	VAA1_V8_S_PHY1	VSSA_SPH_Y1	VDD0_V9	VSS	VSS	VSS	VSS	VDD0_V9	VSS	VDD0_V9	VSS	VDD_DG	VSSA_XO	VAA1_V8_XO	K
L	P1_T_XP	VSSA_SPH_Y1	TDO	TCK	VSSA_SPH_Y1	VAA0_V9_S_PHY1	VSS	VDD0_V9	VSS	VSS	VDD0_V9	VSS	VDD0_V9	VSS	VDD_I_O_2	VSS	VSS	XTAL2	L
M	P1_T_XN	VSSA_SPH_Y1	TMS	VSSA_SPH_Y1	VAA1_V8_S_PHY1	VAA0_V9_S_PHY1	VDD0_V9	VSS	VDD0_V9	VDD0_V9	VSS	VDD0_V9	VSS	VDD0_V9	VSS	VDD_I_O_1	VSS	VSS	M
N	VSSA_SPH_Y1	P1_R_ESREF	GPIO_30	VSS	VSSA_SPH_Y1	VSSA_SPH_Y1	GPIO_22	GPIO_20	VDD_I_O_3	VSS	VSS	VSS	VSS	VSS	VDD_I_O_1	VDD_I_O_1	VSS	GPIO1	N
P	P1_R_XP	VSSA_SPH_Y1	GPIO_31	VAA1_V8_O_TP	VSS	VAA0_V9_P_LL	GPIO_23	GPIO_21	VSS	VAA0_V9_P_LL5	VAA1_V8_S_PHY2	VAA0_V9_S_PHY2	VSS	VSS	VSS	VSS	GPIO8	GPIO3	P
R	P1_R_XN	VSSA_SPH_Y1	GPIO_11	VSS	VAA1_V8_P_LL	VSS	GPIO_26	GPIO_24	VAA1_V8_P_LL5	VSSA_SPH_Y2	VSSA_SPH_Y2	P2_R_ESREF	VSSA_SPH_Y2	VSSA_SPH_Y2	VSS	GPIO_16	GPIO5	GPIO4	R
T	VSSA_SPH_Y1	VSSA_SPH_Y1	GPIO_10	GPIO_29	GPIO_28	OPT_TXEN	GPIO_27	GPIO_25	VSSA_SPH_Y2	P2_T_XN	P2_T_XP	VSSA_SPH_Y2	P2_R_XP	P2_R_XN	VSSA_SPH_Y2	GPIO_15	GPIO_13	VSS	T
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	

Figure 3 Ball Diagram for PG-VF2RBGA-287 (Top View)



2.3 Ball (Pin) Function Description

Use the following abbreviations for I/O table.

Table 3 Abbreviations for Pin Type

Abbreviations	Description
I	Standard Input-only pin. Digital levels.
O	Standard Output-only pin. Digital levels.
I/O	I/O is a bidirectional Input/Output signal.
AI	Input. Analog levels.
AO	Output. Analog levels.
AI/O	Input or Output. Analog levels.
PWR	Power
GND	Ground
MCL	Must be connected to Low (JEDEC Standard).
MCH	Must be connected to High (JEDEC Standard).
NU	Not Usable (JEDEC Standard).
NC	Not Connected (JEDEC Standard). Do not connect in board design.
Prg	Programmable (either input or output).

Table 4 Abbreviations for Buffer Type

Abbreviations	Description
Z	High impedance
Prg	Programmable (OD/PP, PU/PD are programmable)
PU	Pull up (internal, weak)
PD	Pull down (internal, weak)
TS	Tristate capability: The corresponding pin has 3 operational states: Low, high and high-impedance.
OD	Open Drain. The corresponding pin has 2 operational states, active low and tristate, and allows multiple devices to share as a wire-OR. An external pull-up is required to sustain the inactive state until another agent drives it, and must be provided by the central resource.
OC	Open Collector.
PP	Push-Pull. The corresponding pin has 2 operational states: Active-low and active-high (identical to output with no type attribute).
OD/PP	Open-Drain or Push-Pull. The corresponding pin can be configured either as an output with the OD attribute or as an output with the PP attribute.



2.3.1 WAN Signals

The WAN signals are described in [Table 5](#).

Table 5 WAN Signals

Ball No.	Name	Pin Type	Buffer Type	Function
Signal Group Analog				
T11	P2_TXP	AO	–	Transmit Data Pair (Differential)
T10	P2_TXN	AO	–	
T13	P2_RXP	AI	–	Receive Data Pair (Differential)
T14	P2_RXN	AI	–	
R12	P2_RESREF	AI/AO	–	Reference Resistor <i>Note: Connect to a 200 Ω external resistor with +/-1% accuracy.</i>
Signal Group Digital				
T6	OPT_TXEN	I/O	PP	Optical PMD TX Burst Enable Indicating WAN TX Burst Enable to the external optical transceiver.
N8	OPT_RX_LOS	I	Prg	Optical PMD RX LOS It is an input signal from the external optical transceiver. It shares the pin with other functionality on GPIO20.
N7	OPT_TX_FAULT	I	Prg	Optional PMD TX Fault It is an input signal from the external optical transceiver. It shares the pin with other functionality on GPIO22.
P7	OPT_TX_SD	I	Prg	Optional PMD TX Signal Detected It is an input signal from the external optical transceiver. It shares the pin with other functionality on GPIO23.
R8	OPT_TX_PUP	I/O	Prg	Optical PMD TX Power Up It is connected to the external optical transceiver. It shares the pin with other functionality on GPIO24.



2.3.2 SFP XFI/SFI/SGMII Signals

The SFP XFI/SFI/SGMII Signals are described in [Table 6](#).

Table 6 SFP XFI/SFI/SGMII Signals

Ball No.	Name	Pin Type	Buffer Type	Function
Signal Group Analog				
L1	P1_TXP	AO	–	Transmit Data Pair (Differential)
M1	P1_TXN	AO	–	
P1	P1_RXP	AI	–	Receive Data Pair (Differential)
R1	P1_RXN	AI	–	
N2	P1_RESREF	AI/AO	–	Reference Resistor <i>Note: Connect to a 200 Ω external resistor with +/-1% accuracy.</i>

2.3.3 SFP Connector Miscellaneous Signals

Table 7 SFP Connector Miscellaneous Signals

Pin No.	Name	Pin Type	Buffer Type	Function
P8	OPT_TX_DIS_SFP	I/O	Prg	Optical PMD TX Disable SFP It is connected to SFP/SFP+ module connector. It shares the pin with other functionality on GPIO21.
T8	OPT_RX_SD_SFP	I/O	Prg	Optical PMD RX Signal Detected SFP It is connected to SFP/SFP+ module connector. It shares the pin with other functionality on GPIO25.

2.3.4 UART Signals

There are two sets of UART external interfaces. Multiple UART modules are multiplexed over these two external UART interfaces.

Table 8 UART Signals

Pin No.	Name	Pin Type	Buffer Type	Function
J2	URXD0	I	–	SoC UART 0 Data Input SoC UART instance #0 interface data input.
	URXD_QOS			QoS Engine Debug UART Input QoS Engine debug interface Input.
	URXD0_PON			PON Debug UART Data Input PON UART instance #0 (processor debug) interface input.



Table 8 UART Signals (cont'd)

Pin No.	Name	Pin Type	Buffer Type	Function
J1	UTXD0	O	PD	SoC UART 0 Data Output SoC UART instance #0 interface data output <i>Note: This is used for pin strapping BOOT3.</i>
	UTXD_QOS			QoS Module Debug UART Output QoS Module debug interface Output
	UTXD0_PON			PON Debug UART Output PON UART instance #0 (processor debug) interface output
N3	URXD1	I	Prg	SoC UART 1 Data Input SoC UART instance #1 interface data input. It shares the pin with other functionality on GPIO30.
	URXD1_PON			PON UART 1 Data Input PON UART instance #1 interface data input. It shares the pin with other functionality on GPIO30.
	URXD_TEP			Secure Boot Core Debug Data Input Secure boot core UART debug interface data input. It shares the pin with other functionality on GPIO30.
P3	UTXD1	O	Prg	SoC UART 1 Data Output SoC UART instance # 1 interface data output. It shares the pin with other functionality on GPIO31.
	UTXD1_PON			PON UART 1 Data Output PON UART instance #1 interface data output. It shares the pin with other functionality on GPIO31.
	UTXD_TEP			Secure Boot Core Debug Data Output Secure boot core UART debug interface data output. It shares the pin with other functionality on GPIO31.



2.3.5 QSPI Signals

Table 9 QSPI Signals

Pin No.	Name	Pin Type	Buffer Type	Function
N18	QSCK	O	Prg	QSPI Clock QSPI clock output. It shares the pin with other functionality on GPIO0.
N17	QSCS	O	Prg	QSPI Chip Select QSPI chip select. It shares the pin with other functionality on GPIO1.
P18	QSIO0	I/O	Prg	QSPI IO Bit 0 QSPI interface input/output bit 0. It shares the pin with other functionality on GPIO2.
P17	QSIO1	I/O	Prg	QSPI IO Bit 1 QSPI interface input/output bit 1. It shares the pin with other functionality on GPIO3.
R18	QSIO2	I/O	Prg	QSPI IO Bit 2 QSPI interface input/output bit 2. It shares the pin with other functionality on GPIO4.
R17	QSIO3	I/O	Prg	QSPI IO Bit 3 QSPI interface input/output bit 3. It shares the pin with other functionality on GPIO5.

2.3.6 I2C Signals

There are two sets of I2C external interfaces.

Table 10 I2C Signals

Pin No.	Name	Pin Type	Buffer Type	Function
R7	I2C0_SCL	I/O	Prg	I2C Master 0 Serial Clock Line (SCL) I2C master instance #0 clock. It shares the pin with other functionality on GPIO26.
T7	I2C0_SDA	I/O	Prg	I2C Master 0 Serial Data (SDA) I2C instance #0 data. It shares the pin with other functionality on GPIO27.
T3	I2C1_SCL	I/O	Prg	I2C Slave 1 Serial Clock Line (SCL) Both I2C slave instance #1 and slave instance I2C #2 are connected to this interface. It shares the pin with other functionality on GPIO10.
R3	I2C1_SDA	I/O	Prg	I2C Slave 1 Serial Data (SDA) Both I2C slave instance #1 and I2C slave instance #2 are connected to this interface. It shares the pin with other functionality on GPIO11.



2.3.7 Clock

The Clock signals are described in [Table 11](#).

Table 11 Clock Signals

Ball No.	Name	Pin Type	Buffer Type	Function
L18	XTAL1	AI	–	External Crystal Input/Output
L17	XTAL2	AO	–	
K18	CLK_MD	AI	–	Clock Mode Select This pin selects external crystal or direct clock drive mode. 1 _D Crystal mode 0 _D Clock direct clock drive mode
T5	GPC1	Prg	Prg	General Purpose Clock 1 General Purpose clock for Synchronous Ethernet or external devices. It shares the pin with other functionality on GPIO28.
T4	PPS	Prg	Prg	Pulse Per Second Pulse per second. It shares the pin with other functionality on GPIO29.



2.3.8 General Purpose I/O

General Purpose I/O pin input mode can be used as external interrupt inputs.

For each GPIO, there are multiple alternative functionality. Please refer to [GPIO Functions](#) for the multiplexing

Table 12 GPIO Signals

Pin No.	Name	Pin Type	Buffer Type	Function
N18	GPIO0	Prg	Prg	General Purpose IO Bit 0 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
N17	GPIO1	O	Prg	General Purpose IO Bit 1 It is used for output pin. The output characteristic can be selected to be open drain or push-pull. <i>Note: This is used for pin strapping BOOT0.</i>
P18	GPIO2	Prg	Prg	General Purpose IO Bit 2 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
P17	GPIO3	Prg	Prg	General Purpose IO Bit 3 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
R18	GPIO4	Prg	Prg	General Purpose IO Bit 4 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
R17	GPIO5	Prg	Prg	General Purpose IO Bit 5 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
P16	GPIO8	Prg	Prg	General Purpose IO Bit 8 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
T3	GPIO10	Prg	Prg	General Purpose IO Bit 10 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
R3	GPIO11	Prg	Prg	General Purpose IO Bit 11 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.



Table 12 GPIO Signals (cont'd)

Pin No.	Name	Pin Type	Buffer Type	Function
T17	GPIO13	O	Prg	General Purpose IO Bit 13 It is used for output pin. The output characteristic can be selected to be open drain or push-pull. <i>Note: This is used for pin strapping ENDIAN_BOOT.</i>
T16	GPIO15	O	Prg	General Purpose IO Bit 15 The output characteristic can be selected to be open drain or push-pull. <i>Note: This is used for pin strapping 1V8_BOOT.</i>
R16	GPIO16	O	Prg	General Purpose IO Bit 16 It is used for output pin. The output characteristic can be selected to be open drain or push-pull. <i>Note: This is used for pin strapping BOOT1.</i>
N8	GPIO20	Prg	Prg	General Purpose IO Bit 20 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
P8	GPIO21	Prg	Prg	General Purpose IO Bit 21 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
N7	GPIO22	Prg	Prg	General Purpose IO Bit 22 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
P7	GPIO23	Prg	Prg	General Purpose IO Bit 23 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
R8	GPIO24	Prg	Prg	General Purpose IO Bit 23 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
T8	GPIO25	Prg	Prg	General Purpose IO Bit 25 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
R7	GPIO26	O	Prg	General Purpose IO Bit 26 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.



Table 12 GPIO Signals (cont'd)

Pin No.	Name	Pin Type	Buffer Type	Function
T7	GPIO27	O	Prg	General Purpose IO Bit 27 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
T5	GPIO28	Prg	Prg	General Purpose IO Bit 28 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
T4	GPIO29	Prg	Prg	General Purpose IO Bit 29 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
N3	GPIO30	Prg	Prg	General Purpose IO Bit 30 It can be selected as input or output mode. The output characteristic can be selected to be open drain or push-pull.
P3	GPIO31	O	Prg	General Purpose IO Bit 31 It is used for output pin. The output characteristic can be selected to be open drain or push-pull. <i>Note: This is used for pin strapping BOOT2.</i>

2.3.9 EJTAG Signals

The EJTAG interface is described in [Table 13](#).

Table 13 EJTAG Signals

Ball No.	Name	Pin Type	Buffer Type	Function
J3	<u>TRST</u>	I	PU	EJTAG Reset Notes 1. Active low. 2. If the EJTAG interface is not in use this pin must be connected to V_{SS} . 3. A low-to-high transition of the TRST pin latches the JTAG mode (depending on the bootstrap setting of pin TDO).
K4	TDI	I	PU	EJTAG Test Data Input
L3	TDO	O	PD	EJTAG Test Data Output There is a boot strap setting on this pin. This pin is used to decide the interface. 0 = EJTAG Debug; 1 = DFT Test Mode
M3	TMS	I	PU	EJTAG Test Mode Select
L4	TCK	I	-	EJTAG Test Clock



2.3.10 DDR-SDRAM

The Double Data Rate (DDR) SDRAM interface is described in [Table 14](#).

Table 14 DDR-SDRAM Signals

Ball No.	Name	Pin Type	Buffer Type	Function
E18	DDR_DQ31	I/O	—	DDR Data Bus [31:16]
F16	DDR_DQ30	I/O	—	
E16	DDR_DQ29	I/O	—	
H17	DDR_DQ28	I/O	—	
F17	DDR_DQ27	I/O	—	
G16	DDR_DQ26	I/O	—	
E17	DDR_DQ25	I/O	—	
H16	DDR_DQ24	I/O	—	
C17	DDR_DQ23	I/O	—	
A17	DDR_DQ22	I/O	—	
D17	DDR_DQ21	I/O	—	
A15	DDR_DQ20	I/O	—	
C16	DDR_DQ19	I/O	—	
B16	DDR_DQ18	I/O	—	
B17	DDR_DQ17	I/O	—	
A16	DDR_DQ16	I/O	—	
G2	DDR_DQ15	I/O	—	DDR Data Bus [15:0]
E3	DDR_DQ14	I/O	—	
H3	DDR_DQ13	I/O	—	
E2	DDR_DQ12	I/O	—	
H2	DDR_DQ11	I/O	—	
F3	DDR_DQ10	I/O	—	
G3	DDR_DQ9	I/O	—	
F2	DDR_DQ8	I/O	—	
B3	DDR_DQ7	I/O	—	
C2	DDR_DQ6	I/O	—	
A4	DDR_DQ5	I/O	—	
C3	DDR_DQ4	I/O	—	
B2	DDR_DQ3	I/O	—	
D1	DDR_DQ2	I/O	—	
B4	DDR_DQ1	I/O	—	
D2	DDR_DQ0	I/O	—	
C9	DDR_A16	O	—	DDR4: A16 DDR3: Row Address Strobe (RAS) <i>Note: Active low for DDR3 RAS;</i>



Table 14 DDR-SDRAM Signals (cont'd)

Ball No.	Name	Pin Type	Buffer Type	Function
B8	DDR_A15	O	—	DDR4: A15 DDR3: Column Address Strobe (CAS) <i>Note: Active low for DDR3 CAS;</i>
D9	DDR_A14	O	—	DDR4: A14 DDR3: Write enable (WE) <i>Note: Active low for DDR3 WE;</i>
D10	DDR_A13	O	—	DDR Address Bus [13:0]
C10	DDR_A12	O	—	
B10	DDR_A11	O	—	
C11	DDR_A10	O	—	
B11	DDR_A9	O	—	
A11	DDR_A8	O	—	
A12	DDR_A7	O	—	
B12	DDR_A6	O	—	
A13	DDR_A5	O	—	
C13	DDR_A4	O	—	
C12	DDR_A3	O	—	
B13	DDR_A2	O	—	
B14	DDR_A1	O	—	
A14	DDR_A0	O	—	
C6	DDR_ODT1	O	—	DDR ODT
B7	DDR_ODT0	O	—	
G17	DDR_DM3_N	I/O	—	DDR DM, UDM/LDM
D18	DDR_DM2_N	I/O	—	
E1	DDR_DM1_N	I/O	—	
A3	DDR_DM0_N	I/O	—	
G18	DDR_QQS3_T	I/O	—	Differential QQS Pair for upper DQ lines [DQ24 - DQ31]
F18	DDR_QQS3_C	I/O	—	
B18	DDR_QQS2_T	I/O	—	Differential QQS Pair for lower DQ lines [DQ16- DQ23]
C18	DDR_QQS2_C	I/O	—	
G1	DDR_QQS1_T	I/O	—	Differential QQS Pair for upper DQ lines [DQ8 - DQ15]
F1	DDR_QQS1_C	I/O	—	
C1	DDR_QQS0_T	I/O	—	Differential QQS Pair for lower DQ lines [DQ0 - DQ7]
B1	DDR_QQS0_C	I/O	—	
A7	DDR_BG1	O	—	DDR4: Bank/Group Address [1]; DDR3: A14;
C8	DDR_BG0	O	—	DDR4: Bank/Group Address [0] DDR3: Bank Address [2]



Table 14 DDR-SDRAM Signals (cont'd)

Ball No.	Name	Pin Type	Buffer Type	Function
A8	DDR_BA1	O	–	DDR Bank Address [1]
A6	DDR_BA0	O	–	DDR Bank Address [0]
A5	DDR_CS0	O	–	DDR Chip Select 0 <i>Note: Active low. Use this Chip Select, if only one set of DDR device(s) is connected.</i>
B6	DDR_CS1	O	–	DDR Chip Select 1 <i>Note: Active low</i>
A9	DDR_CK_T	O	–	DDR Clock Output
A10	DDR_CK_C	O	–	DDR Clock Inverted
D7	DDR_CKE1	O	–	DDR Clock Enable 1
C7	DDR_CKE0	O	–	DDR Clock Enable 0
F9	DDR_ACT_N	O	–	DDR4: DDR CMD ACTIVATE DDR3: A15
E7	DDR_ZQ	AI	–	DDR Calibration Resistor
D12	DDR_RST	O	–	DDR RESET
G8	DDR_PAR	O	–	DDR4: DDR Parity
F8	DDR_ALERT_N	O	–	DDR4: DDR4 ALERT_N
E12	DDR_ATO	AO	–	Internal Testing Reserved. Leave it open on PCB.
G12	DDR_DTO1	O	–	Internal Testing Reserved. Leave it open on PCB.
F11	DDR_DTO0	O	–	Internal Testing Reserved. Leave it open on PCB.
G9	DDR_VREFCA	O	–	VREF for CA
G6	DDR_VREFDQ0	O	–	VREF for DQ Lines 15 to 0
G13	DDR_VREFDQ1	O	–	VREF for DQ Lines 31 to 16



2.3.11 Reset

The Reset signals are described in [Table 15](#).

Table 15 Reset Signals

Ball No.	Name	Pin Type	Buffer Type	Function
K2	<u>POR</u>	I	PU	Power-on-Reset <i>Note: Active low</i>
K3	<u>HRST</u>	O	OD	Hardware Reset/Power-on-Reset Output This pin is the POR module output and synchronous with the on-chip reset after power-on. <i>Note: Active low</i>



2.3.12 Power

The power supply characteristics are described in the [Electrical Characteristics](#) chapter.

Table 16 Power Supply

Ball No.	Name	Pin Type	Buffer Type	Function
PLL Supply				
R9	VAA1V8_PLL5	PWR	–	Analog power 1.8 V supply for PLL5
R5	VAA1V8_PLL	PWR	–	Analog power 1.8 V supply for PLL3&4, CML clock PAD
K17	VAA1V8_XO	PWR	–	Analog power 1.8 V supply for XO, PLL0a, PLL0b, PLL1, PLL2.
J5	VAA1V8_PVT_L	PWR	–	Analog power 1.8 V supply for PVT in Left Side
L15	VAA1V8_PVT_R	PWR	–	Analog power 1.8 V supply for PVT in Right Side
P10	VAA0V9_PLL5	PWR	–	Analog power 0.9 V supply for PLL5
P6	VAA0V9_PLL	PWR	–	Analog power 0.9 V supply for PLL3&4, CML clock PAD
Digital Supply				
H4, H5	VDD_IO_0	PWR	–	3.3/1.8 V Power supply for Digital I/O
N14, N15	VDD_IO_1	PWR	–	3.3/1.8 V Power supply for Digital I/O
M14	VDD_IO_2	PWR	–	3.3/1.8 V Power supply for Digital I/O
N9	VDD_IO_3	PWR	–	3.3/1.8 V Power supply for Digital I/O
G5, G11, G14, H7, H9, H10, H12, H14, J8, J11, J13, K7, K12, K14, L8, L11, L13, M7, M9, M10, M12	VDD0V9	PWR	–	0.9 V Digital Power Core power supply.
Serdes Supply				
L6, M6	VAA0V9_SPHY1	PWR	–	Analog power supply 0.9 V for SFP Connector Side Serdes
P12	VAA0V9_SPHY2	PWR	–	Analog power supply 0.9 V for WAN Serdes
K5, M5	VAA1V8_SPHY1	PWR	–	Analog power supply 1.8 V for SFP Connector Side Serdes
P11	VAA1V8_SPHY2	PWR	–	Analog power supply 1.8 V for WAN Serdes
DDR Supply				
F7	VAA1V8_DDRPLL_0	PWR	–	Analog power 1.8 V supply for DDR PLL
F10	VAA1V8_DDRPLL_1	PWR	–	Analog power 1.8 V supply for DDR PLL
F12	VAA1V8_DDRPLL_2	PWR	–	Analog power 1.8 V supply for DDR PLL
D4, D6, D13, D15, E5, E6, E8, E11, E13, E14, F4, F15	VDDQ	PWR	–	Power for DDR 1.5/1.35/1.2 V



2.3.13 Ground

The power/ground supply characteristics are described in the [Electrical Characteristics](#) chapter.

Table 17 Ground Signals

Ball No.	Name	Pin Type	Buffer Type	Function
Ground Supply				
J18, K16	VSSA_XO	GND	–	Ground for XO
K1, K6, L2, L5, M2, M4, N1, N5, N6, P2, R2, T1, T2	VSSA_SPHY1	GND	–	Ground for SFP Connector Side Serdes
R10, R11, R13, R14, T9, T12, T15	VSSA_SPHY2	GND	–	Ground for WAN Serdes
A2, A18, B5, B9, B15, C4, C5, C14, C15, D3, D5, D8, D11, D14, D16, E4, E9, E10, E15, F5, F6, F13, F14, G4, G7, G10, G15, H1, H6, H8, H11, H13, H15, H18, J4, J6, J7, J9, J10, J12, J14, J15, J17, K8, K9, K10, K11, K13, L7, L9, L10, L12, L14, L16, M8, M11, M13, M15, M16, M17, M18, N4, N10, N11, N12, N13, N16, P5, P9, P13, P14, P15, R4, R6, R15, T18	VSS	GND	–	Ground Ground for digital core, I/O logic.

2.3.14 Fuse

The fuse pins must be connected as described in [Table 18](#).

Table 18 Fuse Signals

Ball No.	Name	Pin Type	Buffer Type	Function
P4	VAA1V8_OTP	PWR	–	Analog power supply for OTP



2.3.15 Dying Gasp

The dying gasp pin must be connected as described in [Table 19](#).

Table 19 Dying Gasp Signals

Ball No.	Name	Pin Type	Buffer Type	Function
K15	VDD_DG	PWR	–	Dying GASP Reference Input Input for system supply voltage supervision. This pin is connected to an external system reference voltage to detect a system power failure and to enforce a Dying Gasp PLOAM message in upstream direction.

2.3.16 Reserved

The reserved pins must be connected as described in [Table 20](#).

Table 20 Reserved Signals

Ball No.	Name	Pin Type	Buffer Type	Function
J16	Reserved	MSL	–	Tie to VSS



2.4 Digital Signal Ball (Pin) Reset Property and Drive Strength

This section describes the reset values and drive strength of the digital IO pins.

The abbreviations used on the following pages are defined in [Table 21](#).

- The **Reset State** is the logic level of a pin during the power-on-reset (POR) phase.
- The **Reset Release State** is the logic level of a pin after the POR signal transition from low to high.

Table 21 Abbreviations for the Reset Property

Abbreviation	Description
0	Output level Vol, no internal pull-up or pull-down
0-D	Output level Vol, internal pull-down
1	Output level Voh, no internal pull-up or pull-down
1-U	Output level Voh, internal pull-up
Z	High impedance
L	Pin not driven, internal pull-down
H	Pin not driven, internal pull-up

2.4.1 Peripherals and General Purpose I/O

The Peripherals and General Purpose I/O Pins are described here:

Table 22 GPIO Interface Signals

Ball	Name	Reset State	Reset Release State	Drive Strength	Voltage Domain
N18	GPIO0/QSCK	H	H	default 2 mA, programmable (2, 4, 8, 12 mA)	VDD_IO_2 = 3.3 V or 1.8 V
N17	GPIO1/QSCS	H	H		
P18	GPIO2/QSIO0	H	H		
P17	GPIO3/QSIO1	H	H		
R18	GPIO4/QSIO2	H	H		
R17	GPIO5/QSIO3	H	H		
P16	GPIO8	H	H		VDD_IO_1 = 3.3 V or 1.8 V
T3	GPIO10/I2C1_SCL	H	H		VDD_IO_3 = 3.3 V or 1.8 V
R3	GPIO11/I2C1_SDA	H	H		
T17	GPIO13	H	H		VDD_IO_1 = 3.3 V or 1.8 V
T16	GPIO15	H	H	default 2 mA, programmable (2, 4, 8, 12 mA)	
R16	GPIO16	H	H		



Table 22 GPIO Interface Signals (cont'd)

Ball	Name	Reset State	Reset Release State	Drive Strength	Voltage Domain
N8	GPIO20/OPT_RX_LOS	H	H		VDD_IO_3 = 3.3 V or 1.8 V
P8	GPIO21/OPT_GPIO0	H	H		
N7	GPIO22/OPT_TX_FAULT	H	H		
P7	GPIO23/OPT_TX_SD	H	H		
R8	GPIO24/OPT_TX_PUP	H	H		
T8	GPIO25/OPT_GPIO8	H	H		
R7	GPIO26/I2C0_SCL	H	H		
T7	GPIO27/I2C0_SDA	H	H		
T5	GPIO28/GPC1	H	H		
T4	GPIO29/PPS	H	H		
N3	GPIO30/URXD1/URXD1_PON/URXD_TEP	H	H		
P3	GPIO31/UTXD1/UTXD1_PON/UTXD_TEP	H	H		

2.4.2 UART0

Table 23 UART0 Interface Signals

Ball	Name	Reset State	Reset Release State	Drive Strength	Voltage Domain
J2	URXD0/URXD_QOS/URXD0_PON	Z	Z	2 mA	VDD_IO_0 = 3.3 V or 1.8 V
J1	UTXD0/UTXD0_QOS/UTXD0_PON	H	0		

2.4.3 Reset

Table 24 Reset Signals

Ball	Name	Reset State	Reset Release State	Drive Strength	Voltage Domain
K2	POR	H	H	2 mA	VDD_IO_0 = 3.3 V or 1.8 V
K3	HRST	0	H		



2.4.4 OPT_TXEN

Table 25 OPT_TXEN Signal

Ball	Name	Reset State	Reset Release State	Drive Strength	Voltage Domain
T6	OPT_TXEN	Z	Z	2 mA	VDD_IO_3 = 3.3 V or 1.8 V

2.4.5 EJTAG

Table 26 EJTAG Signals

Ball	Name	Reset State	Reset Release State	Drive Strength	Voltage Domain
J3	TRST	H	H	2 mA	VDD_IO_0 = 3.3 V or 1.8 V
K4	TDI	H	H		
L3	TDO	L	0		
M3	TMS	H	H		
L4	TCK	Z	Z		



2.5 Strap Pin Configuration

Strap pins are used to latch in the system specific configuration after reset. All these pins are input during reset. The following table describe the implemented strap on functions and the modes. In addition, there is one more boot strapping to select EJTAG and JTAG mode via TDO pin boot option.

Table 27 Strap Pin Overview

Strap Pin	Function	Description
TDO	JTAG Mode Select	This pin is latched in after reset to define the JTAG interface. Default internal pull down. 1 _D JTAG Test JTAG Test. 0 _D EJTAG EJTAG interface selected to debug the CPS and daisy chained secure boot core.
GPIO13	Endian Mode Select	This pin is latched in after reset to define the Endian setting. Default internal pull up. 0 _D Little CPU operates in Little Endian format. 1 _D Big CPU operates in Big Endian format.
GPIO15	1.8V Supply Select	This pin is latched in after reset to define the power supply for flash interface. Default internal pull up. 0 _D 1V8 Flash interface power supply is 1.8V. 1 _D 3V3 Flash interface power supply is 3.3V.
GPIO1 GPIO16/ GPIO31/ UTXD0	Boot Select	See Table 45 . Default internal pull-up.

3 Data Path

3.1 Data Path Components

Data Path includes the following modules.

- One instance of **WAN Serdes** (Chapter 4.2)
- One instance of **XFI Serdes** (Chapter 4.3)
- Two instances of **Ethernet Physical Coding Sublayer** (Chapter 5)
- Two instances of **Ethernet MAC** (Chapter 6)
- One instance of **PON Subsystem** (Chapter 7)
- One instance of **Packet Switching Engine** (Chapter 8)
- One instance of **Buffer Manager** (Chapter 9)
- One instance of **Carrier Grade Quality of Service Engine** (Chapter 10)

3.2 Data Path Flow

Figure 4 shows the data path flow of one PRX126 configuration.

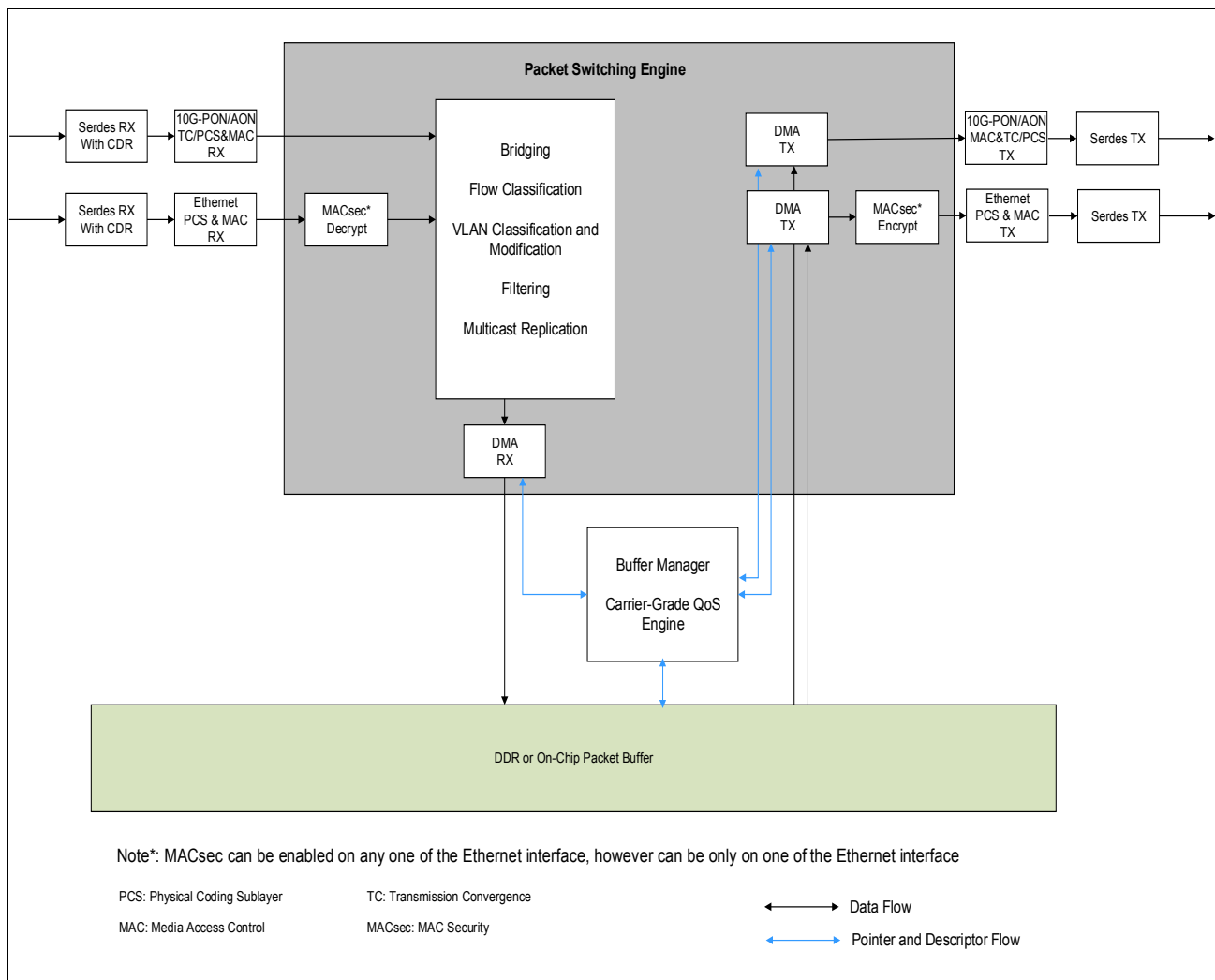


Figure 4 PRX126 Data Path Flow



The data path processes data traffic between the WAN (PON, AON or Copper Wire) interface and the SFP connector side interface (XFI/SFI/SGMII etc).

At the WAN side, the device connects to an external optical transceiver or an external Ethernet transceiver seamlessly via a WAN Serdes interface. PON mode, AON or Copper Wire Mode is configurable.

In PON mode, Ethernet packets are received or transmitted through PON TC/PCS and MAC processing module. In either AON or Copper Wire mode, Ethernet packets are received or transmitted through the Ethernet PCS and MAC processing module.

At the SFP connector side, the interface is XFI/SFI/SGMII.

Data Path Flow

The traffic data path flow as below.

The corresponding ingress data path flow is:

1. Ethernet packet is received via Serdes.
2. Received Ethernet packet is forwarded to Packet switching engine directly by MAC module.
3. If MACsec is enabled, the packet is decrypted first before being processed by Packet switching engine.
4. Packet switching engine parses the packet, bridges, classifies and filters the packet according the multi-fields of the packet, determines the packet's destinations and characteristics (color, QoS class mapping) based on the classification results. VLAN fields, DSCP field and IP checksum field maybe modified according to classification results. If there are multiple destinations, the packet is replicated for each destination. The copy for each destination can be modified differently.
5. DMA inside Packet switching engine requests the free buffer from Buffer manager.
6. Buffer manager manages both on chip packet buffer and off chip packet buffer. On chip packet buffer has 7Mbits. According to configured buffer allocation policy for each traffic flow and QoS class, Buffer manager allocates on chip packet buffer first if it is available and otherwise allocates the off chip packet buffer.
7. DMA inside Packet switching engine writes the modified Ethernet packet to the allocated packet buffer.
8. After DMA stores the complete packet, Buffer manager insert the packet (via packet pointer and meta information for QoS) to the corresponding egress queue managed by QoS Engine.
9. Carrier-grade QoS engine determines if to enqueue or drop the packet according to configured RED/WRED policies.

The corresponding egress data path flow is:

1. Egress interface indicates if it is ready to accept the new packet.
2. If yes, Carrier-grade QoS engine does the rate shaping and schedules the packet from the egress queues mapping to that egress interface.
3. Buffer manager de-queues the packet and passes the packet buffer pointer and meta information to the DMA inside the Packet switching engine.
4. DMA inside the Packet switching engine fetches the packet from on chip or off chip packet buffer and forwards it to MAC processing module.
5. After DMA completes fetching, Buffer manager returns the packet buffer to the free pool.
6. If MACsec is enabled. packet is encrypted first before being forwarded to MAC.
7. The packets are transmitted out via Serdes.

CPU Path

Control packets can be detected by Packet switching engine and be forwarded to the QoS queues to CPU.

CPU path extraction flow is as below:

1. Refer to the above ingress flow. Traffic are queued in QoS queues to CPU.
2. Buffer manager load spreader logic assigns the traffic to one of the virtual CPU according to configurable load spreading algorithm.
3. CPU gets the packet pointer and meta information from the load spreader in Buffer manager.



4. If CPU terminates the packet, after CPU finishes processing the packet, CPU can return the buffer pointer to Buffer manager and Buffer manager returns the buffer to the free pool.
5. CPU can also process the packet, determines the new destination and modifies the packet (updates only the header packet in the packet buffer). CPU inserts the packet to the egress queues of the destination. Refer to below for CPU path insertion flow.

CPU path insertion flow is as below:

1. CPU writes the packet pointer and meta information to the Buffer manager.
2. Buffer manager inserts the packet (via packet pointer and meta information for QoS) to the egress queue managed by QoS Engine.
3. Refer to egress data flow above.

Multicore Processing Engine Path

Packet switching engine can extract data packets which requires the advanced or complementary processing to Multicore Processing Engine (MPE). Packets are stored in the egress queues dedicated to MPE. The data path is same as CPU path flow.

For the traffic to MPE, Packet switching engine can attach the packet parsing information before the start of the packet in order to reduce the MPE processing load.

Refer to [Multi-core Processing Engine](#) for offloading capability description.

4 Serdes

PRX126 supports two Serdes:

- One Serdes (WAN Serdes) is for Line/Optical Side
- One Serdes (XFI Serdes) is for SFP Connector Side

4.1 Serdes Instances

The overview of Serdes is shown in [Figure 5](#).

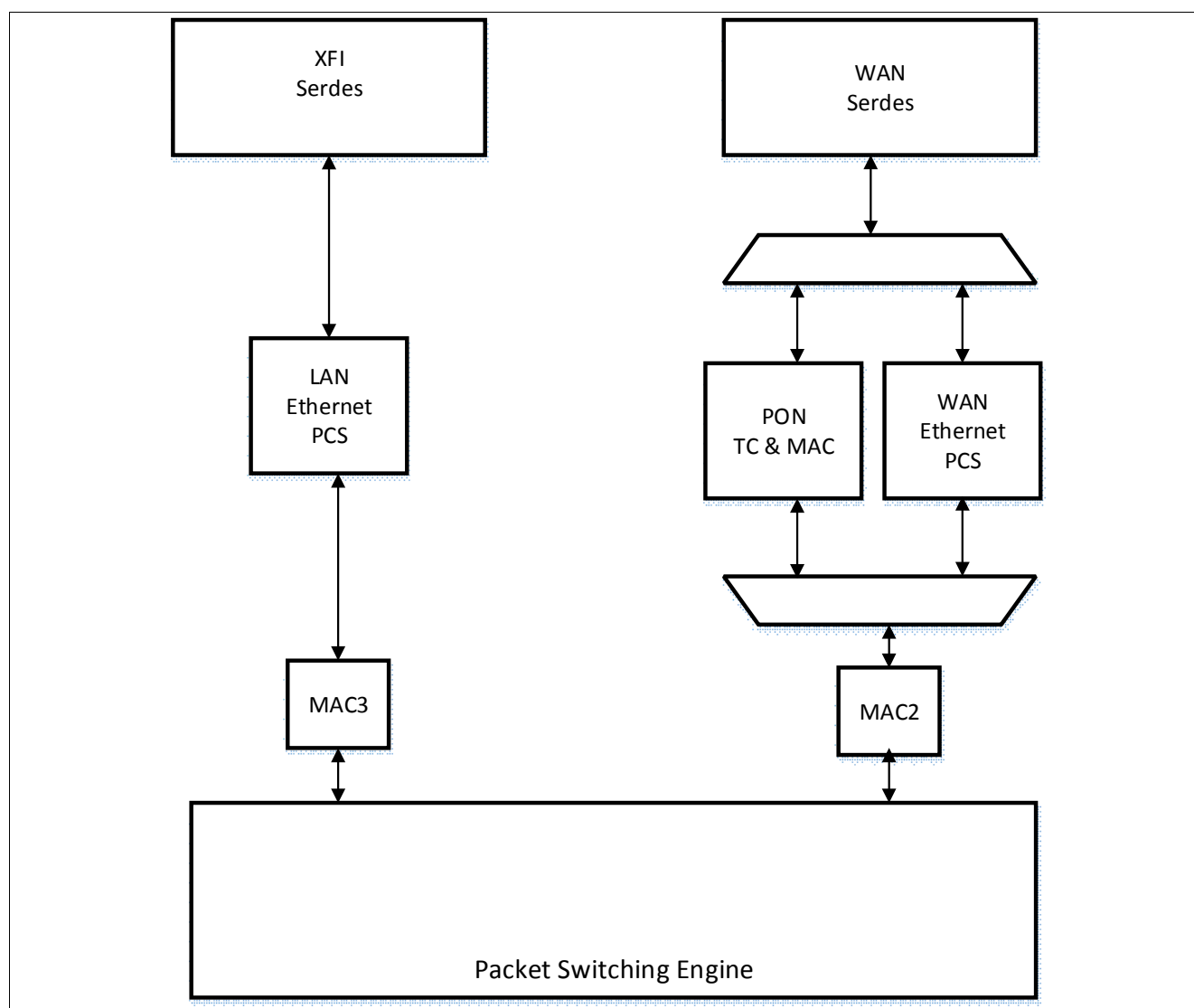


Figure 5 Serdes Instances



4.2 WAN Serdes

The Serdes in WAN side can be connected to an external optical transceiver in PON mode as ONU, an external optical transceiver in AON mode or an external Base-T Ethernet transceiver in Copper Wire mode.

The WAN Serdes support the following features:

- Single Lane
 - A high speed differential transmit pair
 - A high speed differential receive pair
 - An external reference resistor for calibration of TX and RX termination
- PON Mode with an external optical transceiver
 - GPON (RX: 2.48832 Gbit/s and TX: 1.24416 Gbit/s)
 - XG-PON (RX: 9.95328 Gbit/s and TX: 2.48832 Gbit/s)
 - NG-PON2 (RX: 9.95328 Gbit/s and TX: 9.95328 Gbit/s)
 - XGS-PON (RX: 9.95328 Gbit/s and TX: 9.95328 Gbit/s)
 - EPON (RX: 1 Gbit/s and TX: 1 Gbit/s)
 - 10G-EPON asymmetric (RX: 10 Gbit/s and TX: 1 Gbit/s)
 - 10G-EPON symmetric (RX: 10 Gbit/s and TX: 10 Gbit/s)
 - Loop clock from RX CDR to TX in PON mode
- AON Mode with an external optical transceiver
 - 1000BASE-X
 - 10GBASE-R
- Copper Wire Mode with an external Ethernet transceiver via 10G-USXGMII mode, 1G SGMII or 2.5G SGMII mode
 - 10GBASE-T
 - 5GBASE-T
 - 2.5GBASE-T
 - 1GBASE-T
 - 100BASE-TX
 - 10BASE-T(e)
- Support both transmit and receive equalization and adaption
- Support TX amplitude control
- Support independent RX and TX power state controls
- Debug:
 - Support IDDR_DQ test
 - Support diagnostics for characterization and ATE testing: loopback, test pattern generation, BERT, error insertion and detection



4.3 XFI Serdes

The Serdes is connecting to interface at SFP/SFP+ module.

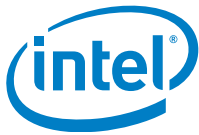
- The Serdes has single lane
 - A high speed differential transmit pair
 - A high speed differential receive pair
 - An external reference resistor for calibration of TX and RX termination
- Support the following modes
 - 10 Gbit/s XFI/SFI
 - 2.5 Gbit/s SGMII
 - 1 Gbit/s SGMII
- Support both transmit and receive equalization and adaption
- Support TX amplitude control
- Support independent RX and TX power state controls
- Debug:
 - Support IDDR_DQ test
 - Support diagnostics for characterization and ATE testing: loopback, test pattern generation, BERT, error insertion and detection



5 Ethernet Physical Coding Sublayer

There are two Ethernet Physical Coding Sublayer (PCS) instances in total. They are connected to WAN Serdes and Client side Serdes. Ethernet PCS supports the following features:

- PCS coding
 - 8b/10b coding (clause 36)
 - 64/66b coding (clause 49)
- Support 10Gbps XFI/SFI
 - 10 Gbit/s (10.3125GT/s)
- Support SGMII
 - 2.5 Gbit/s (3.125GT/s)
 - 1 Gbit/s (1.25GT/s)
- Support Forward Error correction and forwarding in 10G SFI/XFI mode
- SupportEEE mode



6 Ethernet MAC

There are two Ethernet MAC instances in total. They are connected to LAN Ethernet PCS instance and WAN Ethernet PCS. Ethernet MAC supports the following features.

- XGMII interface is compliant to IEEE 802.3 Clause 46
- GMII interface is compliant to IEEE 802.3 Clause 35
- MII interface is compliant to IEEE 802.3 Clause 22
- 10/100/1000/2500/5000/10000 Mbps standard operation speed
- Full-duplex operation mode (for all speeds) and half-duplex operation mode (only for 10/100 Mbps standard operation speed)
- Enhanced frame size support (“Jumbo frames”, programmable limit up to 10 Kbyte)
- Pause frame in full duplex mode. Backpressure (forced collisions) in half-duplex mode
- Frame padding on egress traffic
- Minimum frame length check and maximum frame length check. Maximum frame length are configurable.
- FCS verification and stripping
- FCS generation and insertion
- Support Low Power Idle (LPI) mode as defined by IEEE 802.3az

Ethernet MAC also supports the following PTP/1588 time synchronization features.

- Support Ethernet packet time stamping as IEEE 1588 (v1 and v2), ITU G.8275 and Y.1369
- Support ordinary clock, boundary clock, end to end transparent clock and peer to peer transparent clock modes
- Support PTP over Ethernet and PTP over UDP over IPv4/IPv6.
- Support both external reference time (from PON TC and MAC) and internal reference time. Internal reference time can be time adjusted and frequency adjusted.
- Auxiliary time stamp snapshot with two external events. Support 1 PPS (pulse per second) output.
- Support both fixed and flexible (programmable for start, stop, pulse width, interval) PPS output.
- Snapshot of packet at SFD in both direction. Support programmable ingress time stamp correction and programmable egress time stamp correction.
- Attach captured time stamp to end of the every received packet to CPU
- Support one step and two step time stamping in TX direction



7 PON Subsystem

The PON Subsystem implements all PON related modes of the WAN interface. It resides between the WAN-side Ethernet MAC and the WAN Serdes. The PON interface is designed in a way to allow direct connectivity to a pluggable transceiver SFP/SFP+ or to a PMD transceiver chip (LDD/LA).

Supported PON modes:

- GPON
- XG-PON
- XGS-PON
- NG-PON2
- EPON (1000BASE-PX)¹⁾
- 10G-EPON (10/1GBASE-PRX, asymmetric)¹⁾
- 10G-EPON (10GBASE-PR, symmetric)¹⁾
- Active Ethernet (10GBASE-R, 1000BASE-X)
- Bit/baud rates downstream/upstream:
 - GPON: 2.488 Gbit/s (downstream), 1.244 Gbit/s (upstream)
 - XG-PON: 9.952 Gbit/s (downstream), 2.488 Gbit/s (upstream)
 - XGS-PON: 9.952 Gbit/s (downstream), 9.952 Gbit/s (upstream)
 - NG-PON2: 9.952 Gbit/s (downstream), 9.952 Gbit/s (upstream)
 - EPON: 1 Gbit/s (downstream bit rate), 1 Gbit/s (upstream bit rate), 1.25 GT/s (downstream baud rate), 1.25 GT/s (upstream baud rate)
 - 10G-EPON asymmetric: 10 Gbit/s (downstream bit rate), 1 Gbit/s (upstream bit rate), 10.3125 GT/s (downstream baud rate), 1.25 GT/s (upstream baud rate)
 - 10G-EPON symmetric: 10 Gbit/s (downstream bit rate), 10 Gbit/s (upstream bit rate), 10.3125 GT/s (downstream baud rate), 10.3125 GT/s (upstream baud rate)

Dimensioning:

- 64 T-CONTs (up to 32 in a single 8 kHz frame)
- 256 GEM or XGEM ports (unicast or multicast)
- Up to 16 LLIDs

Other characteristics:

- Supports SFF/SFP/XFP transceivers, fixed or tunable wavelength
- Configurable burst enable timing
- Integrated voltage supervision and automatic Dying Gasp insertion
- Automatic power saving control
- ToD counter with capture/compare unit
- Rx and Tx time stamping in ITU modes

Management:

- Integrates autonomous PLOAM and MPCP handling to decouple real-time low level control functions from higher layer SW tasks
- Functional API and higher layer control protocols like OMCI and SIEPON implemented in SW on main CPU under Linux*
- Message-based mailbox communication between PON interface subsystem and CPU

¹⁾ Available in future release



8 Packet Switching Engine

The packet switching engine subsystem includes dedicated functions for L2 bridging data plane and QoS mapping functions as defined in ITU-T G.988 for ITU based PON systems. The subsystem allows straight-forwarding mapping from Linux* bridge and follows closely G.988 OMCI packet processing model and BBF TR-156 specification.

This chapter provides an overview of the engine. It covers the description of the main features and subsystem functional description.

8.1 Features

MACsec

- One instance of MACsec
 - Configurable to be attached to only one of MAC
- IEEE 802.1AE/AEbn/AEbw. Support both GCM-AES 128-bit and 256-bit. 16 security channels and 32 security associations in each direction
- Support classification based on multiple packet header fields
- Support bypassing for Non-MACsec packets
- 10 Gbit/s in line encryption and decryption

Packet Switching

- Microcode based flexible packet parsing
- Support sub-interface classification (for example, GPON GEM ID, VLAN ID) over the single physical interface
- L2 bridging at sub-interface level
- L3 Multicast processing and replication at sub-interface level
- VLAN filtering
- Extended VLAN tagging operation
- Flexible Traffic flow classification based on multiple fields of the packet
- Packet filtering based on multiple fields of the packet
- Traffic QoS mapping classification
- Priority, DSCP and color marking/remarking
- Traffic metering
- Wake on LAN
- Port trunking of any two ports
- Packet statistics
- Packet mirroring
- Packet loopback
- 802.1ag and Y.1731 Ethernet OAM delay and loss measurement support
- Ethernet OAM loopback support

DMA

- DMA to store the packet to on-chip or off-chip packet buffer
- DMA to fetch the packet from on-chip or off-chip packet buffer
- Request and release the free buffer from/to the Buffer manager

8.2 Functional Description

This chapter provides a detailed functional description of the Packet switching Engine.

8.2.1 Packet Flow

Figure 6 describes a typical packet data flow through various stages of the packet switching engine.

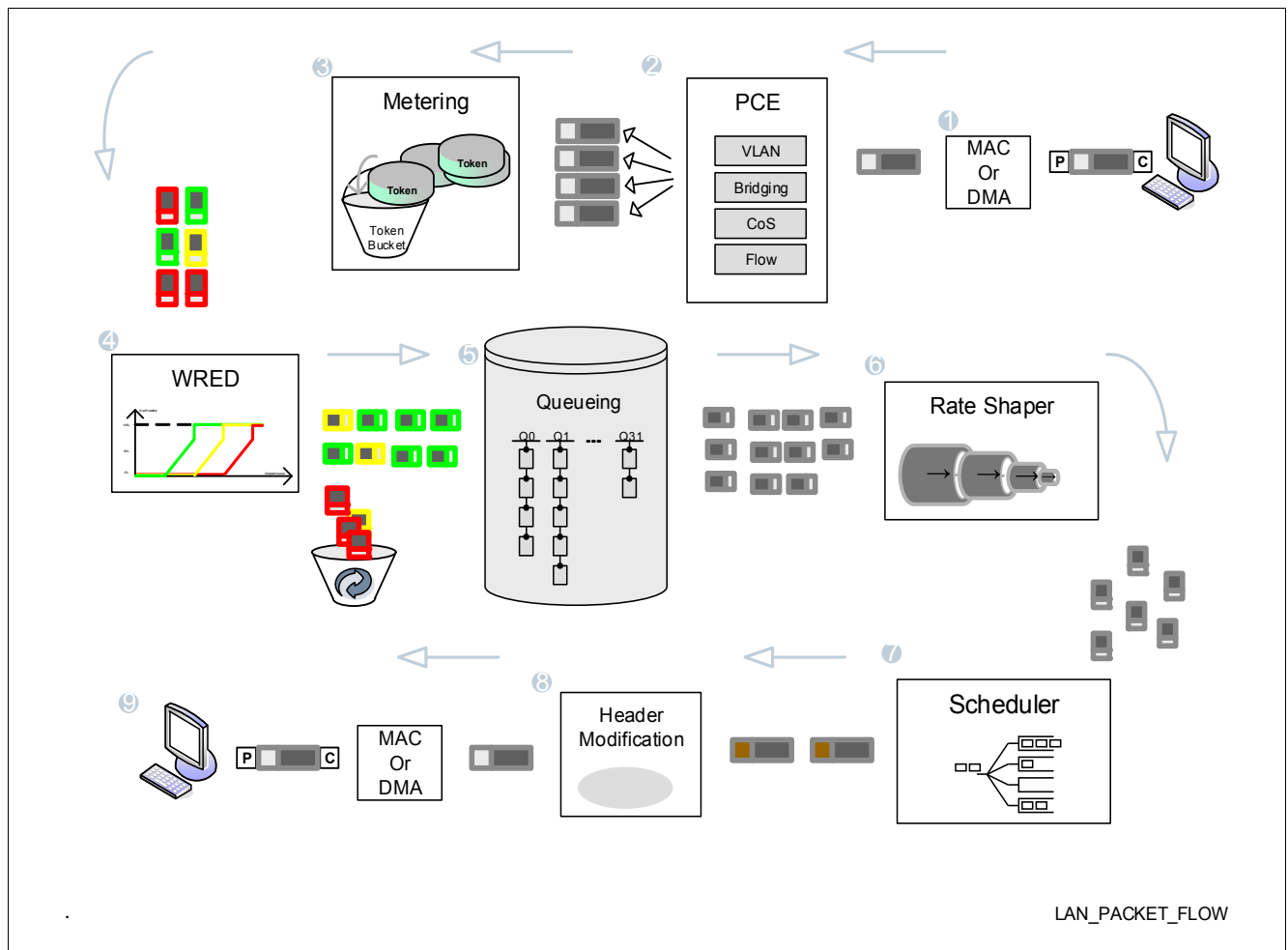


Figure 6 Packet Flow Diagram

Packet received on the ingress interface is stripped from the Ethernet Preamble and checked for correct FCS. In case of any reception errors the packet might be discarded. Received packet is classified in the Packet Classification Engine and assigned to one or several appropriate QoS queue. Prior to accepting the packet to certain queue it is subject to metering and WRED functions. Packets marked as non-conforming by the Metering Engine might be discarded by the WRED algorithm based on the configurable drop precedence. Prior to transmission on the egress side, the packet is subject to Rate Shaping. If transmission of the packet is not delayed by the Rate Shaper it is subject to the Scheduling algorithm (WRR or SP). Packets that are scheduled for transmission are subject to Header Modification, such as, VLAN Tag modification and DSCP remarking. Please note that the shaper and scheduler inside Packet switching engine applies only to processing queues inside the engine. Carrier-grade QoS Engine is responsible for shaping and scheduling of the egress queues.

8.2.2 G.988 OMCI L2 MAC Bridge Reference Model

The OMCI supports two major layer 2 traffic mapping models: MAC bridging and "IEEE 802.1p mapping". MAC bridging is described in [IEEE 802.1D] and [IEEE 802.1Q]. The bridge has many features, and can be used to direct traffic based on a MAC address (that is, true bridging) or on VLAN characteristics (using the VLAN filter feature).

The two basic layer 2 services can be used in various combinations to achieve different overall connectivities.

- N:1 bridging, where a bridge is used to serve multiple UNI ports from a single ANI service.
- 1:M mapping, where a mapper is used to serve a single UNI with multiple ANI connections, based on IEEE 802.1p priorities.
- 1:P filtering, where a bridge with filters is used to serve a single UNI with multiple ANI connections, based on some VLAN information other than IEEE 802.1p priorities.
- N: MP bridging map filtering, the combination of the above.

8.2.3 G.988 OMCI L2 MAC Bridge Implementation in PRX126

The PRX126 packet switching follows closely the G.988 OMCI packet processing model and offers the flexibility of packet processing options the OMCI model and BBF TR-156 specification are requesting. The great advantage of the implemented network processing unit is that each attribute of the received OMCI ME's can be directly mapped to the underlying hardware.

Figure 7 describes the OMCI model implemented by PRX126. Note that arrows in the block diagram (unlike in OMCI ME relationship diagrams) denote the sequence of packet processing in up- and downstream direction. The number in brackets defines the number of occurrences of that particular instance. The hardware instances of the Managed Entities support all attributes of the corresponding G.988 defined ME. Depending on the selected OMCI model the corresponding hardware instances are activated.

Bridging is done either in a single bridge or in several bridges. Up to 64 L2 bridges are supported

- Overall maximum 288 CTP bridge ports are supported (1 of them is reserved for internal processing purpose)
- Overall maximum of 128 bridge ports are supported (1 of them is reserved for internal processing purpose)
- L2 or L3 multicast: L2 multicast is based on destination MAC address; L3 multicast is based on destination and source IP addresses.
- Overall maximum of 1024 Extended VLAN Tagging (EVTG) MEs is supported which can be flexibly assigned as indicated in **Figure 7**.
- Overall maximum of 1024 VLAN filtering MEs is supported which can be flexibly assigned as indicated in **Figure 7**.
- Overall maximum of 128 rate limiter MEs is supported which can be flexibly assigned as indicated in **Figure 7**.
- Egress priority queue management, shaping and schedule are handled by chip **Carrier Grade Quality of Service Engine** outside of Packet switching engine.

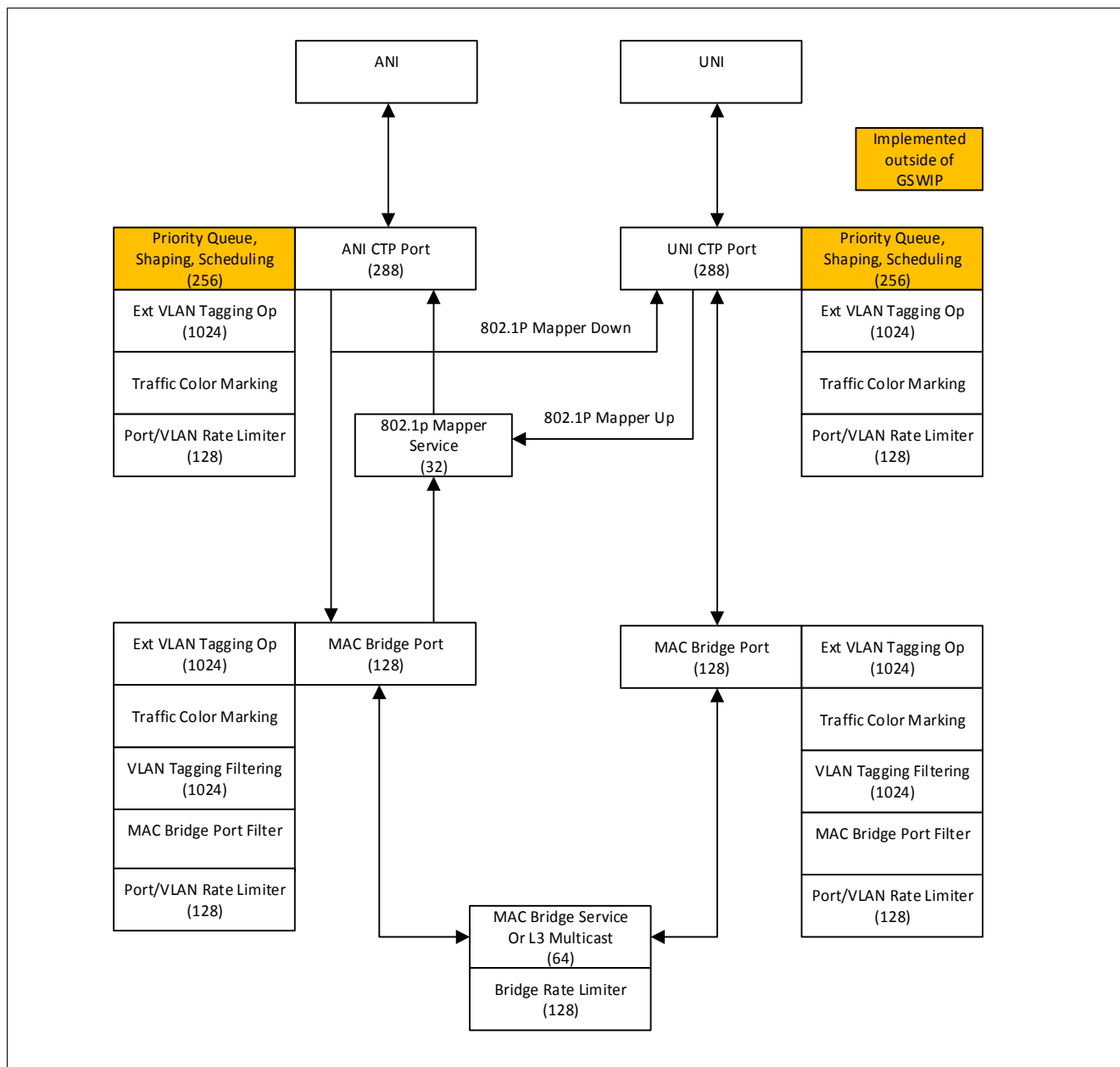


Figure 7 OMCI L2 MAC Bridge Implementation in PRX126

9 Buffer Manager

The Buffer Manager (BM) is composed of the following sub-modules:

- The En-queue Manager (EQM) responsible for the assignment of free buffers to the Ingress ports and En-queuing the filled buffers into the QoS Engine queues.
- The De-queue Manager (DQM) responsible for the Dequeuing of the buffers from the Qos Engine queue to the Egress ports and return the free buffers to the BM
- The Free Segment Queue Manager (FSQM) responsible for the buffer management of the Buffers in the internal Memory and for the segment linking of the segmented internal packets.
- The External Buffer Manager (EBM) responsible for the packet buffer management of the buffers in the external memory.
- The PON-IP Interface Block (PIB) responsible for taking in requests from the PON TC processing module for de-queueing the required number of packets out of de-queue manager for the respective T-CONT.
- The Load Spreader (LS) responsible to spread the egress traffic towards the CPU.

9.1 Buffer Manager Overview

The architecture of the buffer manger is shown in **Figure 8**.

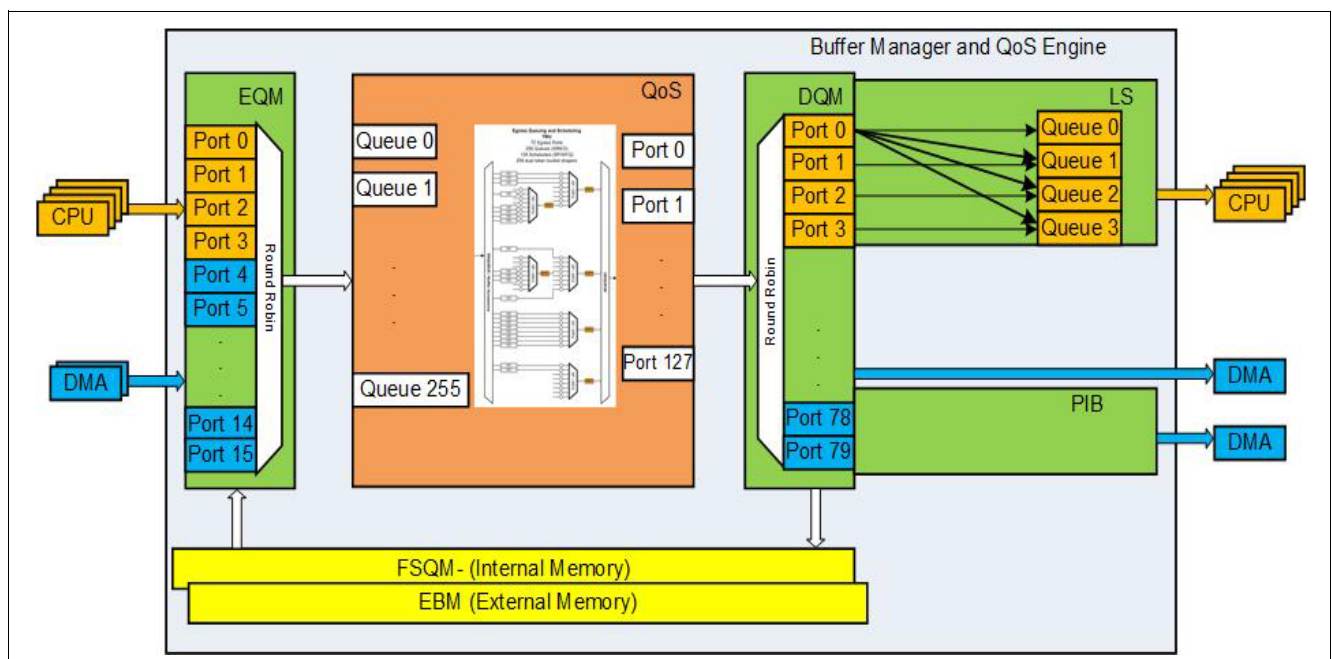


Figure 8 BM and CQE Architecture

The packet can be assigned to either internal packet buffer or one pool of the external packet buffer according to buffer allocation algorithm and the attributes of the packet (packet's destination, packet size and packet traffic class).

9.2 Enqueue Manager

The Enqueue Manager (EQM) supports the following configurations.

- CPU ports for 4 VPE
- DMA ports for data path from Packet switching engine



9.3 Dequeue Manager

The Dequeue Manager (DQM) supports up to 90 ports, comprising of the following configurations.

- 4 CPU ports
 - 4 ports for the 4 VPE
- 19 DMA ports for data path to Packet switching engine
- 64 special PON ports (separate descriptors for Segmented and Non Segmented) for the PON TC module.

9.4 Free Buffer Manager

Free buffer manager provides buffers for 2 regions, namely the internal memory region and the external memory region.

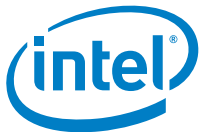
- Internal memory region (on-chip packet buffer size is 7 Mbit): 7k buffer pointer (128B per buffer). One packet may occupy multiple buffers. This region is managed by FSQM.
- External memory region (off chip packet buffer): both packet pointer and the buffer are in external memory. One packet occupy one buffer. This region is managed by EBM. 4 pool sizes are supported: e.g., 256B, 1KB, 2KB and 10KB.
- Allocation of the packet buffer is based on the buffer request of the packet. The allocation is determined by the packet's destination, QoS class, packet size, each type of buffer (internal buffer and four external packet buffer pools) occupancy and the configured watermarks. Internal packet buffer is allocated first in order to reduce the power consumption by external memory interfaces and devices.

9.5 Load Spreader Overview

The Load Spreader (LS) is a module shared between the Dequeue Manager (DQM) and the MIPS cluster. The LS has one port towards the MIPS cluster per input port from the Dequeue Manager.

The LS provides the following basic functions:

- Interrupt based interface for the MIPS processing cluster to read the descriptors for the packets Software or MPE processing.
- Configurable Load Spreading of the traffic directed towards the CPU port.
- Provide a FIFO based mechanism for fetching multiple descriptors with single interrupt request for faster and simplified CPU performance.
- Threshold and timer based interrupts for CPU.
- Depending on their own processing the Virtual CPUs of the MIPS cluster may use only a single port, dedicated per Virtual CPU, on the Load Spreader to read the descriptors from the CQE.



10 Carrier Grade Quality of Service Engine

The QoS Engine is responsible for the **egress** queuing of PDUs stored in the shared buffer and the scheduling of PDUs for transmission on **egress** interfaces.

The QoS (Quality of Service) subsystem is responsible for allocating of Egress bandwidth according to a configurable and controllable scheme (i.e. policy), allowing among other things, prioritization of traffic, fairness, and rate limit.

QoS Engine Features

- Support 30M packets per second throughput.
- Support associating up to 8 queues per egress port.
- Support ingress queue tail drop policies of types – fixed limit, RED and WRED.
- For queues with fixed limit drop policy, QoS has per queue thresholds (in units of packets and bytes) for Green and Yellow colored packets. After the threshold is crossed, packets of the specific color are dropped.
- For queues with RED or WRED drop policy, RED/WRED is first applied on each enqueued packet and the packet is either dropped or enqueued.
- Support color based RED/WRED parameters per ingress QoS queue.
- Support a layered tree of scheduling policies starting with a set of queues and ending with a port. The flow is from the leafs (queues) through nodes and to the root (egress port).
- Support strict priority, and deficit weighted round robin per scheduling node.
- Support per node and per ingress queue bit rate shaping.
- Support group rate shaping where a group of nodes share a single rate shaper.
- Schedule packets based on application of all scheduling logic types together, i.e. apply both policing, individual rate shaping and group rate shaping.
- Support 256 ingress queues.
- Support up to 128 output TX ports.
- Support QoS scheduling for all egress ports.
- Support 8 layers of QoS node schedulers.
- Support 32 shared rate shapers, each assigned to 8 nodes or less.
- Support WRED per queue at the QoS ingress queues.
- Support QoS queue size limiting in both packets and bytes. When a queue exceeds its limit all newly enqueued will be dropped.
- Allow changing scheduler (weights) and shaper parameters (information rates, burst sizes) while scheduler/shaper is active.

11 Coherent Processing Subsystem

The Coherent Processing Subsystem (CPS) as depicted in [Figure 9](#) consists of:

- 2 InterAptiv* Cores
- 12 Thread Contexts
- 4 Virtual CPUs
 - one Watchdog Timer (WDT) per Virtual CPU
 - one Interval Timer per virtual CPU
- Coherence Manager (CM)
- 2x IO Coherence Units (IOCU)
- L2 Cache Controller (L2\$)
- Cluster Power Controller (CPC)
- Global Interrupt Controller (GIC)
- Inter-Thread Communication (ITC) unit
- Debug and Trace System

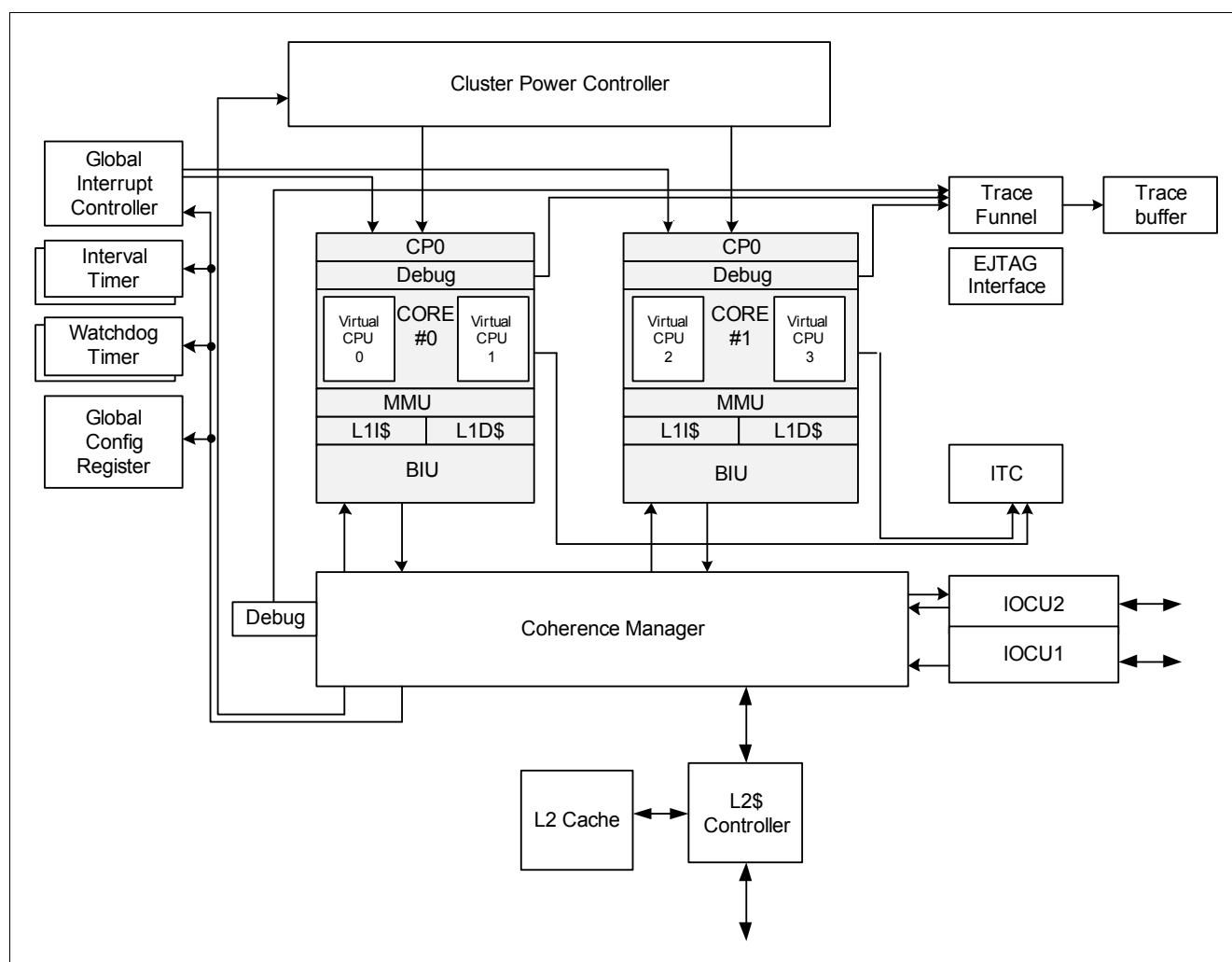


Figure 9 Overview of the Coherent Processing System



11.1 InterAptiv* Processor

The InterAptiv* Processor provides the following features:

- 9-stage pipeline
- 32-bit Address Paths
- 64-bit Data Paths to Caches
- MIPS32* Release 3 Instruction Set and Privileged Resource Architecture
- MIPS MT ASE
 - 2 Virtual CPUs per Core which are also referred to as Virtual Processing Elements (VPEs) [3]
 - 6 Thread Execution Contexts (TCs) per Core
 - Inter-Thread Communication (ITC) memory for efficient communication & data transfer
- MIPS DSP ASE
- Memory Management Unit
 - 32 dual-entry fully associative Joint TLB (JTLB) per VPE
 - JTLBs are sharable under software control
 - Fully associative Instruction micro TLB (ITLB) with 3 shared entries and 1 private entry per TC
 - 8-entry fully associative Data micro TLB (DTLB)
- L1 Instruction Cache (L1I\$) and L1 Data Cache (L1D\$)
 - 32 KByte instruction cache
 - 32 KByte data cache size
 - 4-Way Set Associative
 - 8 outstanding data cache load misses
 - Supports Write-back mode
 - 32-Byte cache line size
 - Virtually indexed, physically tagged
 - Cache line locking support
 - Non-blocking pre-fetches
 - Hardware support of L1\$ aliasing fix
- Core Bus Interface Unit (Core BIU)
 - OCP interface with 32-bit address and 64-bit data
 - OCP interface runs at CPU core clock
 - Burst size of four 64-bit beats
 - 4 x 32-Byte entry write buffer
- Multiply/Divide Unit
 - Maximum issue rate of one 32x32 multiply per clock
 - 5 cycle multiply latency
 - Early-in iterative divide. Minimum 11 and maximum 34 clock latency (dividend (rs) sign extension-dependent)
- Clock Gating based Power Control
 - Minimum frequency: 0 MHz
 - Power-down (clock gated) mode triggered by WAIT instruction
 - Support for software-controlled clock divider: dividers are inside chip CGU, different clock frequency can be used to supply to CPU
 - Extensive use of local clock gating
- EJTAG Debug
 - Support for single stepping
 - Instruction address and data address/value breakpoints
 - TAP controller chained for CPU0-CPU1-CM2 debug



11.1.1 Virtual CPU

Two Virtual CPUs are implemented per core. It is referred as Virtual Processing Element (VPE) in the MIPS Document [3].

11.1.2 Thread Context

Six Thread Contexts (TC0-TC5) are implemented per core.

11.1.3 Execution Unit

The core execution unit implements a load-store architecture with single-cycle Arithmetic Logic Unit (ALU) operations (logical, shift, add, subtract) and an autonomous multiply-divide unit. Each TC on the core contains thirty-two 32-bit general-purpose registers (GPRs) used for scalar integer operations and address calculation. The register file consists of two read ports and one write port and is fully bypassed to minimize operation latency in the pipeline.

11.1.4 Multiply/Divide Unit

The Multiply/Divide Unit (MDU) performs multiply and divide operations. The MDU consists of a pipelined 32x32 multiplier, result-accumulation registers (HI and LO), multiply and divide state machines, and all multiplexers and control logic required to perform these functions. This pipelined MDU supports execution of a multiply or multiply-accumulate operation every clock cycle.

11.1.5 System Control Co-processor CP0

In the MIPS architecture, CP0 is responsible for the virtual-to-physical address translation, cache protocols, the exception control system, the processor's diagnostics capability, operating mode selection (kernel vs. user mode), and the enabling/disabling of interrupts. Configuration information such as cache size, set associativity, and presence of build-time options are available by accessing the CP0 registers.

11.1.5.1 Watch Register

Watch register is used to store an exception address for either instruction fetch or data access. When the watch register content matches the actual access, the watch exception happens. This facilitates an easy software debugging.

11.1.5.2 Processor Identification and Virtual CPU Number

The CPU number in a multi-processor system is specified in register VPEX_CP0_EBase.CPUNum, which can be used by software to distinguish a particular CPU core from the others.

The coding of VPEX_CP0_EBase.CPUNum is shown in [Table 28](#).

Table 28 CPU Number

Processor		Register	Description
CORE0	VCPU0	VPE0_CP0_EBase.CPUNum	00 0000 0000 _B
	VCPU1	VPE1_CP0_EBase.CPUNum	00 0000 0001 _B
CORE1	VCPU2	VPE0_CP0_EBase.CPUNum	00 0000 0010 _B
	VCPU3	VPE1_CP0_EBase.CPUNum	00 0000 0011 _B



11.1.5.3 Endianness

A CPU core can be operated in Big Endian mode or Little Endian mode. The endianness is selected by boot strapping.

The status of the endianness selection signal can be read in VPEx_CP0_Config.BE.

11.1.5.4 CPU Performance Counter

Each TC has two 32-bit performance counters and each core has 12 performance counters altogether.

The performance counter registers are connected to a clock that is stopped when the processor is in sleep mode. This should be considered when analyzing measurements taken on the system.

11.1.6 Memory Management Unit

The following description holds for each VPE of both cores, VPEx = VPE0 and VPE1. The CPU core contains a Memory Management Unit (MMU) per VPE that interfaces between the execution unit and the cache controllers. Each MMU is based on a Translation Lookaside Buffer (TLB). The TLB consists of three (JTLBs are shared) or four translation buffers (JTLBs are independent):

- A 32 dual-entry fully associative Joint TLB (JTLB) per VPE.
- Fully associative Instruction micro TLB (ITLB) with 3 shared entries and 1 private entry per TC
- 8-entry fully associative Data micro TLB (DTLB)

The ITLB and DTLB, also referred to as the micro TLBs, are managed by the hardware and are not software visible. The micro TLBs contain subsets of the JTLB. When translating addresses, the corresponding micro TLB (I or D) is accessed first. If there is not a matching entry, the JTLB is used to translate the address and refill the micro TLB. If the entry is not found in the JTLB, then an exception is taken.

11.1.7 Exception Handling

This chapter describes these Exceptions in detail, which are caused external to the CPU core. They are described in the relative priority, highest to lowest. The CPU core provides several exceptions, which are generated internal to the CPU core e.g. arithmetic overflow. These exceptions are not covered in detail in this chapter, see the appropriate MIPS documentation.

All Exceptions are summarized in [Table 29](#).

Table 29 Priority of Exceptions

Exception	Description
Reset	Reset Exception
DINT	Debug Interrupt Exception
NMI	Non Maskable Interrupt Exception
Interrupt	Interrupt Exception
IBE	Instruction Fetch Bus Error Exception

11.2 Coherence Manager Overview

The Coherence Manager (CM) is responsible for establishing the global ordering of requests as well as collecting the intervention responses and sending the correct data back to the requester. A high-level view of the CM is shown in [Figure 10](#).

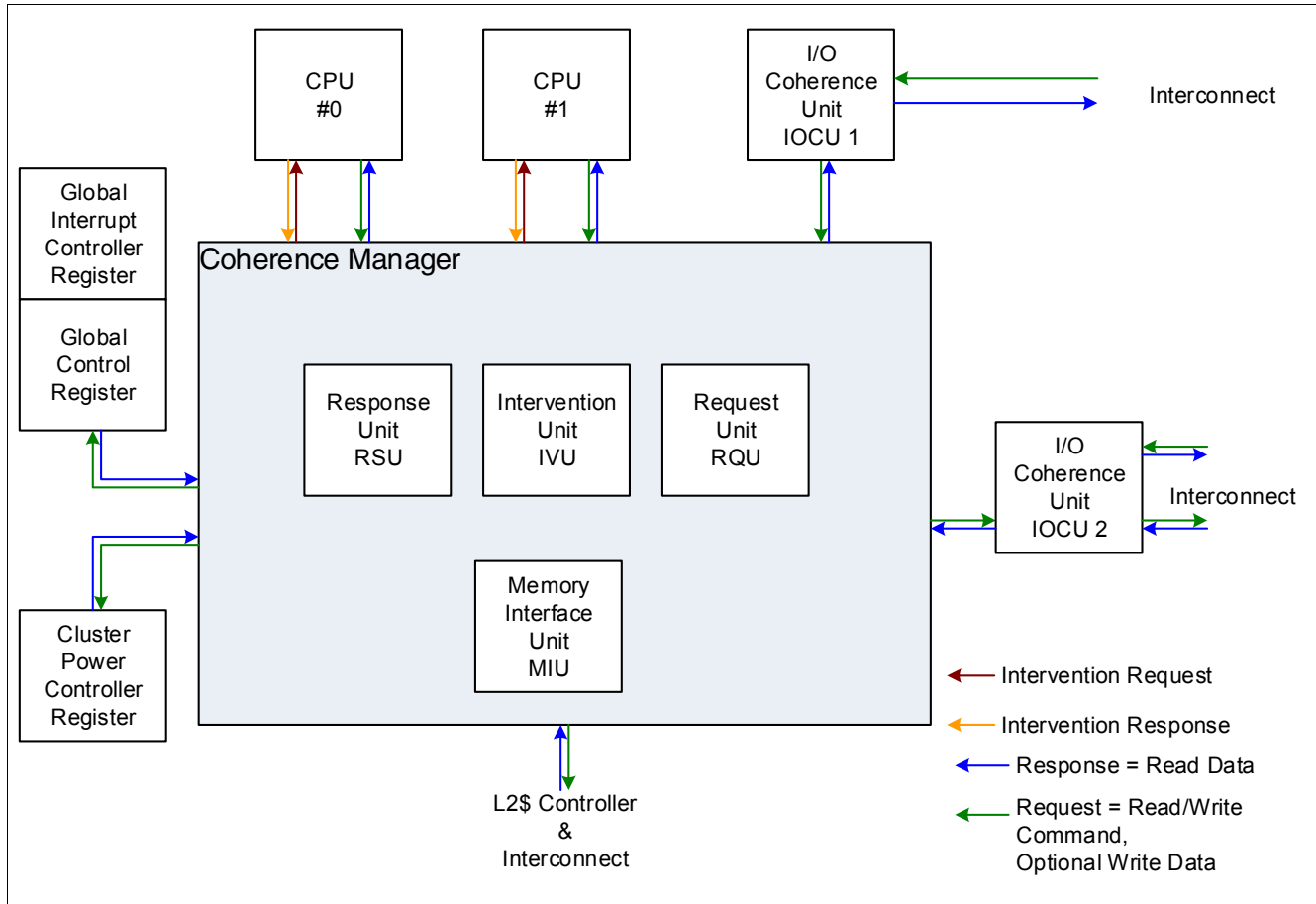


Figure 10 Coherence Manager Overview

Request Unit

The Request Unit (RQU) receives requests from the coherent CPU cores as well as the IO Coherence Unit (also called agents) and serializes them. Non-coherent requests are forwarded either to the Memory Interface Unit (MIU), to the IO Coherence Unit (IOCU) or to the memory-mapped register accesses to the GCR, GIC and CPC. Coherent requests are sent to the Intervention Unit (IVU).

Intervention Unit

This block receives the serialized stream of coherent requests from the Request Unit. These requests are sent as interventions to all the coherent devices, which update their cache states appropriately for the intervention and provide a response. If a requested cache line is in an Exclusive or Modified state, the data is returned with its response on a read type intervention. The Intervention Unit gathers the responses from each of the agents and takes care of the following actions:

- Speculative reads are resolved (confirmed or canceled)
- Memory reads that are required because they were not speculative are issued to the Memory Interface Unit
- Any Modified data returned from a coherent device is sent to the Memory Interface Unit to be written back to memory
- Any data returned from a coherent device is forwarded to the Response Unit to be sent to the requester



- The MESI state in which the line is installed by the requesting CPU is determined (the install state). If there are no other CPUs with the data, a Shared request is upgraded to Exclusive

Memory Interface Unit

The Memory Interface Unit (MIU) handles the interface to the L2/Memory. Non-coherent reads and writes as well as speculative coherent reads are sent to the Memory Interface Unit from the Request Unit. Coherent writes and late reads are generated from the Intervention Unit.

This block is responsible for staging the read data back to each of the agents. There are independent staging registers for each agent, which allows concurrent data return to different agents. This block stores the read data returned from the system if the read is still speculative or if the response unit is busy.

Response Unit

This block is responsible for returning data to the requesting agent. It can send data from the Intervention Unit, the Memory Interface Unit, the IO Coherence Unit (IOCU), or from memory-mapped register accesses to the GCR, GIC and CPC.

11.3 L2 Cache Overview

The L2\$ Controller (L2\$) has two Open Core Protocol (OCP) interfaces. On the OCP slave interface (OCP-S), it communicates with the Coherence Manager (CM), and on the OCP Master interface (OCP-M) it communicates with the Interconnect. The L2\$ also communicates with the Processor through the performance counter interface, error reporting interface, and other side band signals.

11.3.1 L2\$ Attributes

The L2\$ attributes are shown in [Table 30](#).

Table 30 L2\$ Attributes

Attribute	Value	Description
Size	256 KByte	The size of the L2\$ is 256 KByte
Line Size	32 Byte	The L2\$ line size is 32 Byte
Associativity	8 Way	The L2\$ is 8 ways associative
Parity	No	No parity protection for the L2\$ arrays is implemented

11.3.2 L2\$ Cache Protocols

The L2\$ controller supports the following cache protocols.

Uncached

Addresses in a memory area indicated as uncached are not read from the L2\$. Stores to such addresses are written directly to main memory, without changing L2\$ contents.

Write-through, no write-allocate

Read requests first search the L2\$, reading main memory only if the desired data does not reside in the L2\$. On data store operations, the L2\$ is first searched to see if the target address is cache resident. If it is resident, the cache contents are updated, and main memory is also written. If the cache look-up misses, only main memory is written.



Write-back, no write-allocate

Read requests are handled the same as write-through. Stores that hit in the cache will only update the cache contents but are not written to main memory. Stores that miss in the cache will not cause a cache refill and the data is written directly to main memory.

Uncached Accelerated

Treated like uncached access. No additional write merging is performed by the L2 controller. If an uncached accelerated write was originated as a burst on the slave interface, the burst is maintained on the master interface.

Write-back, write-allocate

Read requests and write requests that hit in the L2 cache are handled the same as write-back, no write-allocate. If write requests miss in the cache and the requests are full cache line writes, then the data will be written to the cache. It does not cause a cache refill because only full cache line writes are allocated. However, it can cause an eviction to create a room for the write data.

Allocate on read miss/No allocate on read miss

Normally when read requests miss in the cache, the main memory will be read and the cache will be updated with the data being returned. However, L2 cache also supports a mode that the return data is passed to the core without allocating into L2 cache.

11.4 IO Coherence Unit

The IO Coherence Unit (IOCU2) consists of two channels, the IO Initiator Request/Response Channel (IO) and the Memory Mapped Request/Response Channel (MMIO), the IOCU 1 has one IO Initiator Request/Response channel.

IO Initiator Request/Response Channel

The IO Initiator Request/Response Channel (IO) within the IOCU is responsible for translating non-coherent read/write transactions to the appropriate coherent or non-coherent transactions. Finally, these translated transactions are issued to the Coherence Manager (CM). The non-coherent read/write transactions are initiated by system masters e.g. DMA controller, which are located at the non-coherent interconnect.

The CM routes non-coherent transactions directly to the L2\$. Coherent transactions are snooped in the L1D\$ of all CPUs via their intervention ports. In the case of an L1D\$ hit, read data is returned directly via the IOCU to the system master. Write data is merged and written to the L2\$ or main memory.

Memory Mapped Request/Response Channel

The Memory Mapped Request/Response Channel (MMIO) within the IOCU is responsible to forward read/write transactions initiated by the CPUs to the non-coherent interconnect (SSX7). Corresponding response data is routed from the non-coherent system back to the CM.



12 Secure Boot Core

Secure Boot Core is also called Trusted Execution Processor (TEP).

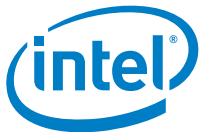
This chapter describes the dedicated MIPS 4KEc* core and its sub system. The core is used as:

- Boot Core in standard Boot products
- Secure Boot Core in Secure Boot products

12.1 MIPS 4KEc* Features

The MIPS 4KEc* Core supports:

- 32-bit Address paths
- 32-bit data paths
- MIPS32* Compatible Instruction Set
 - All MIPS II* Instructions
 - Multiply-Add and Multiply-Subtract Instructions (MADD, MADDDU, MSUB, MSUBU)
 - Targeted Multiply Instruction (MUL)
 - Zero and One Detect Instructions (CLZ, CLO)
 - Wait Instruction (WAIT)
 - Conditional Move Instructions (MOVZ, MOVN)
 - Prefetch Instructions (PREF)
- MIPS32* Enhanced Architecture (Release 2) Features
 - Vector interrupts and support for external interrupt controller
 - Programmable exception vector base
 - Atomic interrupt enable/disable
 - Bit field manipulation instructions
- MIPS16e* Application Specific Extension
 - 16 bit encoding of 32 bit instructions to improve code density
 - Special PC-relative instructions for efficient loading of addresses and constants
 - Data type conversion instructions (ZEB, SEB, ZEH, SEH)
 - Compact Jumps (JRC, JALRC)
 - Stack frame set-up and tear down “macro” instructions (SAVE and RESTORE)
- Instruction and Data Cache
 - 32 KByte Instruction Cache Size in a 4-Way set associative organization
 - 32 KByte Data Cache Size in a 4-Way set associative organization
 - Loads blocks only until critical word is available
 - Supports Write-back with write-allocation and Write-through with or without write- allocation
 - 16-byte cache line size, word sector
 - Virtually indexed, physically tagged
 - Support for cache line locking
 - Non-blocking prefetches
- MIPS32* privileged resource architecture
 - Count/Compare registers for real-time timer interrupts
 - Instruction and Data watch registers for software breakpoints
 - Separate interrupt exception vector
- Memory Management Unit
 - 32 dual entry JTLB with variable page size
 - 4-entry instruction micro TLB
 - 4-entry data micro TLB



- Core Bus Interface Unit (Core BIU)
 - All I/Os fully registered
 - Two 16 Bytes collapsing write buffers
- Multiply Divide Unit
 - Maximum issue rate of one 32 X 16 multiply per clock
 - Maximum issue rate of one 32 X 32 multiply every other clock
 - Early-in divide control. Minimum 11, maximum 34 clock cycles
- Big and Little Endian Support
- Access to NGI as trusted initiator
- Power Control
 - No minimum clock frequency
 - Power Down mode (triggered by WAIT instruction)
- EJTAG Debug Support
 - CPU control with start, stop and single-step feature
 - Software Breakpoints via SDBBP instruction
 - Hardware Breakpoints on virtual addresses
 - Test Access Port (TAP) facilitates high speed download of application

12.2 Secure Boot Core UART

The UART has 16-byte FIFOs on its RX and TX paths. It is possible to configure its parity mode and number of stop bits.

The UART can generate several IRQs – RX/TX FIFO water levels, parity error, Rx idle condition (a case where the RX watermark level was not reached after some idle time, but the UART RX FIFO still has some information inside).

13 Trusted Execution Environment

This chapter describes the Trusted Execution Environment (TEE) from hardware design perspective, including the Secure boot core which is targeted to run the secure OS & software.

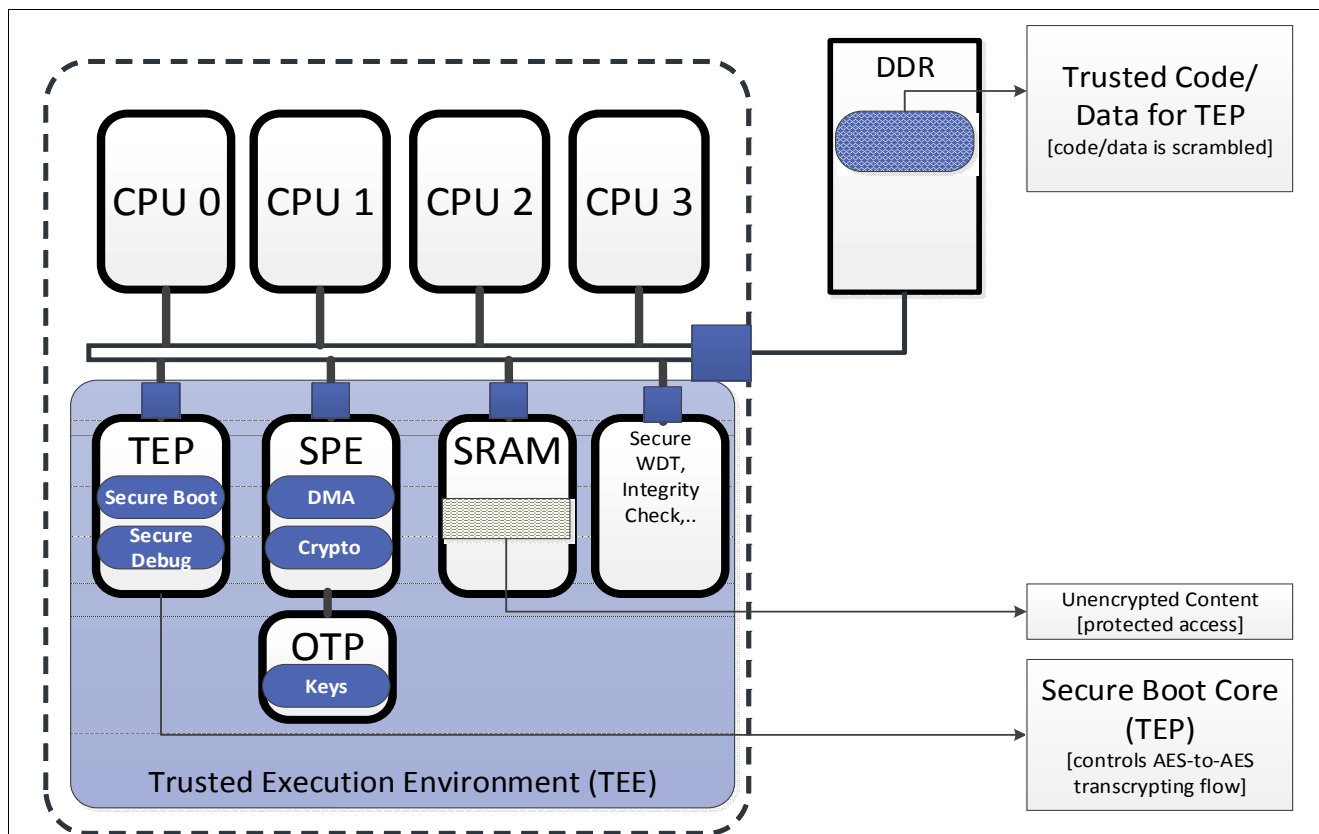


Figure 11 TEE Overview

13.1 Feature List

The TEE provides the following features:

- Dedicated MIPS 4KEc* core as Secure Boot Core (TEP)
- Security and firewall functionality in each Target Agent via protect region
- Setup of access permissions of protected regions only through TEP
- Configuration of the chip GPIO interface functionality only through TEP
- Trusted storage is achieved with trusted access to Secure Platform Engine (SPE) and trusted application execution



13.2 TEP Access to Platform Resource

13.2.1 Root of Trust

The root of trust is stored inside the OTP. The access to OTP is limited to trusted platform engine, while only TEP is allowed access to this engine via mailbox exchange.

13.2.2 Access Protection

The access protection to other on chip/off chip resources is achieved via access initiator to address region mapping.

Each access initiator will be assigned with dedicated initiator ID; based on this ID, the interconnect will allow/stop access to certain address region. The ID to region mapping is defined in interconnect register, and these registers are only allowed set-up or change by the TEP. Refer to [Chapter 13.3](#) for more details.

For chip interface multiplex control register, only TEP is assigned the access rights.

13.2.3 InterActiv* CPU Cluster Initial ID Adaptation

In a Cluster CPU sub system, the CPU tag signal will behave differently for cached L2 and uncached L2 (L1 is cached) to interconnect:

- Cached access, after L2\$, the CPU tag is lost;
- Uncached access, CPU/VPE/TC tag is maintained.

As such, an ID adaptation is needed for uncached access to identify which access is from which TC. For cached access, a default ID of 0 is generated.

Table 31 Initiator ID Adaptation (Hardware Default)

Access Source	CPU0, VPE0	CPU0, VPE1	CPU1, VPE0, all TCs	CPU1, VPE1, all TCs
Cached Access, resulted MConn_ID	0	0	0	0
Uncached access, resulted MConn_ID	0	0	0	0

Table 32 Initiator ID Adaptation (can be re-programmed via GCR registers)

Access Source	CPU0, VPE0, TC0	CPU0, VPE0, not TC0; VPE1, all TCs	CPU1, VPE0, all TCs	CPU1, VPE1, all TCs
Cached Access, resulted MConn_ID	0	0	0	0
Uncached access, resulted MConn_ID	2	1	0	4

Attention: Only the Initiator IDs 0, 1, 2, 4 are possible to be programmed !!



13.3 Interconnect Protection Mechanism

13.3.1 Identification

The Initiator Identification “InitID” can be used for debug purposes at each Target Agent (TA) to identify the initiator of the transaction, the system master(s) connected to a Initiator Agent (IA).

There is a unique Master Identification “MasterID” assigned to each system master. This MasterID is global for all Instances and is mapped onto the IA_MConnID signal. Thus, the MasterID travels along with the transaction from the initiator (system master) to the target (system slave).

13.3.2 Protection Overview

The NGI allows access to specified Target Agents (TA) to be restricted, protecting these TAs and their connect system slaves from unauthorized access. The protection mechanism is based on the use of flexible, pre-programmed protection regions that enable fine-grained access control at the TA based on the following attributes of each request:

- Address
The value of MAddr is used to determine the protection region. Because the IA can do address fill-in and/or multi-channel operations based on the initiator’s MAddr signal, the address received by a TA may not be the same as the address sent by the initiator.
- Initiator ID “InitID”
Used to determine the initiator’s read or write permissions within the protection region.
- MCmd
Used to determine if a read or write is requested.
The ReadEx command is evaluated as both a read and a write.

13.3.3 Interaction with Software

Setting up the protection region register needs to be performed only by TEP. This is done by setting up the XBAR registers into two portions: part 1 registers (which starts with tREG_PM_*) can be programmed by TEP only; part 2 registers can be accessed by all CPUs.

13.4 Secure Platform Engine

The Secure Platform Engine is to protect embedded systems against hackers trying to install hacked OS, copy hacked software image to other system, and protect device manufacturer’s software against illegal copying.

The SPE together with the **OTP Memory** supports Secure Boot/Platform with the following key elements:

- On chip OTP module to store the Root of Trust assets
- Cipher algorithm: AES in CBC mode with 256 bit key
- Hashing: SHA2 with 256 bits key
- True Random Number Generator (TRNG): HW based TRNG for IVs, nonces, public and private keys
- Secure Debug
- Crypto Algorithm specific block data de-/encryption (host must pad/remove padding for non-block size data):
AES: multiples of 16 Bytes

13.4.1 OTP Memory

16 kbit OTP memory will be used as the Non Volatile Memory (NVM). The NVM layout and structure are described in this chapter.

NVM Access Mapping by Secure Boot Engine

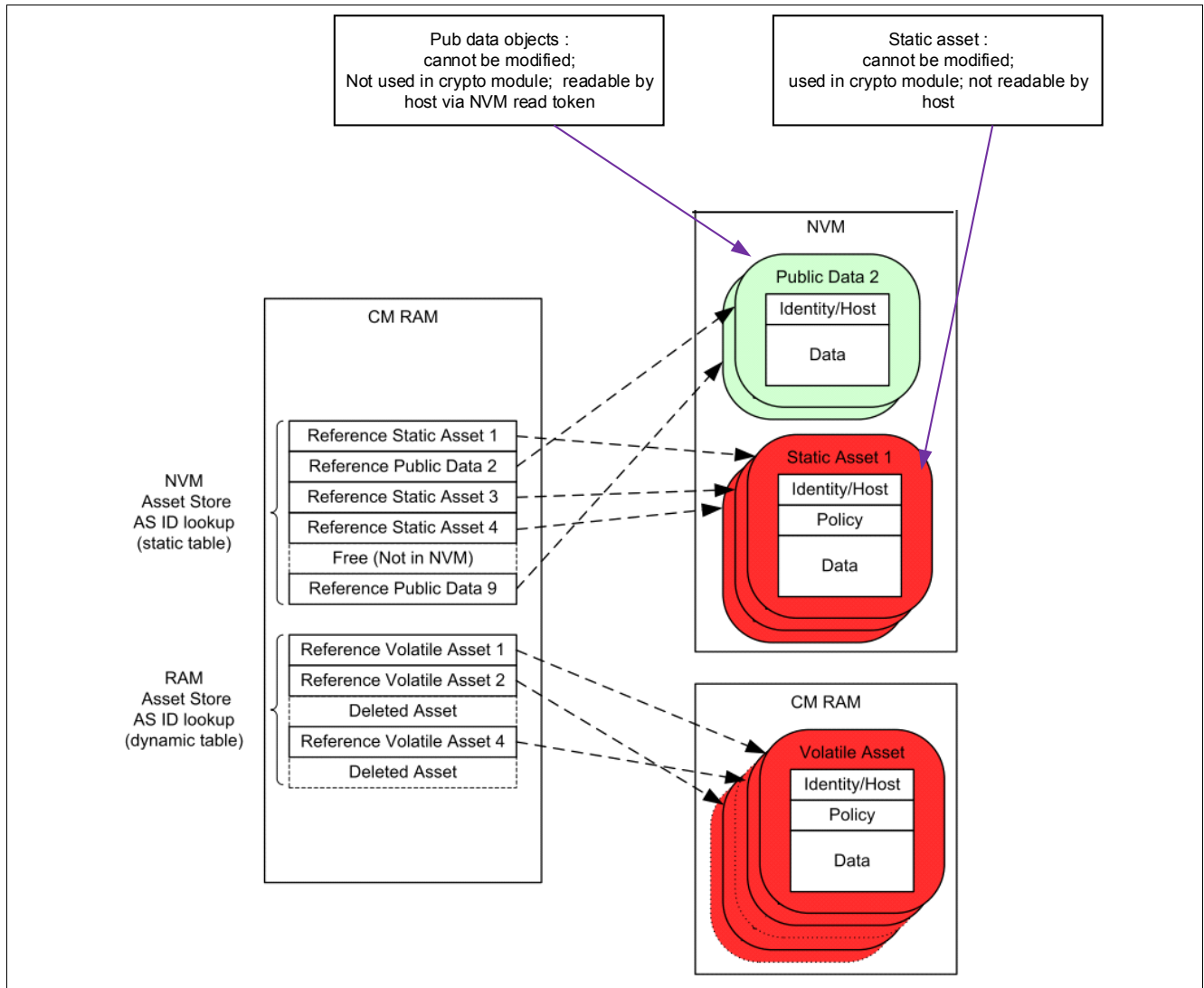


Figure 12 NVM Mapping

It is possible to have up to 62 assets inside the NVM. Each asset is indexed according to the header ID of the data objects. The asset is deemed deleted if the header ID is all programmed, i.e. 0x3F, the secure boot engine will no longer index the deleted asserts or read them out.

A protected asset is not allowed to be deleted and such operation is rejected.

OTP Data Example

An example of OTP data layout is given in [Figure 13](#). Access to the secure data (encryption keys, key hashing, etc.) is strictly limited to the secure boot engine only.

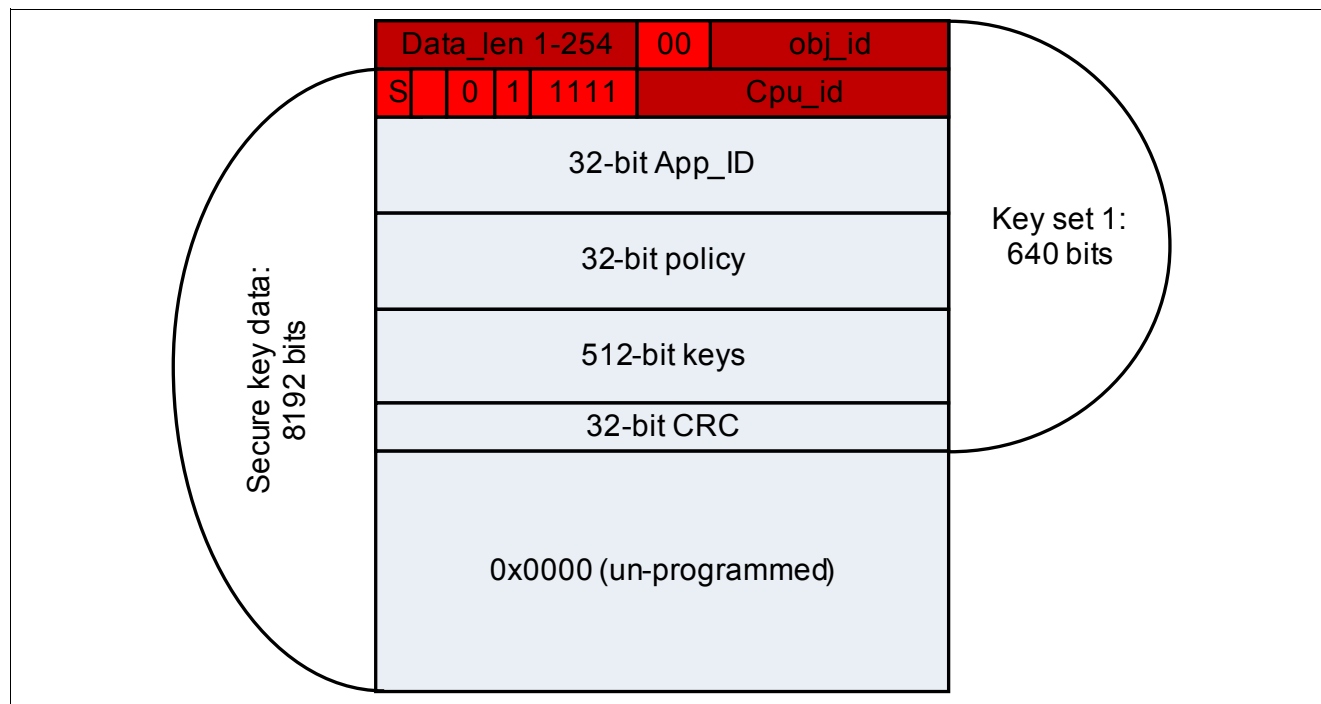


Figure 13 OTP Data Example

Secure Boot Package

Intel provides secure boot package documents that describe how to create NVM data image and program NVM, how to create secure boot application images. The documentation is available on request.



14 Boot System

This chapter describes the features of the boot system.

14.1 Feature List

The Boot system offers the following features:

- Booting from QSPI interface (multiplexing on GPIO0,1,2,3,4,5) with internal ROM
 - 1 bit mode of QSPI NOR, QSPI NAND
 - 4 bit mode of QSPI NOR, QSPI NAND
- Boot from internal ROM using Legacy UART#0 RS232 (ASC) and UART#0 XMODEM.
- Multiple (up to 4) duplicated images for booting system reliably
- DDR Sub-System Configuration and tuning
 - Register programming via encapsulated command from flash
 - White-list register access to prevent wrongly writing into unintended register
- Secure Boot
 - Root-of-Trust Secure Boot: for production provision
 - TrustWorld Secure Boot: for Secure Boot enabled end products
- Boot from NAND with internal ROM using QSPI supporting:
 - Refer to reference board design guide [1] for supported NAND Flash types
 - Boot from 1GB,2GB,4GB size NAND Flash
 - Boot from NAND Flash with Hardware ECC embedded inside flash device

14.2 Functional Description

This chapter describes the hardware conditions after Power-on-Reset (POR) and describes the boot loader for on-chip ROM implementation. This boot loader stores codes in hardware mainly to

1. Pre-configure the system and
2. Handle the (boot) code download from an external non-volatile memory through a target peripheral interface.

14.2.1 Boot Selection after POR

The hardware status after POR is as follows:

- All MIPS hardware module interrupts are disabled.
- A Reset status register in RCU stores the on-board Pin Strapping information.

The above boot selection process is completely done by hardware, no ROM code involvement yet.

Table 33 Boot Mode Settings

Strap Select					1st Boot Source	2nd Boot Source
BOOT 3 (UTXD0)	BOOT 2 (GPIO31)	BOOT 1 (GPIO16)	BOOT 0 (GPIO1)	HEX		
0	0	1	0	0x02	ROM	UART0 XMODEM This mode boot from UART0 with XMODEM protocol.
0	1	0	0	0x04	ROM	Legacy UART0 This mode boot from UART0 using legacy mode.
1	0	0	0	0x08	ROM	QSPI (Single Bit NOR)
1	0	1	1	0x0B	ROM	QSPI (Quad Bit NOR)
1	1	0	0	0x0C	ROM	QSPI (Quad Bit NAND)



Table 33 Boot Mode Settings (cont'd)

Strap Select					1st Boot Source	2nd Boot Source
BOOT 3 (UTXD0)	BOOT 2 (GPIO31)	BOOT 1 (GPIO16)	BOOT 0 (GPIO1)	HEX		
1	1	0	1	0x0D	ROM	QSPI (Single Bit NAND)
				Others	Reserved	

15 Multi-core Processing Engine

The Multi-core Processing Engine (MPE) uses processor resources of the CPS cluster in form of configurable number of Virtual CPUs and Thread Contexts (TCs).

The complete Multi-core Processing Engine solution includes MPE control, search, metering and copy engines. The Carrier Quality of Service Engine (CQE) is the key element in providing the connectivity between the subsystems as well as guaranteeing the required flow priorities.

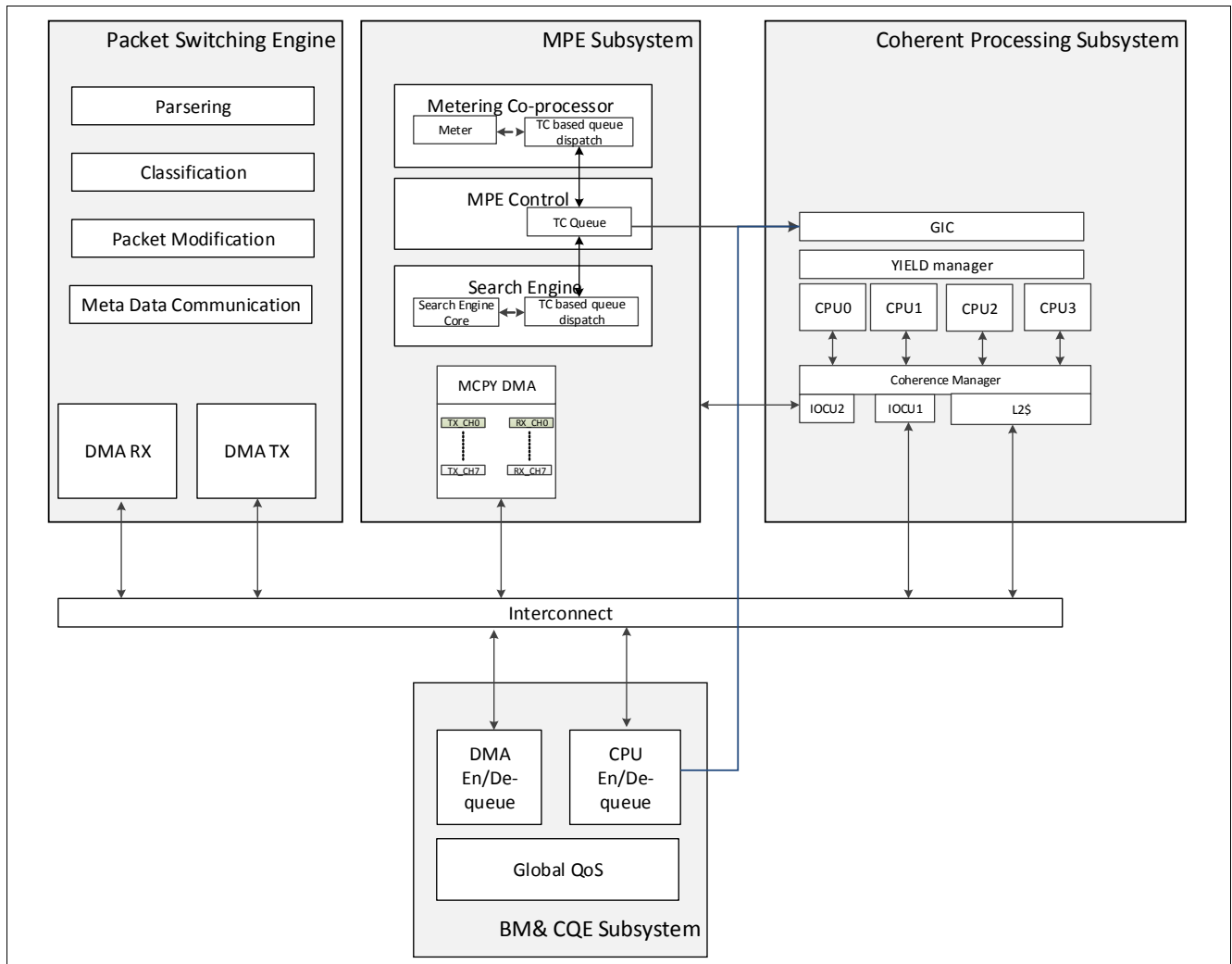


Figure 14 Multi-core Processing Engine



15.1 MPE Features

The following are the MPE features:

- FW based packet processing
- FW interacts with the CQE through interrupts/yields and register reads/writes and is not involved in data path handling
- Header based packet processing architecture; No payload copies into processor cache
- In-band Meta data communication protocol with packet processing engine
- Post processing header construction offload by dedicated cache coherent Memory Copy (MCPY) DMA
- Cache coherent Search Engine
- Meter co-processors
- True multithreading operation with parallel PDU processing context handling that ensure maximum utilization of Virtual CPUs and TCs

15.2 HW Modules

The Search Engine and the Meter Engine are the two co-processors used by the MPE-FW.

The MPE-FW interacts with the HW Engines using the command and response registers and the queues in the MPE Controller.

The MPE-FW also interacts with the MemCopy (MCPY) HW engine to initially load the packet header into the L2 cache and for header copy to the external memory after the header modification is complete.

15.2.1 Search Engine

The MPE-FW sets up the Search Command for search operation and issues the search command to the Search Engine HW using the respective TC command register. There are 2 modes in which the Search Engine can perform the Search operation.

- Manual Mode
- Auto Mode

In the Manual Mode, the MPE-FW needs to create a Search Command Block which has the information on the Key, Mask and the Key size. In the search command, the FW passes the address for this SCB to the search engine. The Search Engine HW reads this block and performs the Search operation.

In the Auto Mode, the MPE-FW just provides the base address of the packet header it received from the CQE. It also provides a field mask in the command. On receiving the Search Command, the Search Engine needs to load the Parser and Header information from the memory and create the Search Key with the predefined fields from the header. After the creation of the key and the mask, the Search Engine HW performs the Search operation.

The MPE Search Engine Compare Tables are maintained in the DDR memory. The Search Engine HW can support up to 8 Compare tables. The table to be used for the particular search operation is chosen by the FW in the Search Command.

15.2.1.1 Search Engine Features

The Search Engine supports the following features:

- Option for Automatic key formation by HW or flexible FW defined key
- Supports up to 8 compare tables
- Configuration to choose between Linear and HASH based tables
- Search can only be triggered by MPE command
- 64k sessions (hash based)



15.2.2 Metering

The Metering supports the following features:

- Total 16 double rate three color metering instances
- Up to two cascaded metering instances can be assigned to measure the rate of a packet
- Metering can be triggered by either the MPE command or by the two parallel CPU control registers

15.2.3 MPE Controller

The MPE controller maintains queues through which the MPE-HW and MPE-FW interact. The MPE-FW (TCs) issues commands to the MPE-HW through the command registers and reads the response from the response registers. There are separate command and response registers for each TC. The MPE controller also provides a mechanism for inter-TC communication which the TCs can use to pass on information to each other during processing.

The MPE Controller has the following queues:

- 1 Free queue for enqueue of commands
- 12 TC queues for communicating with the TCs
- 1 Search Engine Queue to interface with the Search Engine HW
- 1 Metering Engine Queue to interface with the Metering Engine HW

16 DDR SDRAM Controller

The DDR controller features and functionalities are described in this chapter.

16.1 DDR SDRAM System Overview

The DDR SDRAM system illustrated in [Figure 15](#) consists of a Memory Scheduler, a DDR SDRAM controller, a DDR SDRAM PHY Utility Block, a DDR SDRAM PHY (including PADS) and DDR SDRAM devices.

Following DDR speed grades are supported.

- DDR3(L)-1066, DDR3(L)-1333, DDR3(L)-1600, DDR3(L)-1866, DDR3(L)-2133
- DDR4-1866, DDR4-2133, DDR4-2400, DDR4-2666
- LPDDR3-1066, LPDDR3-1333, LPDDR3-1600, LPDDR3-1866, LPDDR3-2133

16.1.1 Block Diagram

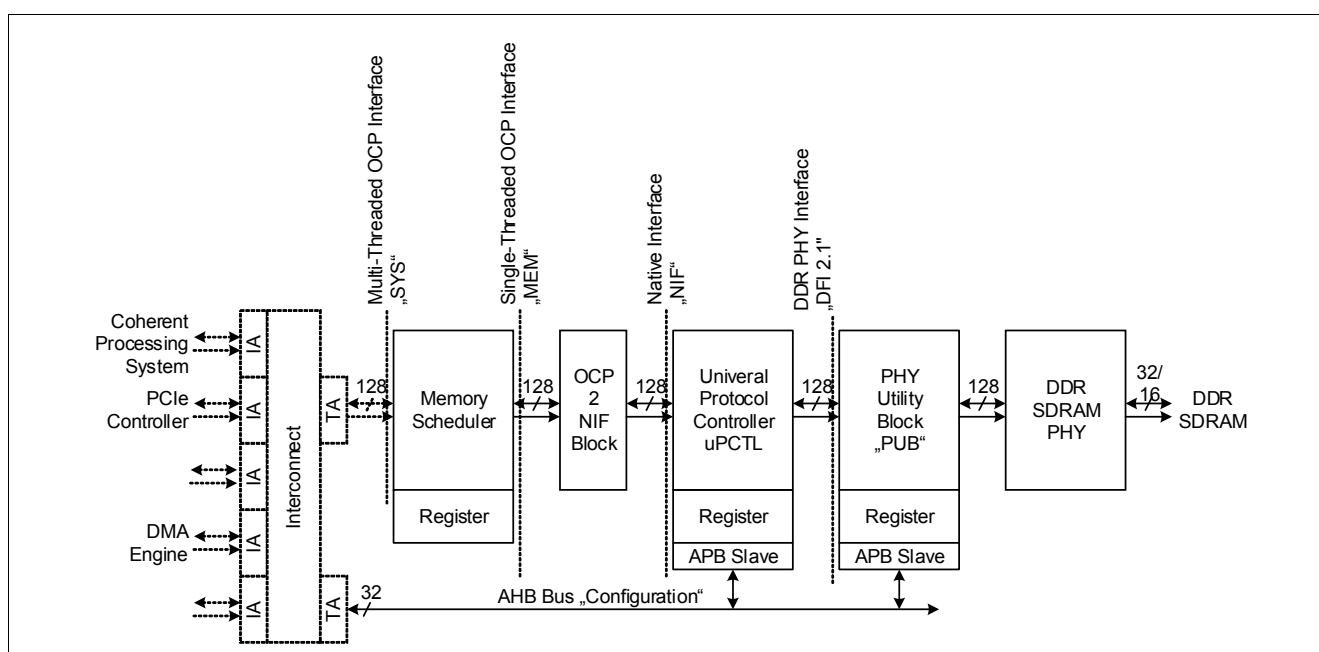
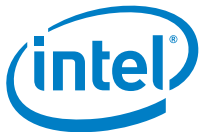


Figure 15 Block Diagram

16.1.2 Feature List

The DDR memory sub system consists of the following building blocks and features:

- Memory Access Scheduler
 - Multi-thread scheduler for QoS data access
 - Requests re-ordering for maximum efficiency
 - User specific address tiling for DRAM bandwidth efficiency
- Universal Protocol Controller (uPCTL, aka DDR Controller)
- PHY Utility Block (PUB)
- DDR PHY (including SSTL PADS)



16.1.3 Memory Access Scheduler

The Memory Access Scheduler is responsible for converting the system internal access into DRAM commands and scheduling optimized access command sequence.

16.1.3.1 Feature List

The Memory Access Scheduler provides the following features:

- Quality-of-Service (QoS)
 - Per-thread QoS scheme selection “priority”, “allocated bandwidth” and “best effort”
 - Higher Priority thread requests achieve extremely low latency
 - Guaranteed BW threads to minimize buffering/receive latency
 - Per-tag starvation avoidance
 - Keep DDR page open as long as possible
 - Groups Reads and Writes towards DDR to achieve maximum utilization
 - Minimizes DDR Rank crossing
- Reorder of requests on different threads and tags to maximize efficiency
- User-configurable address tiling function
 - Transform request addresses
 - Improves DDR SDRAM bandwidth efficiency

16.1.3.2 QoS Arbitration & Levels

Supported arbitration and levels are described in this section.

16.1.3.2.1 Per Tag Arbitration

A thread is assigned with a weight, a counter counts down from the weight, each transfer scheduled will count once, when the counter reaches 0 then wrap around to the initial weight value.

16.1.3.2.2 Per Thread Arbitration

A thread is assigned with different QoS level (priority, bandwidth, and best-efforts), the QoS algorithm ensure priority threads minimize the latency while within bandwidth allocation, bandwidth threads get guaranteed bandwidth.

16.1.3.2.3 QoS Levels

Three levels of QoS are supported: Best-effort, Allocated bandwidth (minimum sustainable BW such as video), and Priority (minimum latency such as cache refill).

16.1.3.2.4 Dynamic QoS Level Adjustment

The adjustment is made based on a per-thread credit value that indicates if a thread is being over- or under serviced. the value is 0 means the thread is being serviced correctly and can stay at its programmed QoS level, a positive means it is under serviced, it will be promoted to the programmed QoS level. Higher QoS level threads always win the arbitration. A serviced thread's credit will be reduced. If its credit becomes negative, then this thread is bearing over-served and it will be demoted to best effort QoS level. The credit value by default will increase at a rate consistent with the user specific bandwidth allocation.



16.2 DDR SDRAM Controller (uPCTL)

The DDR controller has the following features:

16.2.1 Feature List

The DDR controller supports the following features:

- support x8, x16, x32 memories for a memory data path of 32-bits interface
- Two memory ranks, devices within a rank tie to a common chip select
- Up to 8 memory banks
- Efficient DDR protocol with in-order column commands and out-of-order Activate and Precharge commands
- Down to 5 clock cycles command latency
- Automatic power-down and self-refresh entry and exit, driven from software and hardware are supported
- Programmable per rank memory ODT
- Programmable data training assists in training of the data eye of the memory channel
- Programmable page policy
- Optimizes intelligent precharge/activate timings
- Read reorder buffer on the AXI port to holds the data corresponding to a DDR command
- DDR controller Core clock at half of the DDR interface clock (1:2 clock ratio)
- DDR4 features
 - DBI (Data Bus inversion)
 - MPR reads and writes
 - CA parity error with retry
 - Per DRAM addressability (PDA), e.g program ODT or Vref on a DRAM device of a given rank
 - Programmable address latency
 - Write CRC (DRAM device compares the CRC checksum coming from the controller, it flags CRC error)

16.2.2 Data Eye Training Control

The internal data test unit module can generate certain data patterns to assist in bit de-skew, data eye training and long term BER characterization; data patterns are written into memory then read back to automatically adjust the DQS timing.

To minimize the bit error rate (BER), random patterns are continuously written and read back to check external memory bank robustness against VT (voltage, temperature) change.

16.3 DDR PHY Utility Block

The PHY Utility Block (PUB) is used to connect the DDR controller and DDR PHY, it is JEDEC standard based.

16.3.1 Feature List

The PHY PUB supports DDR3/DDR3L (JESD79-3F/79-3-1), DDR4 (JESD79-4) as well as LPDDR3 (JESD209-3B) modes.

16.4 DDR SDRAM PHY

The DDR PHY handles the physical layer protocol including timing, signaling, electrical aspects.

The PHY complies with JEDEC standard.

- Embedded PLL and DDLs for meeting timing specification
- Write leveling delay lines to compensate address and control versus data delays
- Write and read bit delay lines to compensate per-bit delay skew



16.5 SSTL IO Block

The SSTL I/O are interface to the DDR device. The features are described here.

16.5.1 Feature List

The following features are supported:

- Programmable input termination: DDR3(L): 40/60/120 ohms; LPDDR3: 120/240 ohms; DDR4: 34/40/48/60/80/120/240 ohms
- Programmable output impedance
- PVT compensated ODT
- Receiver power down control
- Power on Clear feature to be power supply sequence agnostic

16.5.2 ZQ Calibration Cell

The calibration provides capability for pull-up and pull-down input impedance as well as output impedances of the SSTL I/O. An external 240 Ω +/-1% resistor should be connected from the ZQ pin to ground.

16.6 DDR SDRAM Device Support

The DDR SDRAM controller supports DDR3(L) and DDR4 SDRAM devices as shown in [Table 34](#).

Table 34 Supported DDR3(L) and DDR4 SDRAM Devices

Depth	Width	No. of Banks	Row Addr. Width	Col. Addr. Width
512 Mbit DDR3(L) SDRAM Devices				
32M	x16	8	12	10
64M	x8	8	13	10
1 Gbit DDR3(L) SDRAM Devices				
64M	x16	8	13	10
128M	x8	8	14	10
2 Gbit DDR3(L) SDRAM Devices				
128M	x16	8	14	10
256M	x8	8	15	10
4 Gbit DDR3(L) SDRAM Devices				
256M	x16	8	15	10
512M	x8	8	16	10
8 Gbit DDR3(L) SDRAM Devices				
512M	x16	8	16	10
1G	x8	8	16	11
2 Gbit DDR4 SDRAM Devices				
128M	x16	4	14	10
256M	x8	8	14	10
4 Gbit DDR4 SDRAM Devices				
256M	x16	4	15	10



Table 34 Supported DDR3(L) and DDR4 SDRAM Devices (cont'd)

Depth	Width	No. of Banks	Row Addr. Width	Col. Addr. Width
512M	x8	8	15	10
8 Gbit DDR4 SDRAM Devices				
512M	x16	4	16	10
1G	x8	8	16	10
16 Gbit DDR4 SDRAM Devices				
1G	x16	4	17	10
2G	x8	8	17	10

Table 35 Supported LPDDR3 Devices

Depth	Width	No. of Banks	Row Addr. Width	Col. Addr. Width
1 Gbit LPDDR3 SDRAM Devices				
64M	x16	8	13	10
32M	x32	8	13	9
2 Gbit LPDDR3 SDRAM Devices				
128M	x16	8	14	10
64M	x32	8	14	9
4 Gbit LPDDR3 SDRAM Devices				
256M	x16	8	14	11
128M	x32	8	14	10
8 Gbit LPDDR3 SDRAM Devices				
512M	x16	8	15	11
256M	x32	8	15	10
16 Gbit LPDDR3 SDRAM Devices				
1G	x16	8	15	12
512M	x32	8	15	11



17 Clock Generation Unit

The chapter describes the Clock Generation Unit

17.1 Features

The Clock Generation Unit consists of:

- An oscillator which receives either a crystal or a digital clock
- PLL0a with SSC (Spread Spectrum Clocking) capability for
 - **Coherent Processing Subsystem**
 - **Quad Serial Peripheral Interfaces** Reference Clock
- PLL0b with SSC (Spread Spectrum Clocking) capability for
 - **Packet Switching Engine**
 - **Buffer Manager** and **Carrier Grade Quality of Service Engine**
 - **Peripherals**
 - **Trusted Execution Environment**
- PLL2 with SSC (Spread Spectrum Clocking) capability for
 - **DDR SDRAM Controller**
- Low Jitter PLL3 for
 - **WAN Serdes** and **PCIe/XFI Serdes** in Ethernet mode
 - Precise time protocol reference time
 - Synchronous Ethernet functionality
- Low Jitter PLL5 generates PON mode Serdes reference clock
- Clock Dividers and Multiplexers

17.2 Functional Description

The CGU provides all necessary clock frequencies for the SoC, using a single crystal and multiple PLLs/dividers/multiplexers to generate the required clocks.

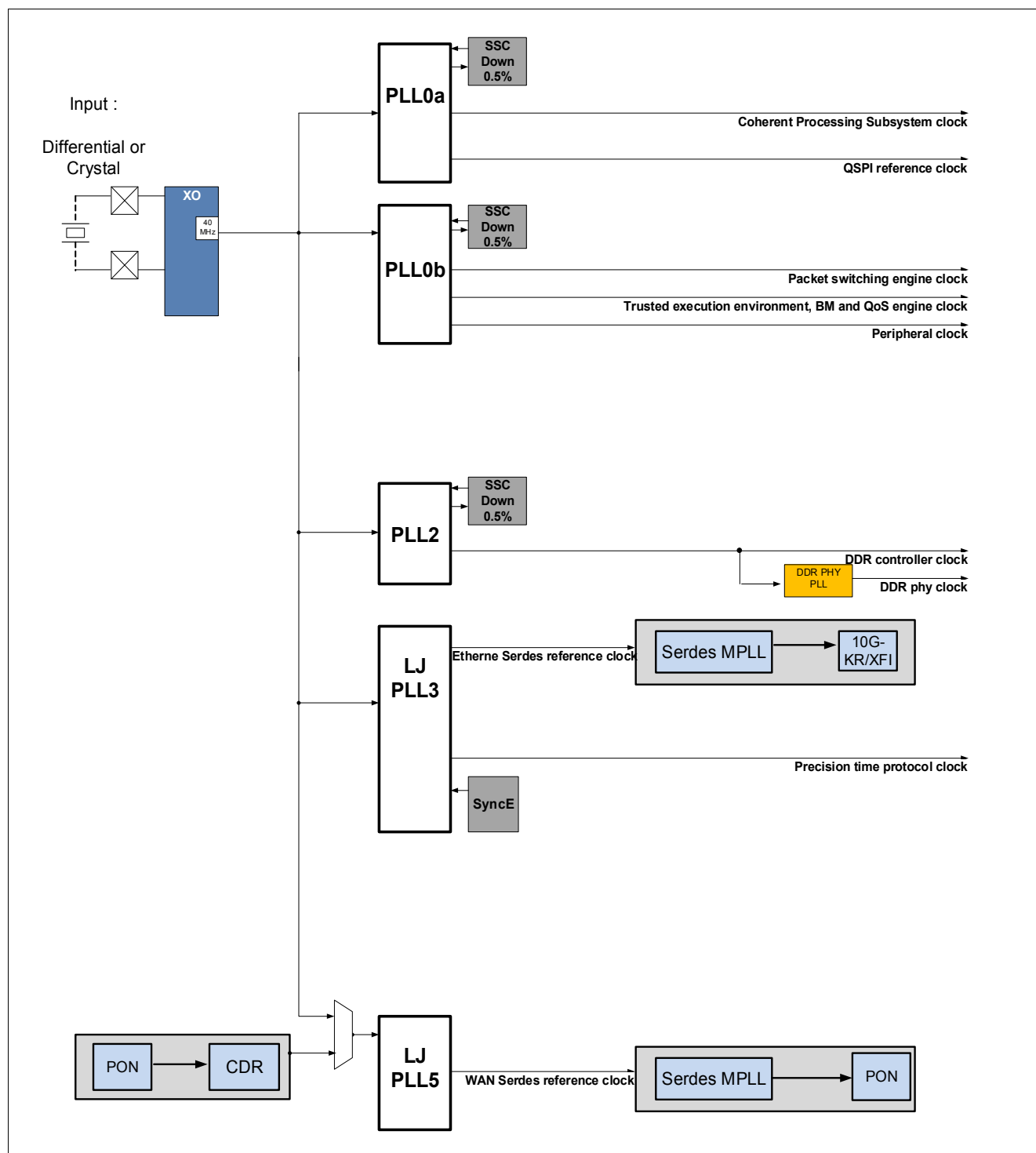


Figure 16 Overview of the CGU

The oscillator circuitry is described in [Chapter 17.2.1](#).

17.2.1 Clock Input Option and Oscillator Circuit

The oscillator circuit is designed to work either with an external crystal or an external clock source. An oscillator circuit basically consists of an inverting amplifier with XTAL1 as input and XTAL2 as output. An AC-coupled shaper stage provides 50% duty cycle. The CLK_MD pin defines which option to use during power on reset.

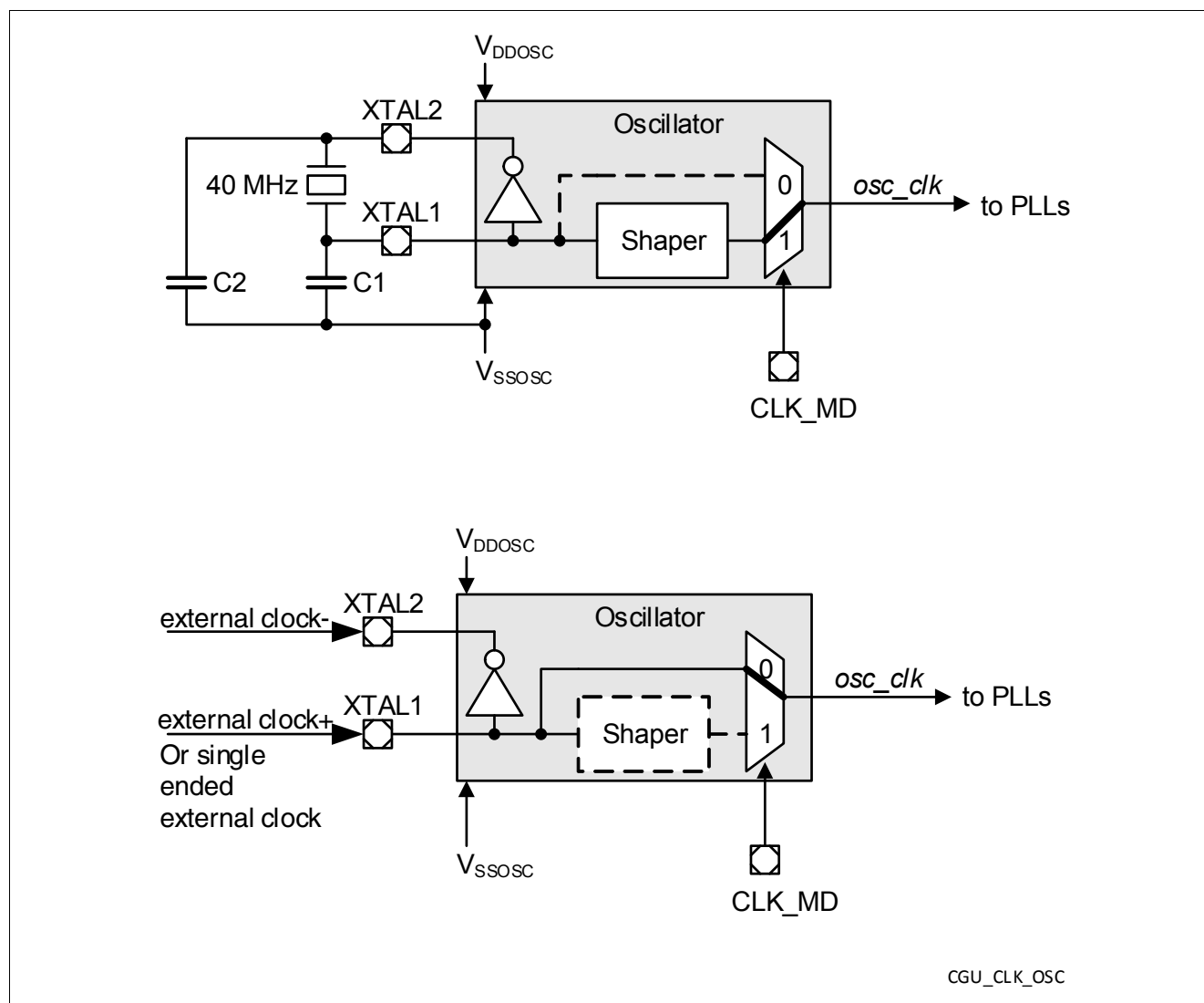


Figure 17 Clock Input Options

17.2.2 External Clock Output or Input Interface

The Clock Generation Unit provides clocks at the external pins used for multiple purpose. The clocks are selected inside the CGU by register **CGU_IF_CLK** and are available as alternative functions at GPIO pins.

The following clocks can be selected and provided at the GPIOs:

- GPC1: configurable to be one of the below
 - 40 MHz output mode: Buffered XTAL clock
 - 25 MHz or 10 MHz output mode: this is a 25 MHz or 10 MHz clock output from LJ-PLL3.
 - 1.544 MHz or 2.048 MHz or 8 kHz clock or PON NTR clock: the clock is divided from LJ-PLL3 or from PON TC layer.
 - Input mode: it is used for clock source of SyncE or as PPS input
- PPS: configurable to be one of the below
 - Output mode: 1 Hz PPS pulse or CPU triggered pulse
 - Input mode: it is used for reference time synchronization with other chips.

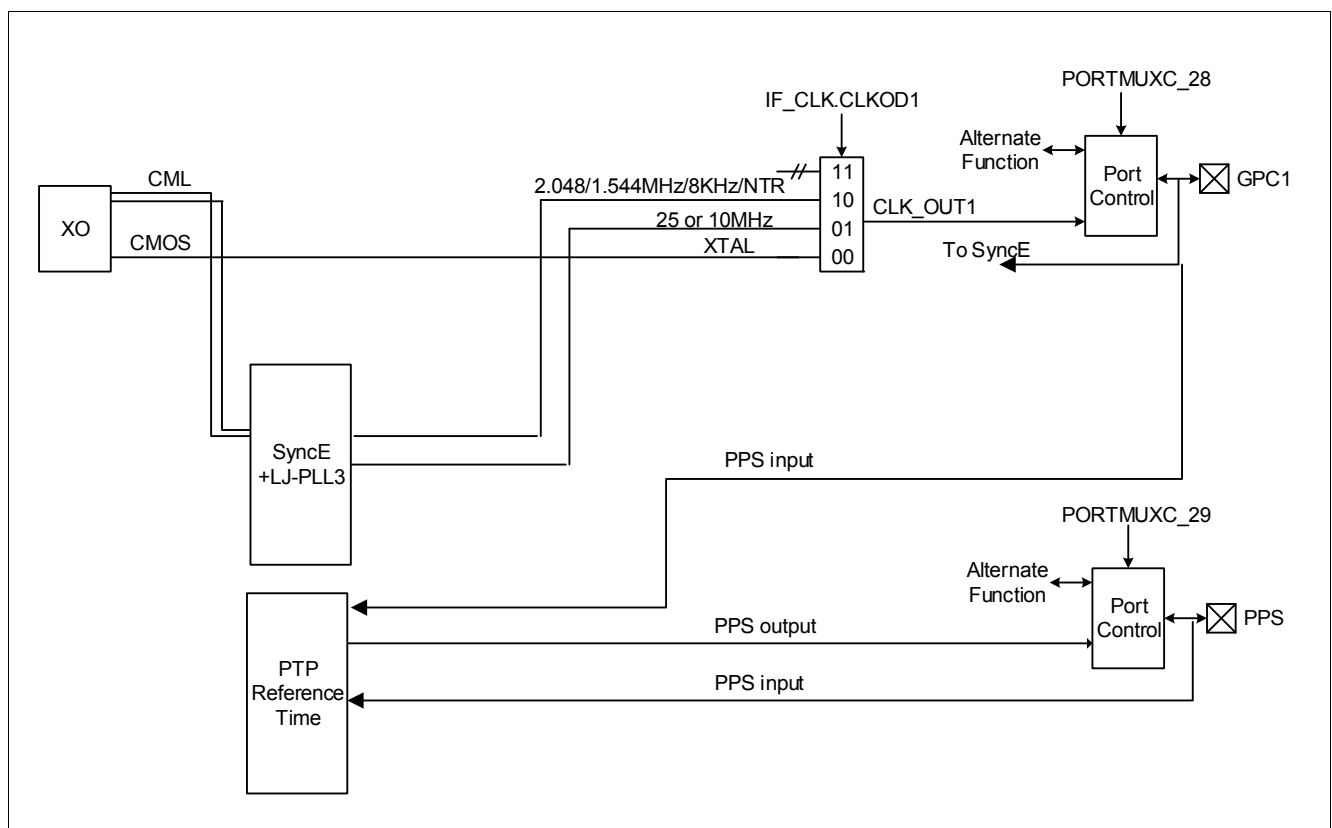


Figure 18 External Clock Output and Input Interface



17.2.3 Synchronous Ethernet

PRX126 support Synchronous Ethernet.

It is configurable to synchronize to one of the below clock sources:

- Recovered received clock by WAN Serdes (PON mode, AON mode and Copper Wire mode)
- Recovered received clock by Client side Serdes
- Input clock via GPC1
- Programmable bandwidth 0.1 Hz, 3 Hz and 100 Hz

The following functions are clocked from the synchronized clock

- WAN Serdes (PON mode, AON mode and Copper Wire mode)
- Client side Serdes
- 1588/PTP timers
- Output clock via GPC1

Loss of signal detection is integrated on chip and all the modules are supplied seamlessly with clocks generated from the local source when the external master clock signal is not available.

18 Reset System

18.1 Features

The Reset System supports:

- Power-On Reset Generation
 - Power-On Reset input pin
 - Under Voltage Detection
- Software Reset Requests and Reset Domains
- One 32-bit non-resetable register for software usage
- Watchdog Timer (WDT) Reset Request and Reset Domain
- Reset Control Unit with Finite State Machine controlling the reset flow

18.2 Functional Description

The reset system mainly consists of two parts, the power-on reset logic as well as the Reset Control Unit (RCU). The power on reset logic controls the pin strapping functionality. Additionally, the reset system provides a reset output pin. Further, the system on chip implements components, which can trigger the RCU. This is the Watchdog Timer (WDT). The software can force the RCU to generate a global reset request as well as reset requests individually per peripheral, see [Chapter 18.2](#).

18.2.1 Overview

An overview of the reset system is given in [Figure 19](#).

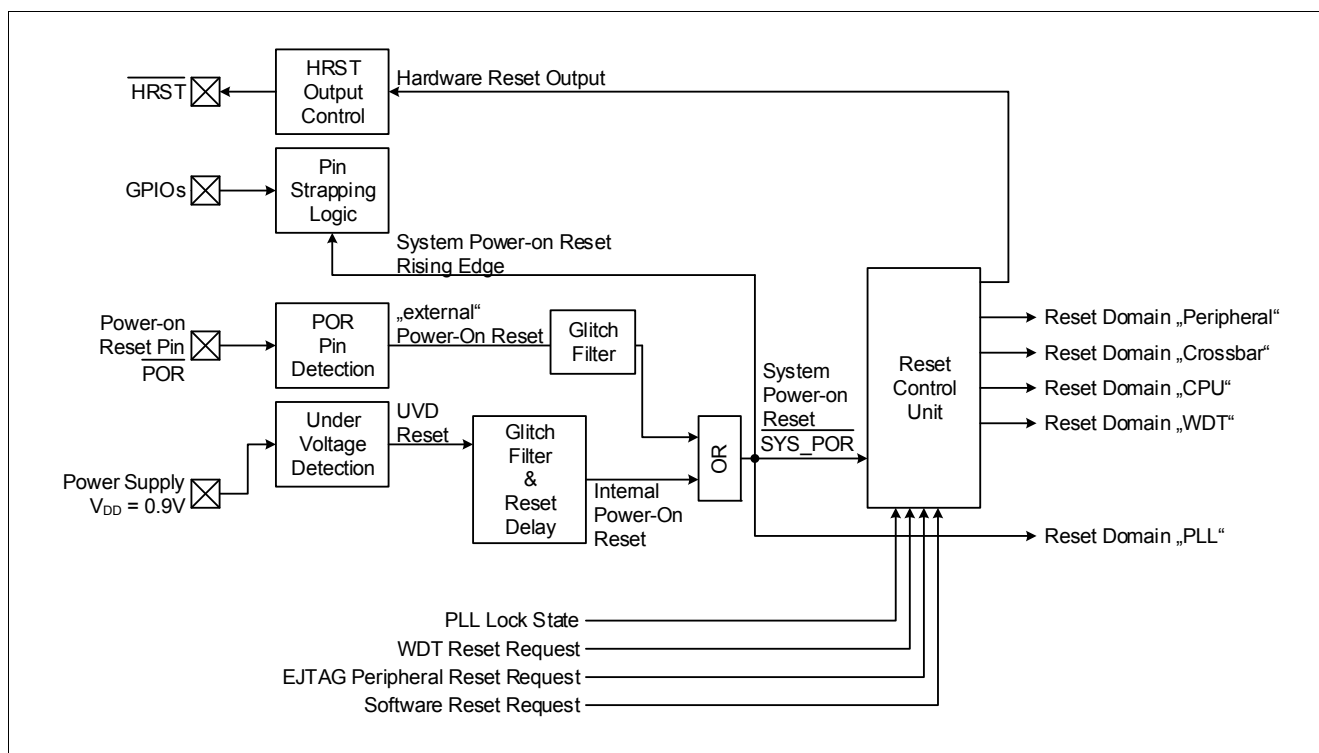


Figure 19 Reset System



18.2.2 Power-On Reset Generation

The active-low system power-on reset $\overline{\text{SYS_POR}}$ is generated by the following events:

- An active-low-level at the external power-on reset input pin $\overline{\text{POR}}$
- Internally by an Under Voltage Detection (UVD)

The system power-on reset $\overline{\text{SYS_POR}}$ forces the following actions:

- Activates the reset domain PLL
- Latches the pin strap information
- Activates the General Hardware Reset
- Activates the Hardware Reset Output Pin $\overline{\text{HRST}}$

When the system power-on reset, $\overline{\text{SYS_POR}}$ has been executed, the following status bits are affected:

- The power-on reset cause indication bit RST_STAT.POR is set
- All other reset cause indication bits in RST_STAT.XX are cleared
- All peripheral reset status bits in register RST_STAT.XX for all reset domains are set

18.2.2.1 Power-On Reset Generation by Input Pin $\overline{\text{POR}}$

The external power-on reset input pin $\overline{\text{POR}}$ is an active-low input pin with an internal pull up resistor. The glitch filter logic suppresses glitches at input pin $\overline{\text{POR}}$. A low-level at input pin $\overline{\text{POR}}$ generates the system power on reset $\overline{\text{SYS_POR}}$.

The external power-on reset input pin $\overline{\text{POR}}$ is left open, when using the internal sources Under Voltage Detection (UVD). Furthermore, a need for external resistor and capacitor components at input pin $\overline{\text{POR}}$ is eliminated. The input pin $\overline{\text{POR}}$ is pulled up internally, and therefore inactive in this case.

18.2.2.2 Power-On Reset Generation by Under Voltage Detection

The UVD logic monitors the core voltage level, which is typ. V_{DD0V9} and works as follows:

- An on-chip comparator compares a reference voltage, V_{REF} , and VDD. If VDD falls below a predefined threshold, the UVD Reset signal is asserted. The glitch filter suppresses spikes of less than 20 μs .

The internal power on reset output of the UVD logic is always enabled. Finally, the reset delay element prolongs the UVD reset request for up to 120 ms.

18.2.2.3 Pin Strapping Functionality

The configuration input pins of the system-on-chip, the strap pins, are sampled and latched with the rising edge of the system power-on reset $\overline{\text{SYS_POR}}$. This affects the following registers and status bits associated with the pin strapping functionality:

- Latches the values at the GPIOs (including flash interface power supply information) into GP_STRAP
- Latches the boot information into RST_BOOT.INFO
- Latches the endianness information into RST_STAT.ENDIAN

18.2.3 Hardware Reset Output Pin

The hardware reset output pin $\overline{\text{HRST}}$ is an open drain output with internal pull up resistor. Pin $\overline{\text{HRST}}$ can be used to force a reset of external board components, while the system-on-chip is in reset state. Additionally, pin $\overline{\text{HRST}}$ can be activated by software. The hardware reset output pin $\overline{\text{HRST}}$ is driven active-low upon the following events:

- The system power-on reset $\overline{\text{SYS_POR}}$ is activated (e.g. a system power up event happened)
- Setting the global software reset request bit RST_REQ.SWRST or WDT events
- Setting the reset request bit RST_REQ.HRST



19 Chip ID

PRX126 Chip ID is 00004113_H.

Table 36 Chip ID Fields

Fields	Bits	Description
V0	31	This bit is fixed to 0.
VERSION	30:28	Chip Version Number 000 _B V11 First Version
PNUM	27:12	Part Number 0004 _H PRX126 PRX126 Part Number
MANID	11:1	Manufacturer ID The JEDEC normalized manufacturer ID code is 89 _H .
V1	0	This bit is fixed to 1.

20 Peripherals

This chapter describes the following:

- [Universal Asynchronous Receiver/Transmitter \(Chapter 20.1\)](#)
- [Quad Serial Peripheral Interface \(Chapter 20.2\)](#)
- [I2C Controller \(Chapter 20.3\)](#)
- [General Purpose Timer Counter \(Chapter 20.4\)](#)
- [General Purpose I/O \(Chapter 20.5\)](#)
- [Dying Gasp Detection Module \(Chapter 20.6\)](#)
- [UART Debug Interface \(Chapter 20.7.1\)](#)
- [EJTAG Interface \(Chapter 20.7.2\)](#)
- [On-Chip Packet Buffer \(Chapter 20.8\)](#)

20.1 Universal Asynchronous Receiver/Transmitter

There are two identical ASC (Asynchronous Serial Channel) modules integrated in the chip. Each of the ASC module is primarily used as UART and consists of the modules shown in [Figure 20](#).

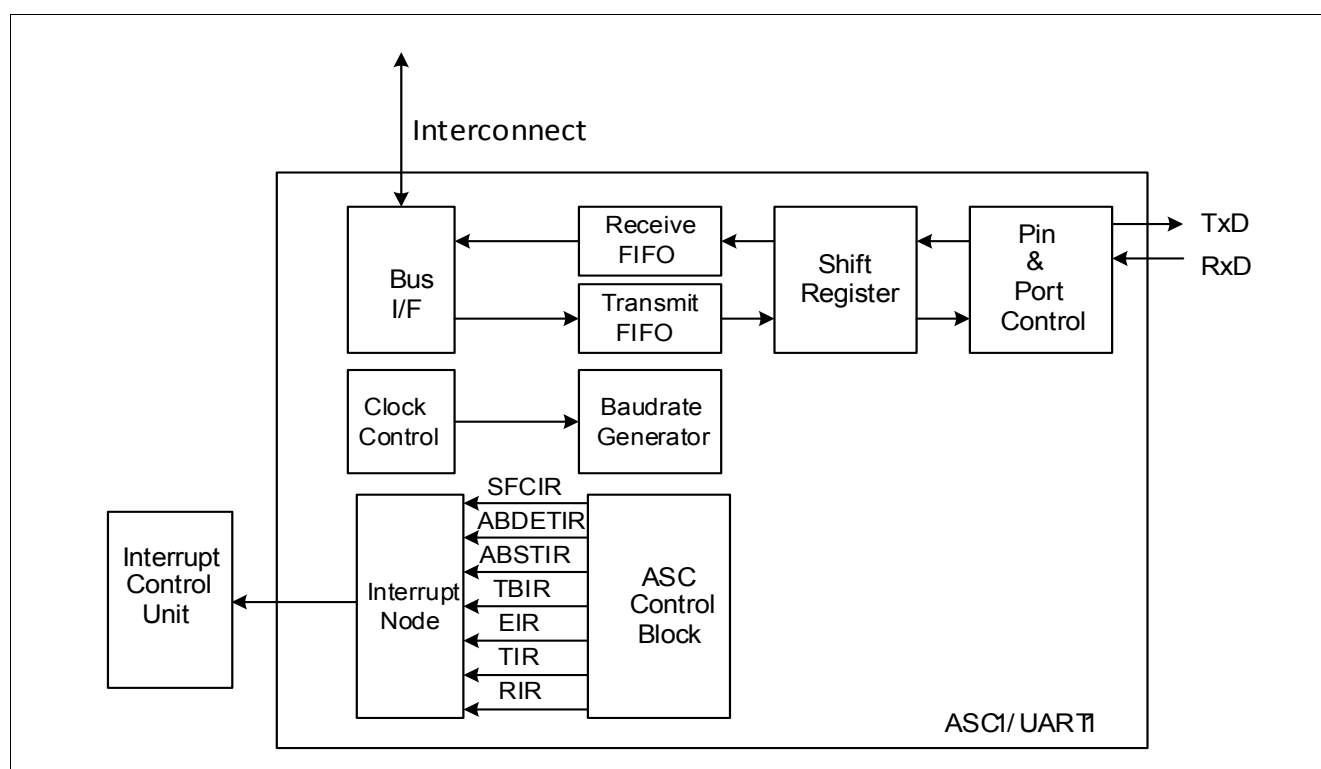


Figure 20 ASC/UART System Block Diagram



20.1.1 Feature List

ASC module provides the following features:

- Automatic clock shut on/off, depending on line and/or bus activity
- Full-duplex asynchronous operating modes
 - 7- or 8-bit data frames, LSB first
 - Parity bit generation/checking
 - One or two stop bits
 - Baudrate from 12.6 MBaud to 426 Baud (@200 MHz module fixed clock)
- Programmable XON/XOFF flow control
 - XON and XOFF characters programmable
 - Automatic flow control can be programmed, so that XON/XOFF characters are automatically inserted into/removed from the data stream, depending on the FIFO status
- Line break detection
- Programmable End-of-Message detection based on character matching or time-out
- Multiprocessor Mode for automatic address/data byte detection
- Loopback capability
- Double buffered transmitter/receiver
- Interrupt generation
 - On a transmitter buffer empty condition
 - On a transmit last bit of a frame condition
 - On a receiver buffer full condition
 - On an error condition (frame, parity, overrun error, line break)
 - On XOFF reception On Modem Control/Status signal changes
- Autobaud detection unit for asynchronous operating modes
 - Detection of standard baudrates
1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200, and 230400 Baud
 - Detection of non-standard baudrates
 - Detection of Asynchronous Modes
 - 7-bit, even parity; 7-bit, odd parity
 - 8-bit, even parity; 8-bit, odd parity; 8-bit, no parity Automatic initialization of control bits and baudrate generator after detection
 - Detection of a serial two-byte ASCII character frame
- FIFO
 - 16 X 8 receive FIFO (RXFIFO), 8-bit organized for ASC1
 - 16 X 8 transmit FIFO (TXFIFO), 8-bit organized for ASC1
 - Independent control of RXFIFO and TXFIFO depth and width
 - 8-bit FIFO data width without segmentation or reassembly in 8-bit configuration
 - Programmable Receive/Transmit Interrupt Trigger Level
 - Receive and transmit FIFO filling level indication
 - Overrun error generation
 - Underflow error generation



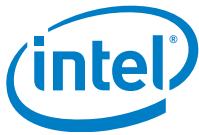
20.2 Quad Serial Peripheral Interface

The Quad Serial Peripheral Interface (SPI) is equipped with a receive as well as a transmit FIFO with eight entries each 32-bit width. The QSPI control and status registers are explicitly accessed via the bus. The CPU handles bidirectional data traffic on the QSPI interface via another bus.

20.2.1 Feature List

The QSPI supports:

- 1 bit SPI interfaces, compliance to legacy SPI standard, mode 0
- 2 bit, 4 bit mode single data rate operation
- 2 bit, 4 bit double data rate operation
- Maximum 100 MHz for single data rate
- Maximum 100 MHz for double data rate
- Adaptive delay tabs for high speed interface access timing with proven integrated DLL designed for double data rate interface
- Programming sequence to adapt different SPI flash type
- eXecution In Place, fully memory mapped access for CPU



20.3 I2C Controller

There is one I2C master controller (I2C0) and two I2C slave controllers (I2C1 and I2C2) integrated in chip.

20.3.1 Features of the I2C Master Controller

The I2C bus is a simple 2-wire serial communication bus. A I2C master is the device which initiates a data transfer on the bus and generates the clock signals to permit the transfer. At any given time, only one master is present on the bus, any device addressed is considered a slave. Master also terminates the transfer.

- I2C bus specification V2.1 complied
- 100 kbit/s and 400 kbits/s speed modes are supported
- Total bus load up to 400 pF
- 8-bit per byte and data is transferred MSB first
- 7-bit address
- 8 stages FIFO depth (32-bit width)
- Master Mode: bus access arbitration protocol is supported
- Only master mode is supported

Limitation of Multiple Masters in Single I2C Bus

However, when there are multiple masters in the same I2C bus, the embedded I2C master will issue dummy address cycle after a re-start phase when SOPE=0 and last data of current packet is received. This could potentially hang the bus if the dummy address is incidentally same as one of the I2C slave addresses on the same bus.

20.3.2 Features of the I2C Slave Controller

The I2C bus is a simple 2-wire serial communication bus. A I2C slave is the device which respond to a data transfer on the bus.

The two I2C slave controllers are connected to the same I2C bus.

- I2C bus specification V2.1 complied
- 100 kbit/s speed modes (0~100 Kbaud range) and 400 kbit/s speed modes (0~400 Kbaud range) are supported
- Total bus load up to 400 pF
- 8-bit per byte and data is transferred MSB first
- 7-bit address
- 8 stages FIFO depth (32-bit width)



20.4 General Purpose Timer Counter

There are three identical GPTC modules integrated.

20.4.1 Feature List

Each GPTC provides the following features:

- 6 16-bit Timers
- 3 single timer/counter blocks (timer1,2,3) consisting of two 16-bit timer/counter blocks (A, B)
- Operation as two separate timer/counter blocks in 16-bit mode or a single concatenated timer/counter in 32-bit mode
- Programmable counting direction
- Run/stop control allows to stop and restart the timer/counter by software at any time
- Software controlled reload allows to restart at the current value or at the reload value
- Auto reload function enables cyclic timer/counter operation
- Operation as timer or as counter
- Programmable edge detection of external count input or level sensitive counting
- Programmable inversion of the external input can be synchronized (to count asynchronous events at 1/2 of the GPTC clock frequency) or unsynchronized (for clock-synchronous events up to the GPTC clock frequency)
- Interrupt generation at overflow

20.4.2 Overview

The overview and GPTC alternate input sources of 8 kHz clock is as shown in [Figure 21](#).

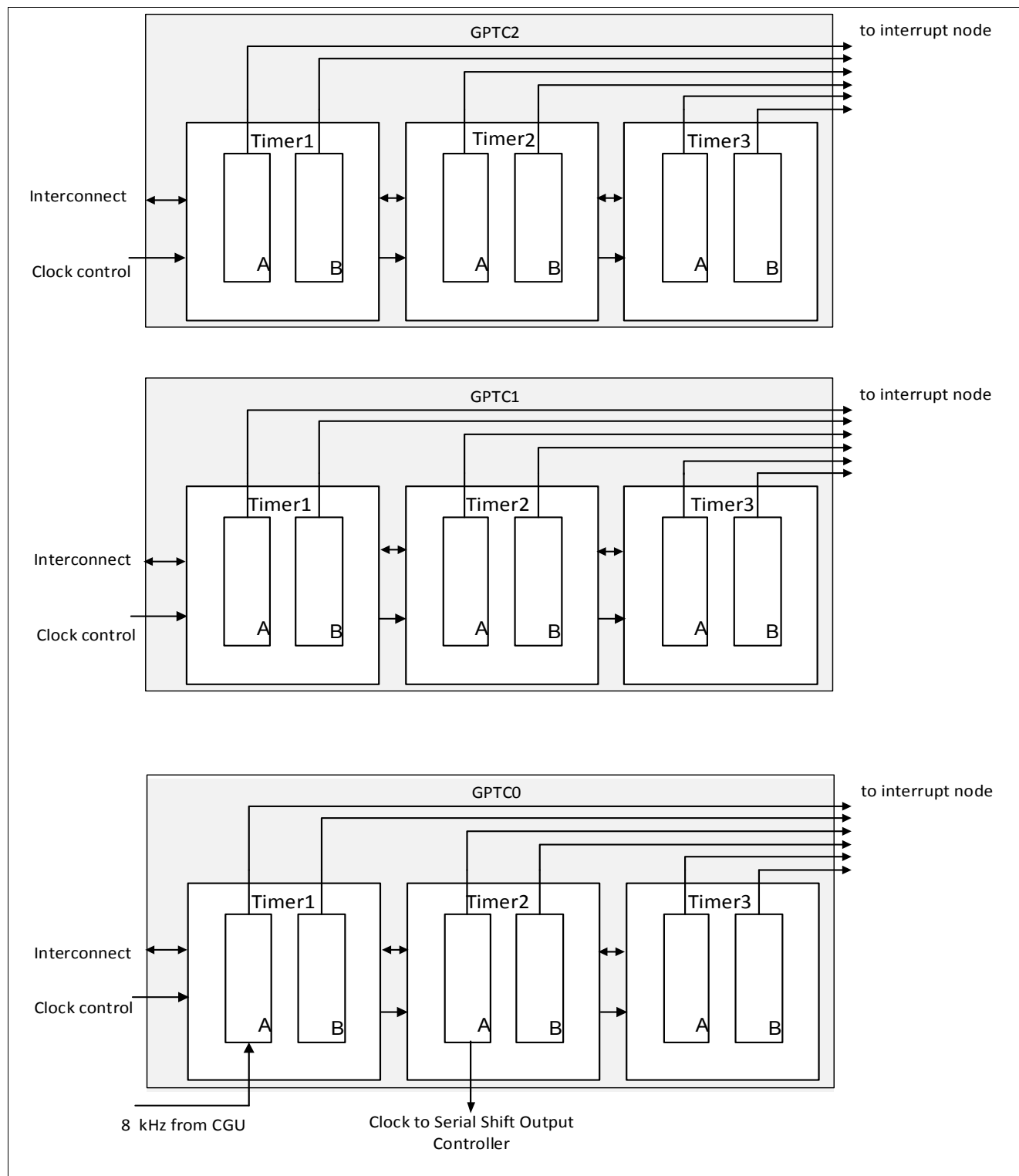


Figure 21 GPTC0/1/2 Alternate Input Source

20.5 General Purpose I/O

The module includes four parallel general purpose input/output (GPIO) ports.

20.5.1 Features

The GPIO Ports support:

- Up to 24 general purpose I/Os
- Each GPIO pin can be interrupt input
- GPIO programming support bit-wise write (no need read-modify-write)
- Input/Output selectable
- Input Function
 - Tristate
 - High-impedance with a weak pull-up device
 - High-impedance with a weak pull-down device
 - Schmitt-Trigger detection (always enabled)
- Output Functions
 - Push/pull (optional pull-up/pull-down)
 - Open drain with internal pull-up
 - Open drain with external pull-up

20.5.2 GPIO Functionality

Figure 22 shows a general block diagram of a port line. Each port line is equipped with a number of control and data bits, enabling very flexible usage of the line.

Each port pin can be configured for input or output operation. In input mode (default after reset), the output driver is switched off (high-impedance). The actual voltage level present at the port pin is translated into a logic 0 or 1 via a Schmitt-Trigger device and can be read via the read only register Px_IN. In output mode, the output driver is activated and drives the value supplied through the multiplexer to the port pin. For GPIO function, Switching between input and output mode is accomplished through the Px_DIR register, which enables or disables the output driver. If the same pin used for other alternative function, the output driver may be controlled by the function module.

The output multiplexer in front of the output driver enables the port output function to be used for different purposes. If the pin is used as general purpose output, the multiplexer is switched by software to the Output Data Register Px_OUT. Software can set or clear the bit in Px_OUT, and therefore it can directly influence the state of the port pin. If the on-chip peripheral units use the pin for output signals, alternate output lines can be switched via the multiplexer to the output driver circuitry.

Latch Px_IN is provided for input functions of the on-chip peripheral units. Its input is connected to the output of the input Schmitt-Trigger. Further, an input signal can be connected directly to the various inputs of the peripheral units (AltDataIn). The function of the input line from the pin to the input latch Px_IN and to AltDataIn is independent of the port pin operates as input or output. This means that when the port is in output mode, the level of the pin can be read by software via latch Px_IN or a peripheral can use the pin level as an input. This offers additional advantages in an application.

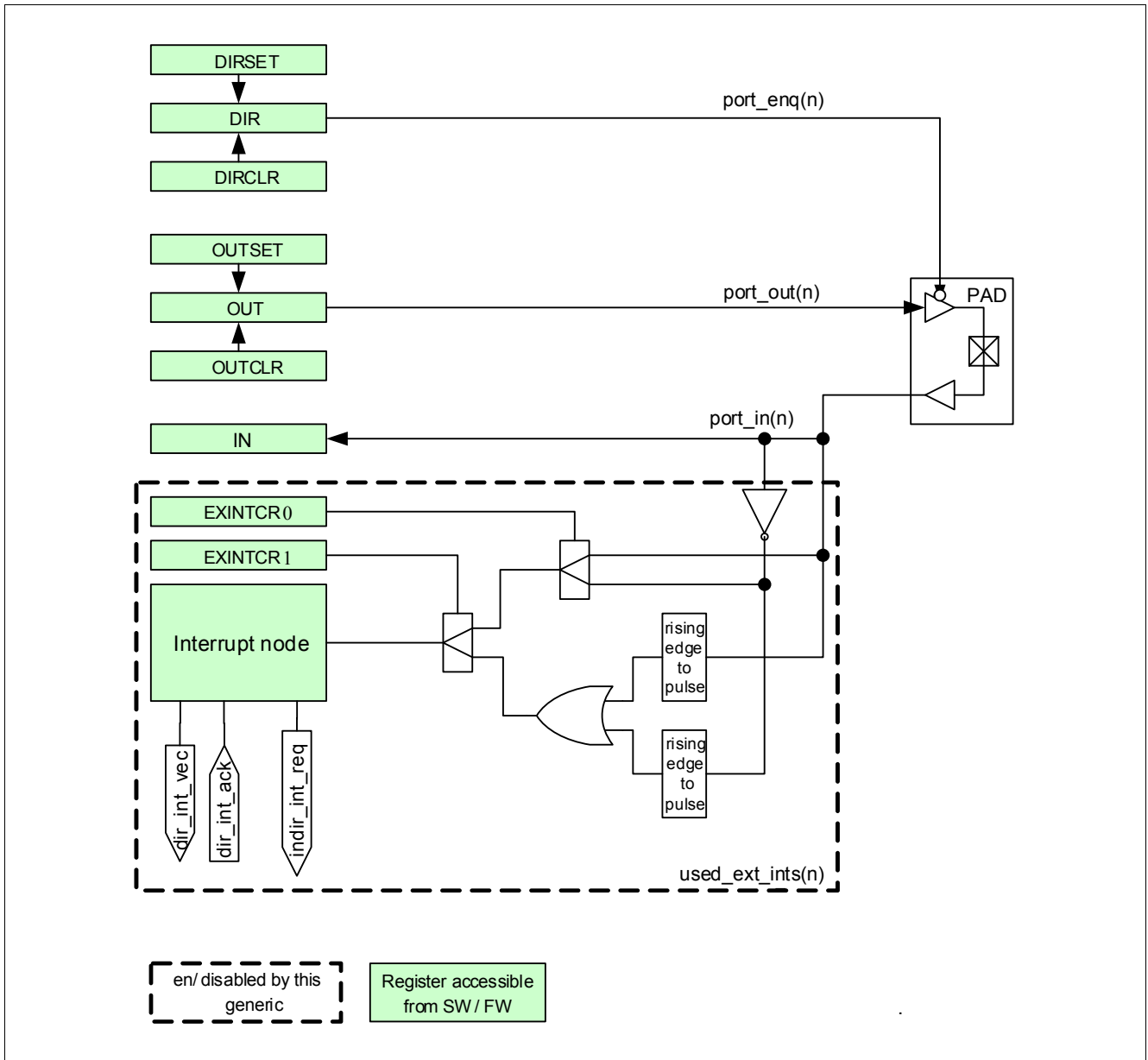


Figure 22 GPIO bit

20.5.3 Pad Control Overview

This block provides the SW/FW interface for the static pad control for up to 32 pads.

Per Pad Features

- Pull up enable
- Pull down enable
- Slew Rate control (1 bit)
- Drive current control (2 bit)
- Pad multiplexer control (up to 3 bit)



20.5.4 External Interface

Attention: The pin functionality in [Table 37](#) highlighted in bold indicates the pin functionality after reset.

Table 37 GPIO Functions

GPIO Pin	Pin Functionality	Associated Register/Module	Alternate Function	Direction Control
0	GPIO used as input	GPIO	PORTMUXC_00:000	DIR_0
	GPIO used as output			
	QSCK used as output	QSPI	010	Fixed
1	GPIO used as input	GPIO	PORTMUXC_01:000	DIR_0
	GPIO used as output			
	QSCS used as output	QSPI	010	Fixed
2	GPIO used as input	GPIO	PORTMUXC_02:000	DIR_0
	GPIO used as output			
	QSIO0 used as inout	QSPI	010	Controlled by QSPI
3	GPIO used as input	GPIO	PORTMUXC_03:000	DIR_0
	GPIO used as output			
	QSIO1 used as inout	QSPI	010	Controlled by QSPI
4	GPIO used as input	GPIO	PORTMUXC_04:000	DIR_0
	GPIO used as output			
	QSIO2 used as inout	QSPI	010	Controlled by QSPI
5	GPIO used as input	GPIO	PORTMUXC_05:000	DIR_0
	GPIO used as output			
	QSIO3 used as inout	QSPI	010	Controlled by QSPI
8	GPIO used as input	GPIO	PORTMUXC_08:000	DIR_0
	GPIO used as output			
10	GPIO used as input	GPIO	PORTMUXC_10:000	DIR_0
	GPIO used as output			
	I2C1_SCL as inout	I2C1 & I2C2	010	Controlled by I2C1 and I2C2
11	GPIO used as input	GPIO	PORTMUXC_11:000	DIR_0
	GPIO used as output			
	I2C1_SDA as inout	I2C1 & I2C2	010	Controlled by I2C1 and I2C2



Table 37 GPIO Functions (cont'd)

GPIO Pin	Pin Functionality	Associated Register/Module	Alternate Function	Direction Control	
13	GPIO used as input	GPIO	PORTMUXC_13:000	DIR_0	
	GPIO used as output				
15	GPIO used as input	GPIO	PORTMUXC_15:000	DIR_0	
	GPIO used as output				
16	GPIO used as input	GPIO	PORTMUXC_16:000	DIR_0	
	GPIO used as output				
20	GPIO used as input	GPIO	PORTMUXC_20:000	DIR_0	
	GPIO used as output				
	OPT_RX_LOS as input	PON TC	010	Fixed	
21	GPIO used as input	GPIO	PORTMUXC_21:000	DIR_0	
	GPIO used as output				
	OPT_TX_DIS_SFP as inout	PON TC	010	Controlled by PON TC	
22	GPIO used as input	GPIO	PORTMUXC_22:000	DIR_0	
	GPIO used as output				
	OPT_TX_FAULT as input	PON TC	010	Fixed	
23	GPIO used as input	GPIO	PORTMUXC_23:000	DIR_0	
	GPIO used as output				
	OPT_TX_SD used as input	PON TC	010	Fixed	
24	GPIO used as input	GPIO	PORTMUXC_24:000	DIR_0	
	GPIO used as output				
	OPT_TX_PUP used as inout	PON TC	010	Controlled by PON TC	
25	GPIO used as input	GPIO	PORTMUXC_25:000	DIR_0	
	GPIO used as output				
	OPT_RX_SD_SFP used as inout	PON TC	010	Controlled by PON TC	
26	GPIO used as input	GPIO	PORTMUXC_26:000	DIR_0	
	GPIO used as output				
	I2C0_SCL used as inout	I2C0	010	Controlled by I2C0	
27	GPIO used as input	GPIO	PORTMUXC_27: 000	DIR_0	
	GPIO used as output				
	I2C0_SDA used as inout	I2C0	010	Controlled by I2C0	
28	GPIO used as input	GPIO	PORTMUXC_28:000	DIR_0	
	GPIO used as output				
	GPC1 as input	CGU	010		
	GPC1 as output				

Table 37 GPIO Functions (cont'd)

GPIO Pin	Pin Functionality	Associated Register/Module	Alternate Function	Direction Control
29	GPIO used as input	GPIO	PORTMUXC_29:000	DIR_0
	GPIO used as output			
	PPS as input	CGU	010	
	PPS as output			
30	GPIO used as in/output	GPIO	PORTMUXC_30:000	DIR_0
	GPIO used as output			
	UART1_RX as input	UART1	001	Fixed
	UART1_RX_PON as input	PON TC	010	Fixed
	UART_RX_TEP as input	Secure TEP	100	Fixed
31	GPIO used as in/output	GPIO	PORTMUXC_31:000	DIR_0
	GPIO used as output			
	UART1_TX as output	UART1	001	Fixed
	UART1_TX_PON as output	PON TC	010	Fixed
	UART_TX_TEP as output	Secure TEP	100	Fixed

20.6 Dying Gasp Detection Module

The dying gasp monitors the external input pin (VDD_DG) and generates the interrupt to processor and to PON TC module whenever the voltage level lower than the predefined level.

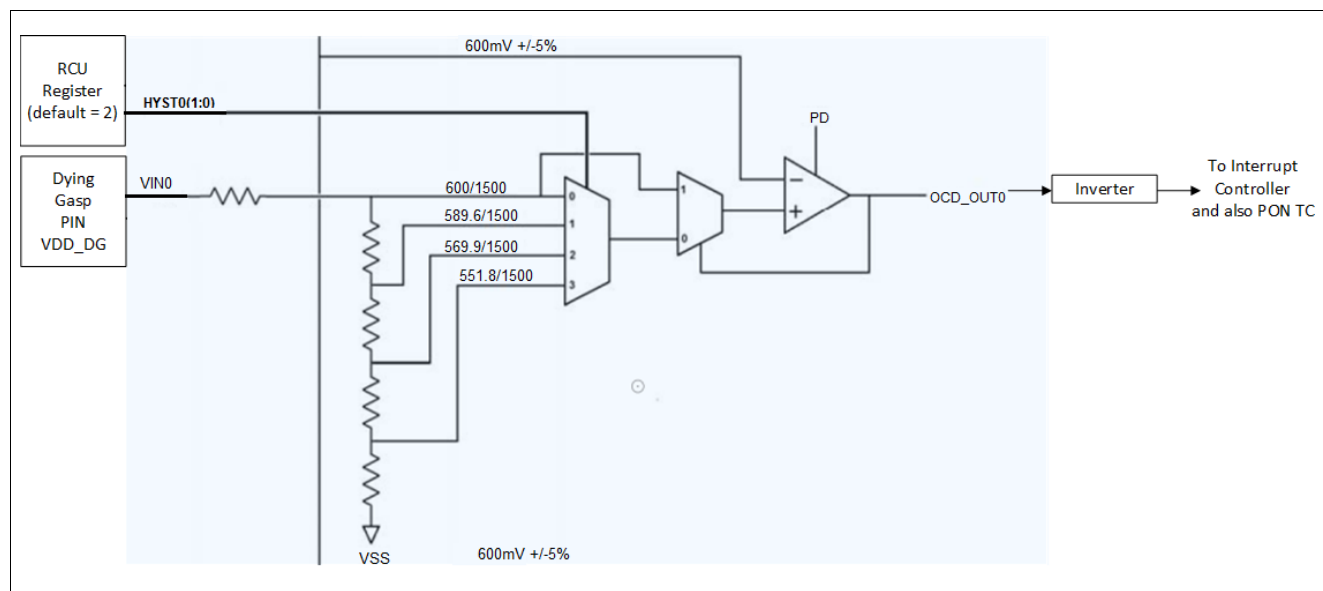
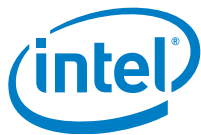


Figure 23 Block Diagram of Dying Gasp Detection

20.7 Debug Interface

There are two types of debug interfaces: UART debug interface and EJTAG debug interface.



20.7.1 UART Debug Interface

Two instances in SoC can be used for Coherent processor cores debugging.

Besides the two instances in SoC, the following modules has embedded UART for debugging purpose.

- Secure boot core
- QoS engine
- PON TC&MAC

There are two sets of UART external interfaces. Two SoC UART instances and the UART instances in the above modules are multiplexed to these two UART external interfaces. At the same time, only two processors can be debugged.

Please refer to [UART Signals](#) for all UART interface multiplexing.

20.7.2 EJTAG Interface

The TDO line is pin strapped in order to latch test or debug select information, whether to perform DFT test or CPU level debugging. Upon de-asserting $\overline{\text{TRST}}$, the level applied at TDO is latched and controls the multiplexer.

The strap pin for JTAG/EJTAG multiplex uses:

- 0 = EJTAG
- 1 = DFT test

For debug mode, EJTAG controllers of two coherent processor cores, coherent manager, secure boot core, the processor inside 2.5GBase-T Ethernet PHY and the processor inside QoS engine are daisy chained.

20.8 On-Chip Packet Buffer

Total 7Mbits on-chip packet buffer are integrated.

All 7Mbits of the memory or only partial of the memory can be used for packet storage which is managed by FSQM inside the Buffer manager.

Partial of memory can be configured for other purposes.

21 Power Management

PRX126 supports advanced power management to control both active and idle mode power consumption. The following features are supported:

- Dynamic voltage scaling
- Two instances of PVT sensors to support Dynamic voltage Scaling
- Adaptive and static clocking frequency scaling and clock gating
- Smart on-chip and off chip buffer allocation to save chip and system power consumption

21.1 PVT (Process, Voltage and Temperature) Monitor

There are two PVT monitors integrated in the chip.

The digital process speed monitor real time tracks the digital cell performance and converts the performance into 10 bits counter per process corner. With proper co-relationship between the measured counters and actual silicon performance, a range value for the counters co-responding for fast/normal/slow processing speed can be established, software reads out the counter value and compares to the range setting, decides the achievable system clocking option, and set the clock frequency accordingly.

It also monitors the voltage (core voltage) and temperature.

The output from PVT sensor are captured to the register (readable by software).

There is overheat detection logic. The overheat detection logic can be enabled or disabled via the configuration. When output of PVT monitor is higher than the configured “overheat on” value (programmed via a 10-bit register), overheat signal is asserted. When output of PVT monitor is lower than the configured “overheat_off” value (programmable via a 10-bit register), overheat signal is de-asserted. The overheat signal is readable via a register bit. The overheat signal can be maskable with the overheat interrupt mask (configurable via a register bit).

The outputs of the overheat signals of each PVT monitor are “or” and is connected to interrupt controller.

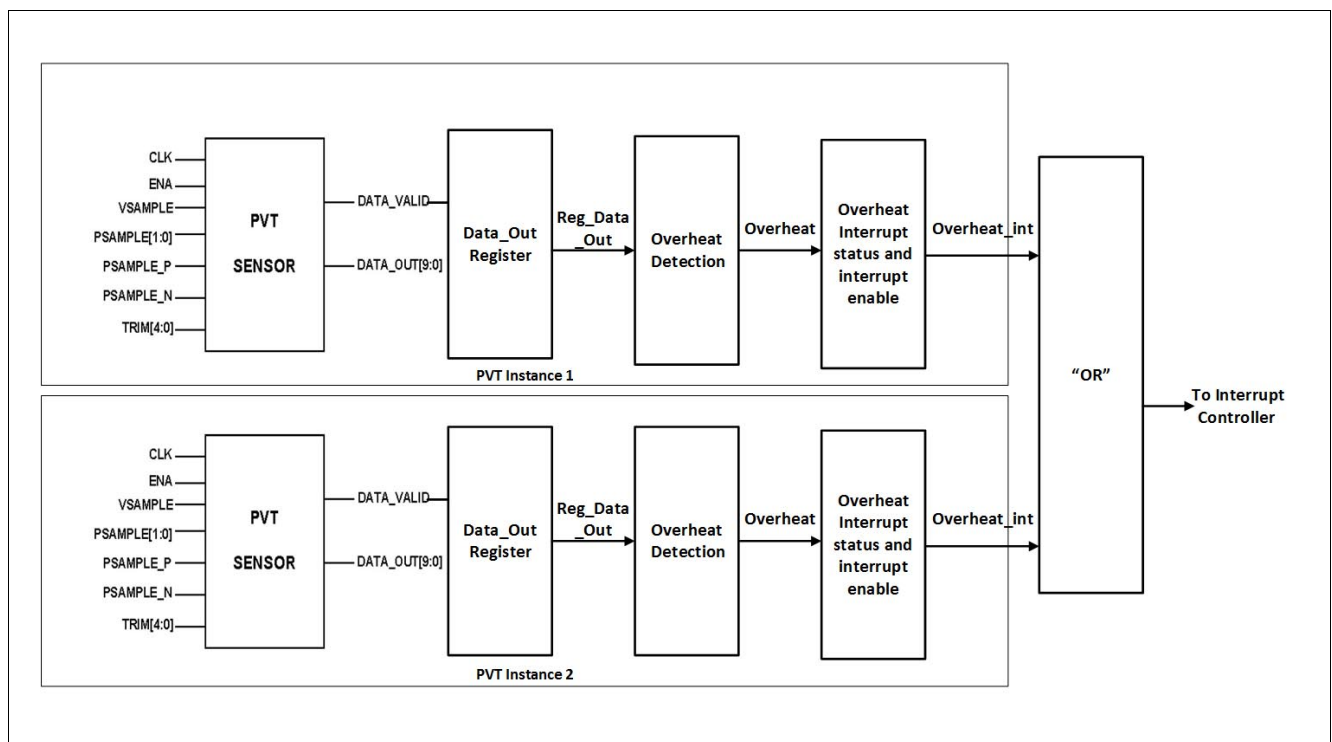
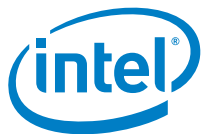


Figure 24 PVT Monitor and Overheat Detection



21.2 Dynamic Voltage Scaling

With the indications from integrated PVT sensors and the performance status from the data path hardware modules, CPU can adjust the external power supply via the command over the I2C master interface. Dynamic voltage scaling allows the reduction of both static leakage power and dynamic power consumption.

21.3 Adaptive and Static Frequency Scaling and Clock Gating

For unused functions in each application, the clocks can be gated off via the static configuration.

The maximum frequency of the major modules can be configured statically according to the performance requirements of each application.

The following modules support adaptive frequency and clock gating:

- 2.5GBASE-T EEE modes
- Ethernet Serdes EEE modes
- PCIe Power saving modes
- Power saving modes defined by XGSPON/XGPON1/NGPON2/GPON standards
- Packet switching engine according to the interface speed and activity
- Coherent processor core auto clocking gating and auto wake up by enabled interrupts

21.4 Smart On Chip and Off Chip Packet Buffer Allocation

The 7 Mbits segmented on chip packet buffer are integrated. Each segment is 128B and this ensures the very high efficient utilization with all packet sizes. For most of time, this buffer is sufficient to store and absorb the bursts and there is no or very low activity access to DDR memory. This saves both chip and system dynamic power consumption.



22 Electrical Characteristics

The electrical characteristics describe the interface timings, power supply ranges, interface driving strength and ESD robustness.

22.1 Absolute Maximum Ratings

The absolute maximum ratings are listed in [Table 38](#). The nominal values of voltages are

$$V_{DD_IO} = 1.8 \text{ V or } 3.3\text{V}$$

$$V_{AA1v8_PLL} = V_{AA1v8_XO} = V_{AA1v8_SPHY} = V_{AA1v8_PVT} = V_{AA1v8_OTP} = V_{AA1v8_DDRPLL} = 1.8 \text{ V}$$

$$V_{AA0v9_PLL} = V_{AA0v9_XO} = V_{AA0v9_SPHY} = 0.9 \text{ V}$$

$$V_{DDQ} = 1.5/1.35/1.2 \text{ V (DDR type dependent)}$$

$$V_{DD0v9} = 0.9 \text{ V}$$

Table 38 Absolute Maximum Ratings (Chip Pad Level)

Parameter	Symbol	Values			Unit	Note/ Test Condition
		Min.	Typ.	Max.		
Storage temperature	T_{stg}	-55	—	125	°C	—
Moisture Level 3 temperature	T_{ML3}	—	—	260	°C	According to IPS J-STD 020
Supply voltage Core	V_{DD0v9}	-0.5	—	+1.05	V	—
Supply voltage IO	V_{DD_IO}	-0.5	—	+3.63	V	—
Supply voltage Analog Low	V_{AA0v9}	-0.5	—	+0.99	V	—
Supply voltage Analog High	V_{AA1v8}	-0.5	—	+1.98	V	—
Supply voltage DDR PAD	V_{DDQ}	-0.5	—	+1.65	V	—
Voltage on any digital IO pin with respect to ground	V_{max}	-0.5	—	$V_{DD_IO} + 0.5$	V	—
ESD HBM Robustness	$V_{ESD,HBM}$	—	—	1000	V	According to ANSI/ESDA/JEDEC JS-001-2014
ESD CDM Robustness	$V_{ESD,CDM}$	—	—	250	V	According to JEDEC JESD22-C101

Attention: Stresses above the max. values listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods may affect device reliability. Maximum ratings are absolute ratings; exceeding only one of these values may cause irreversible damage to the integrated circuit.



22.2 Operating Range

Table 39 defines the maximum values of voltages and temperature which may be applied to guarantee proper operation. These values specify the supply voltage at package pin. It considers IR drop of package bond wire and substrate trace to ensure required voltage at pad level.

$$V_{DD_IO} = 1.8 \text{ V or } 3.3\text{V}$$

$$V_{AA1v8_PLL} = V_{AA1v8_XO} = V_{AA1v8_SPHY} = V_{AA1v8_PVT} = V_{AA1v8_OTP} = V_{AA1v8_DDRPLL} = 1.8 \text{ V}$$

$$V_{AA0v9_PLL} = V_{AA0v9_XO} = V_{AA0v9_SPHY} = 0.9 \text{ V}$$

$$V_{DDQ} = 1.5/1.35/1.2 \text{ V (DDR type dependent)}$$

$$V_{DD0v9} = 0.9 \text{ V}$$

Table 39 Operating Range (Package Pin)

Parameter	Symbol	Values			Unit	Note/ Summary of the voltage range
		Min.	Typ.	Max.		
Ambient temperature	T_A	-40	—	85	°C	—
Junction temperature	T_j	—	—	110	°C	—
Supply voltage IO 3.3 V	V_{DD_IO}	3.13	3.3	3.47	V	3.3 V $\pm 5\%$ ¹⁾
Supply voltage IO 1.8 V	V_{DD_IO}	1.71	1.8	1.89	V	1.8 V $\pm 5\%$ ¹⁾
Supply voltage Core	V_{DD0v9}	0.855	0.9	0.945	V	0.9 V $\pm 5\%$ ¹⁾
Supply voltage Analog High	V_{AA1v8}	1.71	1.8	1.89	V	1.8 V $\pm 5\%$ ¹⁾
Supply voltage Analog Low	V_{AA0v9}	0.855	0.9	0.945	V	0.9 V $\pm 5\%$ ¹⁾
Supply voltage DDR3 PAD 1.5 V	V_{DDQ}	1.425	1.5	1.575	V	1.5 V $\pm 5\%$ ¹⁾
Supply voltage DDR3L PAD 1.35 V	V_{DDQ}	1.283	1.35	1.418	V	1.35 V $\pm 5\%$ ¹⁾
Supply voltage LPDDR3/DDR4 PAD 1.2 V	V_{DDQ}	1.14	1.2	1.26	V	1.2 V $\pm 5\%$ ¹⁾
DDR Reference supply voltage	V_{REF}	$0.49 \cdot V_{DDQ}$	$0.5 \cdot V_{DDQ}$	$0.51 \cdot V_{DDQ}$	V	
DDR External termination voltage	V_{TT}	$V_{VEF} - 0.04$	V_{VEF}	$V_{VEF} + 0.04$	V	—
Ground	V_{SS}	0	—	0	V	—

1) Voltage ripple (AC Noise P2P) must be less than +/-2.5%.

Power Up Sequence

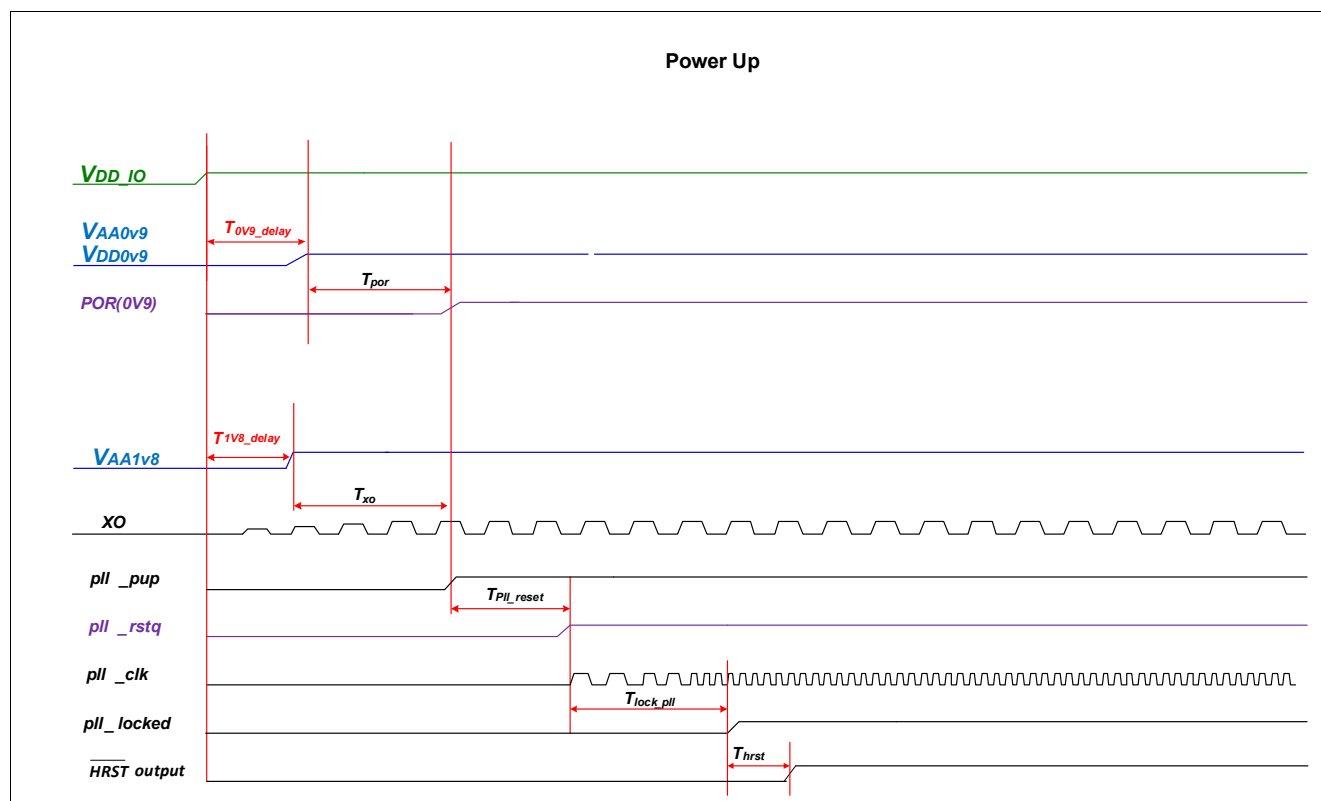


Figure 25 Power Up Sequence

Attention: a) It is recommended that the voltage domains power up simultaneously. $T_{0V9_delay} \geq 0$ and $T_{0V9_delay} \geq T_{1V8_delay}$, i.e. IO Voltage, Core Voltage, XO/PLL Voltage ramp up at least at same time or IO Voltage and XO/PLL voltage ramp up earlier than core voltage.

b) T_{por} is POR reset delay, on-chip POR delay is up to 120 ms.

c) T_{xo} , T_{pll_reset} , T_{lock_pll} , T_{hrst} are SoC internal and independent of the system design. System design does not require to achieve them.

Attention: Minimum Supply Voltage ramp up time for all power supply shall $\geq 10\mu s$.



22.3 DC Characteristics

This chapter includes the DC characteristics of the I/O interfaces.

Table 40 DC Characteristics

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
3.3 V I/O Region						
High-Level Input Voltage	V_{IH}	2.0	–	$V_{DD_IO} + 0.3$	V	
Low-Level Input Voltage	V_{IL}	– 0.3	–	0.8	V	$V_{out} \leq V_{OL(max)}$
High-Level Output voltage	V_{OH}	2.4	–	–	V	
Low-Level Output voltage	V_{OL}	–	–	0.4	V	
Input Hysteresis voltage	V_{HYS}	0.2	–	–	V	
1.8 V I/O Region						
High-Level Input Voltage	V_{IH}	0.65* V_{DD_IO}	–	$V_{DD_IO} + 0.3$	V	
Low-Level Input Voltage	V_{IL}	– 0.3	–	0.35* V_{DD_IO}	V	$V_{out} \leq V_{OL(max)}$
High-Level Output voltage	V_{OH}	$V_{DD_IO} - 0.4$	–	–	V	
Low-Level Output voltage	V_{OL}	–	–	0.4	V	
Input Hysteresis voltage	V_{HYS}	0.1* V_{DD_IO}	–	–	V	
DDR4						
I/O Supply Voltage DDR4 @1.2 V	V_{DDQ}	1.14	1.2	1.26	V	
High-Level Input Voltage	V_{IH}	$V_{REF} + 0.065$	–	V_{DDQ}	V	–
Low-Level Input Voltage	V_{IL}	-0.3	–	$V_{REF} - 0.065$	V	–
High-Level Output voltage	V_{OH}	0.9* V_{DDQ}	–	–	V	
Low-Level Output voltage	V_{OL}	–	–	0.1* V_{DDQ}	V	
LPDDR3						
I/O Supply Voltage LPDDR3 @1.2 V	V_{DDQ}	1.14	1.2	1.26	V	
High-Level Input Voltage	V_{IH}	$V_{REF} + 0.1$	–	V_{DDQ}	V	–
Low-Level Input Voltage	V_{IL}	-0.3	–	$V_{REF} - 0.1$	V	–
High-Level Output voltage	V_{OH}	0.9* V_{DDQ}	–	–	V	



Electrical Characteristics

Table 40 DC Characteristics (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Low-Level Output voltage	V_{OL}	–	–	0.1* V_{DDQ}	V	

DDR3L

I/O Supply Voltage DDR3L @1.35 V	V_{DDQ}	1.283	1.35	1.418	V	
High-Level Input Voltage	V_{IH}	$V_{REF} + 0.09$	–	V_{DDQ}	V	–
Low-Level Input Voltage	V_{IL}	-0.3	–	$V_{REF} - 0.09$	V	–
High-Level Output voltage	V_{OH}	0.8* V_{DDQ}	–	–	V	
Low-Level Output voltage	V_{OL}	–	–	0.2* V_{DDQ}	V	

DDR3

I/O Supply Voltage DDR3 @1.5 V	V_{DDQ}	1.425	1.5	1.575	V	
High-Level Input Voltage	V_{IH}	$V_{REF} + 0.09$	–	V_{DDQ}	V	–
Low-Level Input Voltage	V_{IL}	-0.3	–	$V_{REF} - 0.09$	V	–
High-Level Output voltage	V_{OH}	0.8* V_{DDQ}	–	–	V	
Low-Level Output voltage	V_{OL}	–	–	0.2* V_{DDQ}	V	

Typical. Power Supply Current

10G-SFP	$I_{IO}^{1)}$	–	tbd	–	mA	3.3 V IO, $T_A = 25^\circ\text{C}$
	$I_{0v9_CORE}^{2)}$	–	tbd	–	mA	0.9 V Digital Core, $T_A = 25^\circ\text{C}$
	$I_{0v9_Analog}^{1)}$	–	tbd	–	mA	0.9 V Analog, $T_A = 25^\circ\text{C}$
	$I_{1v8_Analog}^{1)}$	–	tbd	–	mA	1.8 V Analog, $T_A = 25^\circ\text{C}$
	$I_{DDR}^{1)}$	–	tbd	–	mA	1.2 V DDR, $T_A = 25^\circ\text{C}$

1) Please consider 25% increase on top of the typical current value as maximum current for power regulator design.

2) Please consider 35% increase on top of the typical current value as maximum current for power regulator design.

Note: Please use tbd as maximum Thermal Design Power (TDP) of the PRX126 for system thermal design.

22.3.1 Power-on Reset Detection

The following table includes the electrical parameters of the power-on reset (POR) detection.



Table 41 POR Characteristics

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Detection Threshold (0.9 V)	$V_{\text{DET_POR}}$	0.64		0.68	V	5% accuracy
Maximum Ramp Up Rate	dV_{DD}/dt	—	—	0.1	V/us	
Delay for activation of reset	$t_{\text{ACT_POR}}$	10	20	30	us	50% accuracy
Delay for deactivation of reset	$t_{\text{DEACT_POR}}$	41	82	123	ms	50% accuracy

22.3.2 Dying Gasp Detection

The following table includes the electrical parameters of the dying gasp detection.

Table 42 Dying Gasp Detection Characteristics

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Detection Threshold	$V_{\text{DET_DG}}$		1.5		V	5% accuracy
Hysteresis	$V_{\text{HYS_DG}}$	0	79	131	mV	5% accuracy

22.3.3 Built-in Temperature Sensor

The following table includes the electrical parameters of the integrated temperature sensor.

Table 43 Temperature Sensor Characteristics

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Temperature Range	T_{range}	-40		125	°C	
Resolution			10		bits	
Accuracy (Post Trim)		-1	—	+1	°C	
Accuracy (Untrimmed)		-5	—	+5	°C	

22.4 AC Characteristics

$T_A = 0$ to 70°C , $V_{DD_IO} = 3.3/1.8 \text{ V} \pm 5\%$, $V_{SS} = 0 \text{ V}$

Timing measurements are made at minimum V_{IH} for a logical 1 and at maximum V_{IL} for a logical 0.

The AC testing input/output waveforms are shown in Figure 26. The load capacitors are according to the specific interface standard, all non-specified interfaces use 30 pF as assumed loading.

Attention: All timing parameters specified in this release of the document may be updated after finished chip characterization.

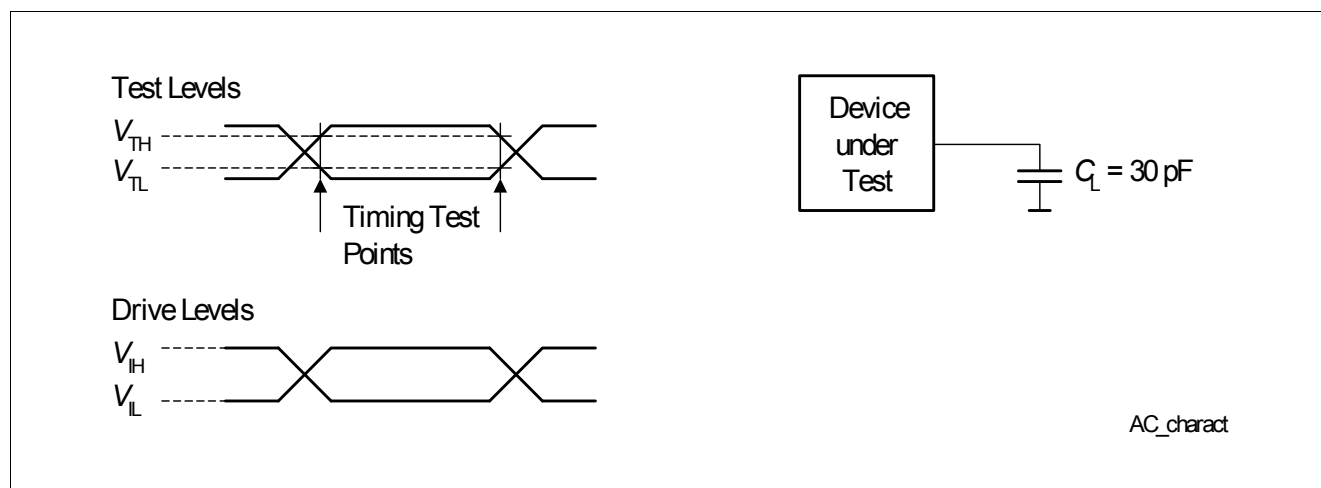


Figure 26 Input/Output Waveform for AC Tests

22.5 Timing Characteristics

The timing characteristics describes the detailed interface timings.

22.5.1 QSPI Interface

This chapter defines the QSPI Interface timings.

Attention: All timing parameters specified in this release of the document may be updated after finished chip characterization.

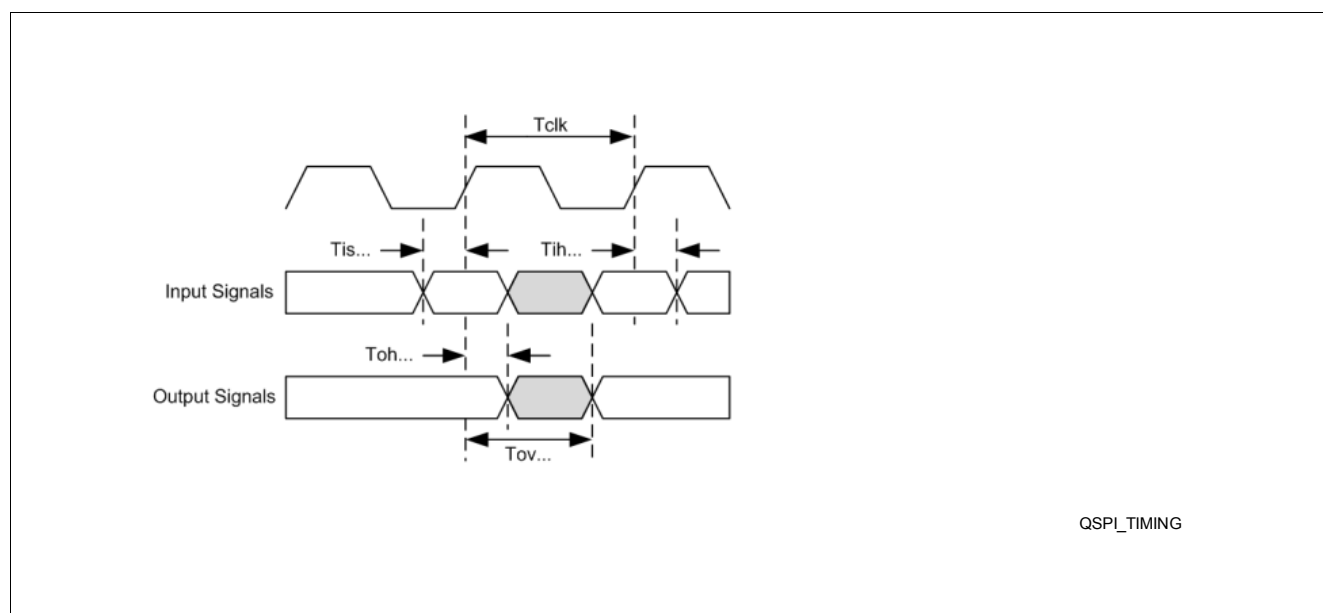


Figure 27 QSPI Interface Timing

Table 44 Timing Values QSPI Interface

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
CLK period ¹⁾	t_{CLK}	10		–	ns	–
CLK duty cycle	$t_{CLKh}/t_{CLK}, t_{CLKl}/t_{CLK}$	40	50	60	%	–
RX setup	t_{is}	$35\% \cdot t_{CLK}$	–	–	ns	–
RX hold	t_{ih}	$5\% \cdot t_{CLK}$	–	–	ns	–
TX delay	t_{oh}	$5\% \cdot t_{CLK}$	–	$50\% \cdot t_{CLK}$	ns	–

1) QSPI clock supports range of frequencies, up to 100 MHz.

22.5.2 I2C Interface

The I2C interface timing is shown in [Figure 28](#).

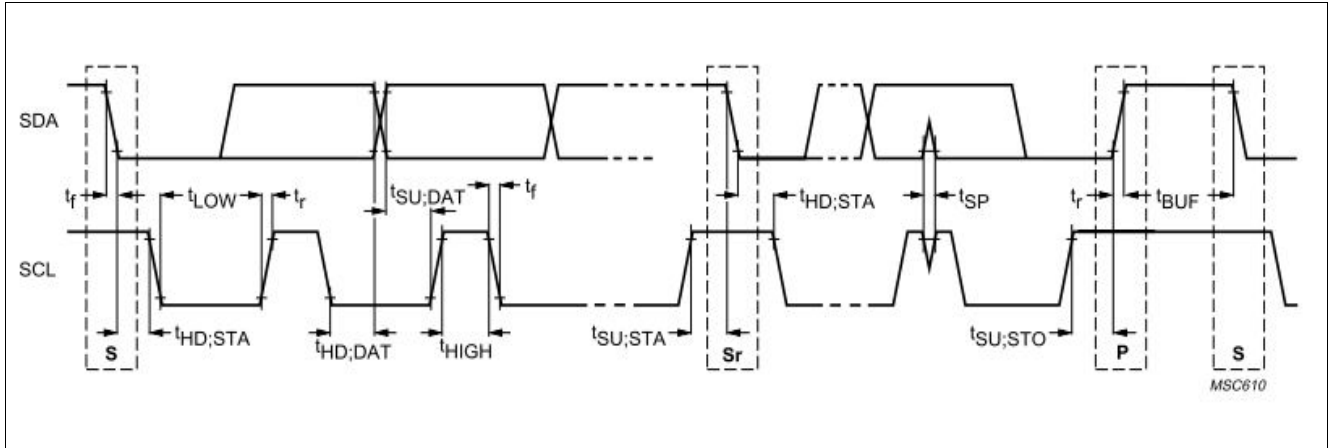


Figure 28 I2C Timing

The timing values are described in [Table 45](#).

Attention: All timing parameters specified in this release of the document will be updated based on the chip characterization.

Table 45 I2C Timing

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Standard Mode						
SCL frequency	f_{SCL}	0	–	100	kHz	–
Setup time data to shift clock	$t_{\text{SU,DAT}}$	250	–	–	ns	–
Hold time data to shift clock	$t_{\text{HD,DAT}}$	0	–	3450	ns	–
Setup time START to shift clock	$t_{\text{SU,STA}}$	4700	–	–	ns	–
Hold time START, STOP to shift clock	$t_{\text{HD,STA/STO}}$	4000	–	-	ns	–
Low time	t_{LOW}	4700	–	–	ns	–
High time	t_{HIGH}	4000	–	–	ns	–
Rising time	t_{r}	–	–	1000	ns	–
Falling time	t_{f}	–	–	300	ns	–
Bus free time	t_{BUF}	4700	–	–	ns	–
Fast Mode						
SCL frequency	f_{SCL}	0	–	400	kHz	–
Setup time data to shift clock	$t_{\text{SU,DAT}}$	100	–	–	ns	–
Hold time data to shift clock	$t_{\text{HD,DAT}}$	0	–	900	ns	–
Setup time START to shift clock	$t_{\text{SU,STA}}$	600	–	–	ns	–
Hold time START, STOP to shift clock	$t_{\text{HD,STA/STO}}$	600	–	-	ns	–



Table 45 I2C Timing (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Low time	t_{LOW}	1300	–	–	ns	–
High time	t_{HIGH}	600	–	–	ns	–
Rising time	t_r	20	–	300	ns	–
Falling time	t_f	12	–	300	ns	–
Bus free time	t_{BUF}	1300	–	–	ns	–

22.5.3 Serdes Interface

This section describes the AC characteristics of the Serdes Interface on the PRX126. The interface can operate at 1.25 GT/s, 3.125 GT/s, and 10.3125 GT/s.

22.5.3.1 Transmit Timing Characteristics

Figure 29 shows the timing diagram of the transmit interface at the PRX126. It is referred to by **Table 46**, which specifies the timing requirements.

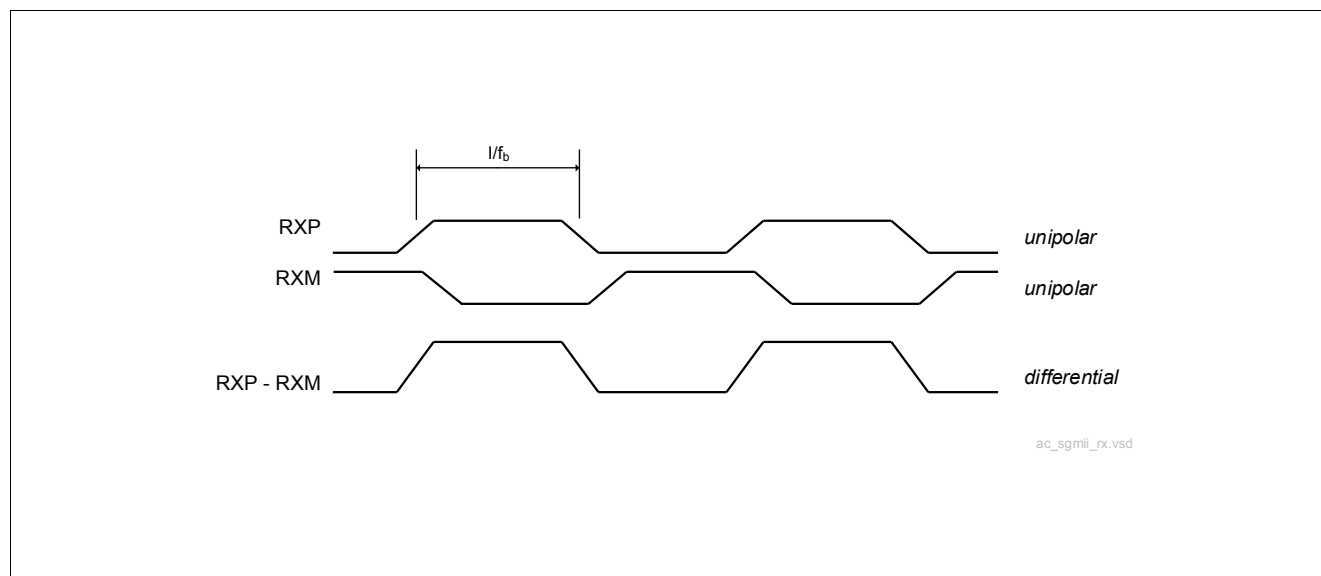


Figure 29 Transmit Timing Diagram (shows alternating data sequence)

Table 46 Transmit Timing Characteristics

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Transmit baud rate	T_{baud}	-100 ppm	f_b	+ 100 ppm	Mbaud	$f_b = 1.25 \text{ GT/s}$ or 3.125 GT/s or 10.3125 GT/s
Output impedance (differential)	$T_{R_{\text{diff}}}$	90	100	110	Ω	–
Length variation within the differential pair	$T_{L_{\text{diff}}}$	–	–	5	mils	–
Output impedance (common)	$T_{R_{\text{cm}}}$	45	50	55	Ω	–
Differential mode return loss	$T_{S_{\text{dd}22}}$	–	–	tbd	dB	–
Common mode return loss	$T_{S_{\text{cc}22}}$	–	–	tbd	dB	–
Common mode noise	$T_{N_{\text{cm}}}$	–	–	5% of $T_{V_{\text{diff}}}$	mVppd	–

22.5.3.2 Receive Timing Characteristics

Figure 30 shows the timing diagram of the receive Serdes interface on the PRX126. It is referred to by Table 47, which specifies the timing requirements.

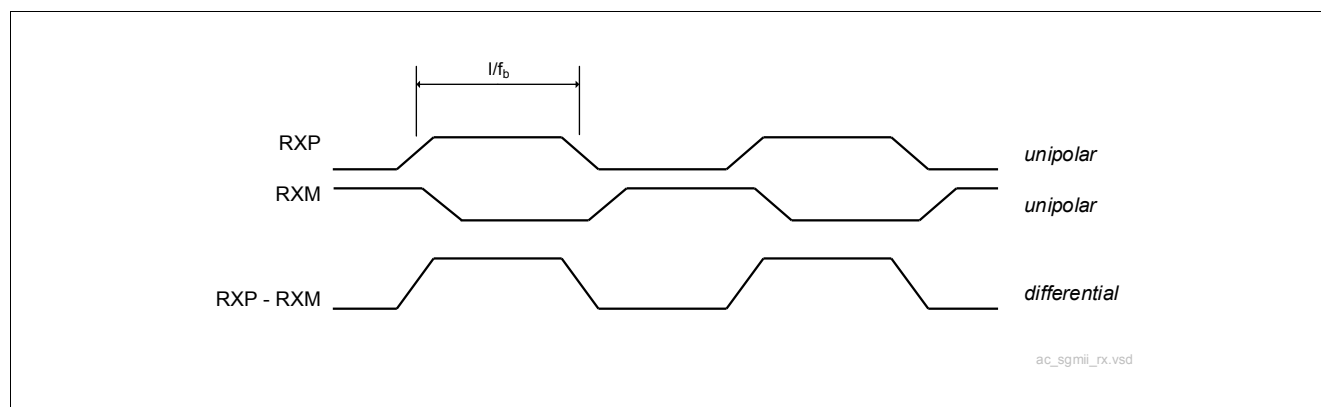


Figure 30 Receive Timing Diagram (alternating data input sequence)

Table 47 Receive Timing Characteristics of the Serdes

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Receive baud rate	R_{baud}	-100 ppm	f_b	+ 100 ppm	Baud	$f_b = 1.25 \text{ GT/s}$ or 3.125 GT/s or 10.3125 GT/s
Input impedance (differential)	$R_{R_{diff}}$	90	100	110	Ω	—
Input impedance (common)	$R_{R_{cm}}$	45	50	55	Ω	—
Differential mode return loss	$R_{S_{dd11}}$	—	—	tbd	dB	—
Common mode return loss	$R_{S_{cc11}}$	—	—	tbd	dB	—

22.5.4 Test Interface

The test interface is used to debug the CPU.

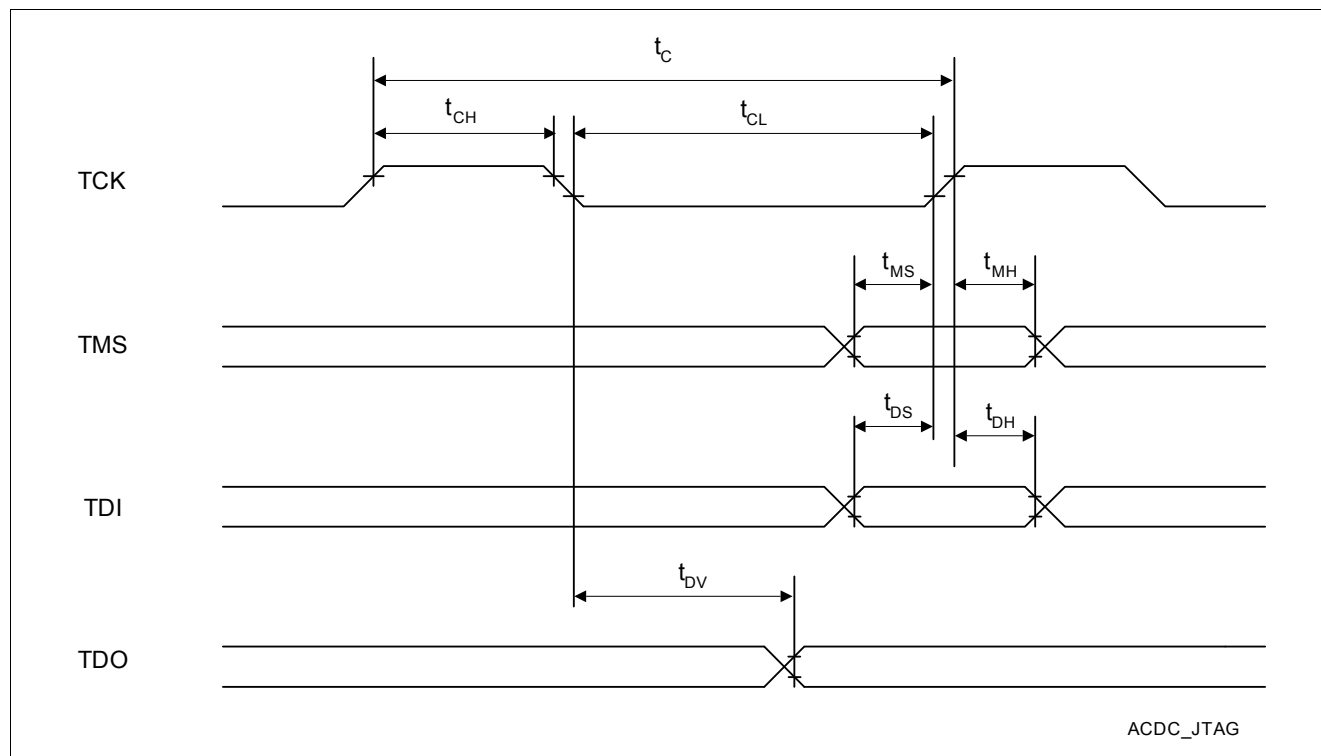


Figure 31 Test Interface Timing

The timing values are described in [Table 48](#) and [Table 49](#).

Attention: All timing parameters specified in this release of the document may be updated after finished chip characterization.

Table 48 Test Interface Clock

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
TCK Clock Period	t_C	100	—	—	ns	—
TCK High Time	t_{CH}	40	—	—	ns	—
TCK Low Time	t_{CL}	40	—	—	ns	—

Attention: All timing parameters specified in this release of the document may be updated after finished chip characterization.

Table 49 JTAG Timing

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
TMS setup time	t_{MS}	40	—	—	ns	—
TMS hold time	t_{MH}	40	—	—	ns	—
TDI setup time	t_{DS}	40	—	—	ns	—
TDI hold time	t_{DH}	40	—	—	ns	—



Table 49 JTAG Timing (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Hold: $\overline{\text{TRST}}$ after TCK	t_{HD}	10	–	–	ns	–
TDO valid delay	t_{DV}	–	–	40	ns	–



22.5.5 Crystal Specification

The crystal is attached to the PRX126 SoC and must follow the specification in [Table 50](#).

Table 50 Specification of the 40 MHz Crystal

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Nominal Frequency	—	—	40	—	MHz	Oscillation mode is fundamental.
Total Frequency Stability	—	-30	—	+30	ppm	Refer to sum of all effects: eg. general tolerance, aging, temperature dependency
Series Resonant Resistance	—	—	—	40	Ω	—
Drive Level	—	0.08	0.10	0.2	mW	—
Load Capacitance	C_L	16	—	30	pF	—
Shunt Capacitance	C_0	—	—	7	pF	—
XTAL1 Voltage Input Level	—	0.5	—	2.1	V	—



23 Package Outlines

The product is assembled in PG-VF2RBGA-287-9.4x8.6-C-1A package which complies with regulations requiring lead free material. It is ROHS 6/6 compliant.

PG-VF2RBGA: Plastic Green Very Thin Profile Flip Chip Fine Pitch Rectangle Ball Grid Array Package

Table 51 Thermal Resistance Package Parameter

Item	Description/Value
Package Type	PG-VF2RBGA-287-9.4x8.6-C-1A
Thermal Resistance Junction to Ambient	$R_{th, JA} = 21.1 \text{ K/W}$
Thermal Resistance Junction to Case - Top	$R_{th, JCTop} = 0.08 \text{ K/W}$
Thermal Resistance Junction to Case - Bottom	$R_{th, JCbott} = 11.4 \text{ K/W}$

Note: The above values are based on JEDEC standard thermal simulation condition.

Table 52 Thermal Resistance Package Parameter under System Design Condition

Item	Description/Value
Package Type	PG-VF2RBGA-287-9.4x8.6-C-1A
Thermal Resistance Junction to Case - Top	$R_{th, JCTop} = \text{TBD K/W}$
Thermal Resistance Junction to Case - Bottom	$R_{th, JCbott} = \text{TBD K/W}$

Note: Please refer to Hardware Design Guide [\[5\]](#) for the detailed system design simulation setup.

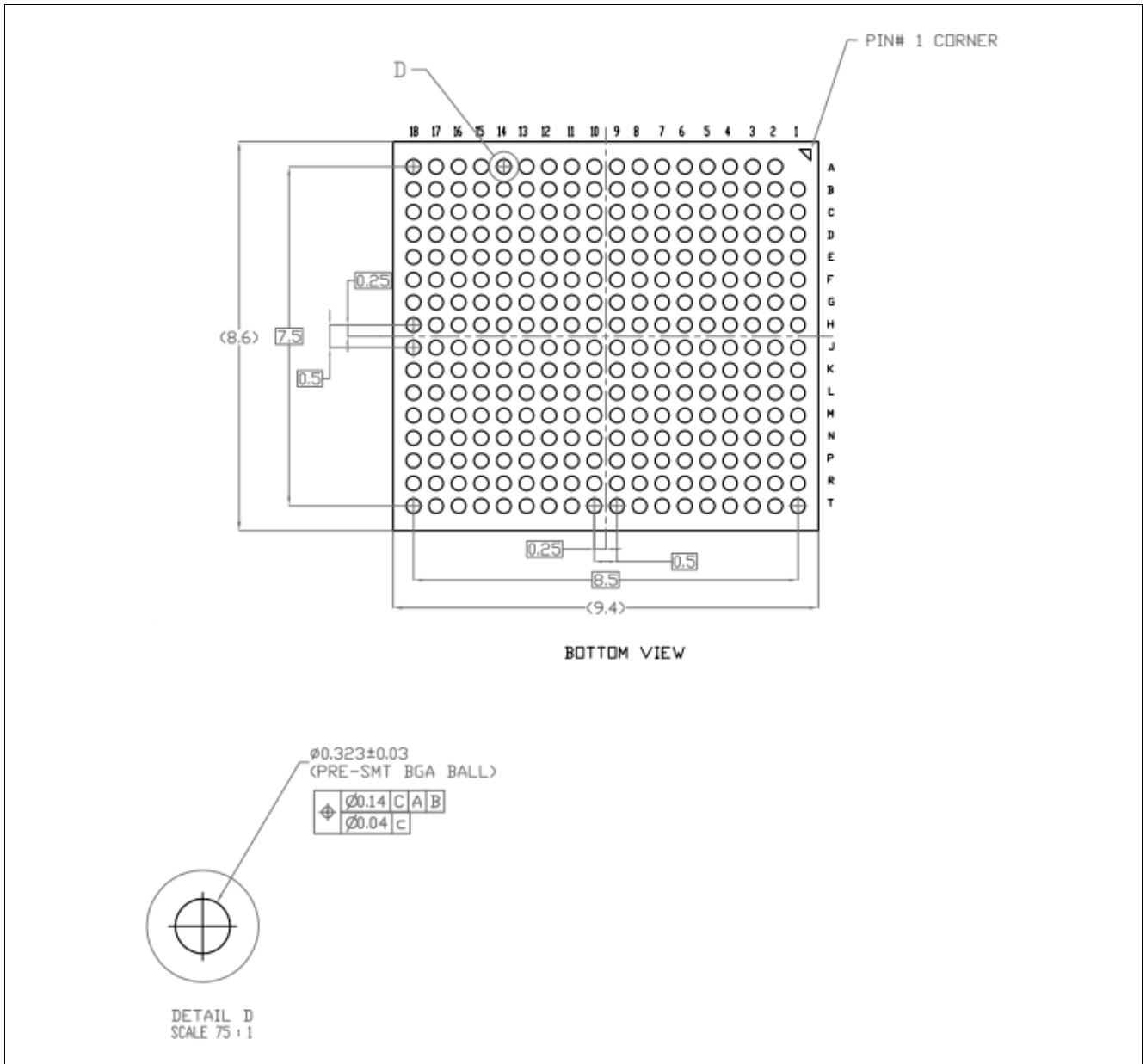


Figure 32 PG-VF2RBGA-287 (Bottom View)

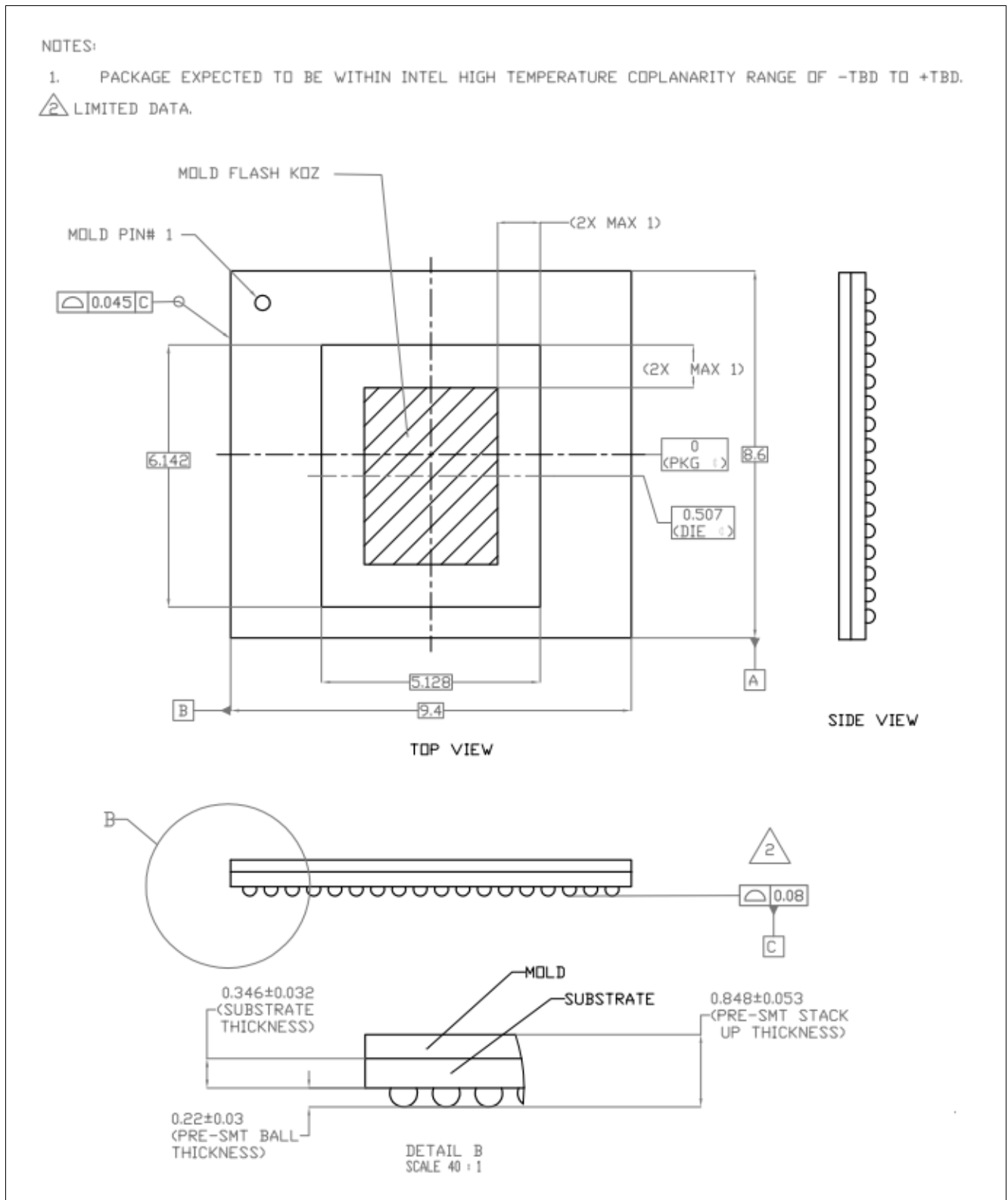


Figure 33 PG-VF2RBGA-287 (Top & Side View)

Note: The package is sourced from multiple vendors.

Package description, package handling, PCB and board assembly information is available on request.



References

- [1] Intel® 10G PON Chipset Preliminary System Overview Rev. 1.0
- [2] Intel® 10G PON Chipset TrustWorld Secure Boot Application Note Rev. 1.0 (in preparation)
- [3] MIPS32 interAPTIV* Multiprocessing System Software User's Manual, Revision 01.04 or later
- [4] Intel® Universal Gateway Software 7.2 Software Overview Rev. 1.0
- [5] Intel® 10G PON Chipset PRX126 (PRX126A1BI) Reference Boards Hardware Design Guide Rev. 1.0 (in preparation)
- [6] Intel® 10G PON Chipset Preliminary Software Overview Rev. 1.0 (in preparation)
- [7] Intel® 10G PON Chipset Preliminary User's Manual Programmer's Reference Rev. 1.0 (in preparation)



Terminology

A

AES	Advanced Encryption Standard
ANSI	American National Standards Institute

B

BL	Boot Loader
----	-------------

C

CM	Crypto Module
CM	Coherence Manager
CPS	Coherent Processing Subsystem
CQE	Carrier Grade QoS Engine

D

DC Port	Direct Connect Port
DECT	Digital Enhanced Cordless Telecommunications (Digital European Cordless Telecommunications), usually known by the acronym DECT, is a standard primarily used for creating cordless telephone systems
DMA	Direct Memory Access
DSA	Digital Signature Algorithm

E

ECC	Error Correction Code
EEE	Energy-Efficient Ethernet
EMI	Electromagnetic interference
EP	PCIe End Point
EJTAG	Enhanced Joint Test Action Group

F

FMT	Fixed Memory Translation
FSB	First Stage Bootloader
FSQM	Free Segment Queue Manager
FTTx	Fiber to the X
FXS	Foreign eXchange Subscriber interface is the port that actually delivers the analog line to the subscriber

G

10G-KR	Backplane Ethernet 10 Gbit/s over printed circuit boards
GEM	GPON Encapsulation Method
GPIO	General Purpose Input/Output
GPTC	General Purpose Timer Controller
GPON	Gigabit Passive Optical Network
G.INT	Interface between the link layer and the physical layer for digital subscriber line (DSL) transceivers

H



HSTL	High-Speed Transceiver Logic
I	
I2C	Inter-Integrated Circuit
IA	Image Authenticator
iAptiv	interAptiv
J	
JTAG	Joined Test Action Group
L	
LAN	Local Area Network
LED	Light-emitting diode
LPI	Low Power Idle
M	
MAC	Media Access Controller
MACsec	IEEE MAC Security standard
MDIO	Management Data Input/Output
MPE	Multi-Core Processing Engine
N	
NPU	Network Processing Unit
NGPON	Next Generation Passive Optical PON
NTR	Network Time Reference
NVM	Non-Volatile Memory
O	
OAM	Operations, Administration, and Maintenance
OMCI	ONU management and control interface
OS	Operating System
OTP	One Time Programmable Memory
ONU	Optical Network Unit
P	
PCB	Printed Circuit Board
PCIe	Peripheral Component Interconnect Express
PCM	Pulse-code modulation
PCS	Physical Coding Sublayer
PMAC	Pseudo Media Access Controller
PLL	Phase-locked loop
POE	Power over Ethernet
PON	Passive Optical Network
POR	Power on Reset
PTP	Precise Transport Protocol
PPS	Pulse Per Second
Q	



QoS	Quality of Service
QSPI	Quad Serial Peripheral Interface
R	
RC	PCIe Root Complex
RED	Random Early Detection
RoT	Root of Trust
RTP	Real time transport protocol
RX	Receive
RXAUI	10 Gigabit Reduced Attachment Unit. Interface
S	
SB	Secure Boot
SBCR	Secure Boot Confidentiality Root
SFP	Small form factor pluggable
SFU	Single Family Unit
SLIC	Subscriber line interface circuits
SHA	Secure Hash Standard
SMP	Symmetric Multi-Processing
SoC	System on Chip
SPE	Secure Platform Engine
SPI	Serial Peripheral Interface
SSC	Spread-Spectrum Clocking
SSID	Service set identifier
SSO	Serial Shift Output
SyncE	Synchronous Ethernet
T	
TC	Thread Context
TC	Transmission Convergence
TDM	Time-division multiplexing
TEE	Trusted Execution Environment
TEP	Trusted Execution Processor
TRNG	True Random Number Generator
TX	Transmit
U	
UART	Universal Asynchronous Receiver/Transmitter
UGW	Universal GateWay
USXGMII	Universal Serial Media Independent Interface
V	
VoIP	Voice over IP
W	
WAN	Wide Area Network



WDT	Watchdog Timer
Wi-Fi	Wireless Local Area Network
WRED	Weighted Random Early Detection
WoL	Wake on LAN
X	
XFI	10G serial electrical interface
XGMII	10 Gigabit Media Independent Interface
XGPON	10-Gigabit Capable Passive Optical PON
XGSPON	10-Gigabit Capable Symmetrical Passive Optical PON
XO	Crystal oscillator